BIRMINGHAM CITY University

SignLink: A Smart Sign Language Translation Framework Using EMG Signal Processing, Machine Learning & Mobile Application

Sombatla Truc Deydeepya Lakshmi

(25195661/1)

CMP6200/DIG6200 Individual Undergraduate Project 2025-2026

B.Sc. (Honors) Computer Science

Submitted by June 2026

A thesis submitted to **Birmingham City University** in partial fulfilment of the requirements of the degree of Bachelor of Science.

Abstract

This dissertation presents the design, development and evaluation of SignLink, an intelligent sign language translation system that integrates wearable Electromyography (EMG) sensing technology, machine learning-based gesture recognition and a mobile application platform to assist. The final objective of this initiative is to translate hand gestures into understandable text and speech in real time, thereby facilitating more effective communication between sign language users and non-sign language users.

The proposed system combines EMG signal acquisition through a low-cost wearable alternative manually built and tested to commercial EMG devices (such as the Myo Armband), artificial intelligence-driven gesture classification and a dedicated Android application to provide an accessible communication solution. The mobile application serves as the primary interface of the system, offering features such as text-to-speech, real-time translation, communication assistance and local data management. To enable accurate gesture recognition, the machine learning model was trained and evaluated using electromyographic signal data, allowing the system to identify and classify sign language gestures for translation into meaningful outputs.

To establish a reliable baseline, we first validated the machine learning component of this study using an example, publicly available electromyography dataset of ASL fingerspelling collected via a Myo Armband. This approach provides a reliable benchmark for assessing the effectiveness of the proposed artificial intelligence framework while demonstrating the potential capabilities of the intended wearable hardware platform currently under development.

The findings of this study highlight the potential of integrating wearable sensing technologies, mobile computing and artificial intelligence to improve communication accessibility for individuals with hearing and speech impairments. Furthermore, the project contributes a scalable framework for sign language translation that combines intelligent gesture recognition with mobile application technologies. Future work includes expanding the supported gesture vocabulary from individual alphabet gestures to complete words and phrases, refining the wearable sensing platform, collecting a dedicated EMG dataset and implementing more advanced machine learning models to improve recognition accuracy and real-time translation performance.

Acknowledgement

I would like to express my sincere gratitude to all those who have contributed to the successful completion of this dissertation and the SignLink project. First and foremost, I would like to thank my supervisor, Dr. Vi Chi Thanh, for his guidance, support and valuable feedback throughout this research. His advice and encouragement were instrumental in shaping the direction and development of this project.

I would also like to express my sincere thanks to Mr. Nguyen Huu Nghia for his guidance and assistance with the hardware development of the project. His insights and technical advice regarding the design and implementation of the wearable EMG sensing platform were greatly appreciated and contributed significantly to the progress of this work. I would also like to extend my sincere appreciation to my friend Satwik for his valuable advice and insights regarding the machine learning aspects of this project. His guidance on data preparation, training approaches and model development provided useful perspectives that helped shape the artificial intelligence component of SignLink.

My heartfelt gratitude goes to my friends and volunteers who generously dedicated their time to participating in usability and user experience testing. Their valuable feedback, suggestions and observations provided important perspectives that helped improve the overall quality, usability and functionality of the SignLink application. I would also like to thank my friend Andrew for his constant encouragement and support throughout the development of this project. His positivity and motivation were greatly appreciated during challenging stages of the research and implementation process.

Finally, I am especially grateful to my family for their unwavering support, patience and encouragement throughout my project. Their belief in my abilities and their constant motivation has been a source of strength during both the challenges and achievements encountered throughout this journey. To all those who contributed, whether directly or indirectly, I extend my sincere appreciation and gratitude.

Table of Contents

Abstract 2
Acknowledgement 3
Glossary 6
List of Figures 6
List of Tables 7
List of Code Snippets 7
1.0 Introduction 8
1.1 Problem Statement 8
1.2 Scope 9
1.3 Significance 9
1.4 Project Aim and Objectives 9
1.4.1 Project Aim 9
1.4.2 Project Objectives 10
2.0 Review of Existing Knowledge 12
2.1 Sign Language and Sign Language Translation Technologies 12
2.2.1 Vision-Based Translation Systems 13
2.2.2 Sensor-Based Translation Systems 13
2.2 Electromyography (EMG) for Gesture Recognition 14
2.3 Machine Learning for Sign Language Recognition 15
2.4 Research Gap 16
3.0 Method and Implementation 17
3.1 UI/UX Design and User Testing 17
3.1.1 Evaluation Framework and Participants 19
3.1.2 Data Collection Methods 21
3.2 Mobile Application Development 21
3.3 Wearable Hardware Development 25
3.4 Machine Learning Development 30
3.4.1 Dataset Identification 30

3.4.2 Data Cleaning and Preprocessing 30
3.4.3 Exploratory Data Analysis 31
3.4.4 Feature Selection and Engineering 31
3.4.5 Splitting of Data into Training and Testing 32
3.4.6 Training the model 32
3.5 Final UI and UX App Testing 32
4.0 Evaluation 33
4.1 UI/UX Prototype Evaluation 33
4.1.1 Quantitative Results 33
4.3.2 Qualitative Results (Think Aloud & Interviews) 34
4.2 Machine Learning Model Evaluation 34
4.3 Final UI/UX Prototype Evaluation 36
4.3.1 Quantitative Results 36
4.3.2 Qualitative Results (Think Aloud & Interviews) 37
4.4 Final Application Evaluation 37
5.0 Conclusions 38
5.1 Limitations 38
5.2 Future Work 39
6.0 References 40
7.0 Appendix 43
7.1 Appendices of UI/UX Design Section 43
7.1.1 Gantt Chart 43
7.1.2 Low Fidelity Diagrams 44
7.1.3 High Fidelity Diagrams 46
7.1.4 Testing Consent Forms 51
7.1.5 Results from Form 56
7.2 App Screens 66
7.3 Model Training Code 71
7.4 Appendices of Final UI/UX Design Section 75
7.4.1 Testing Consent Forms 75

7.1.5 Results from Form 91

Glossary

AI	Artificial Intelligence
ML	Machine Learning
EMG	Electromyography
sEMG	Surface Electromyography
IMU	Inertial Measurement Units
ASL	American Sign Language
UI	User Interface
UX	User Experience

List of Figures

Figure 1: Vision-Based vs. Sensor-Based Approaches 13
Figure 2: Hand Muscles (Retrieved from Backyard Brains: Signal Classification Experiment) 14
Figure 3: Example Electrode Placement for EMG Acquisition (Same source as figure 2) 15
Figure 4: Gantt Chart 17
Figure 5: Hierarchical Task Analysis 19
Figure 6: Screens set 1 22
Figure 7: Screens set 2 23
Figure 8: Screens set 3 24
Figure 9: Screens set 4 25
Figure 10: Setup of V1 26
Figure 11: V1 Signal 26
Figure 12: V1 Rough Code 27
Figure 13: Setup of V2 28
Figure 14: Code and Signal 28
Figure 15: V2 In Real Life 29

Figure 16: Setup of V3 30
Figure 17: Performance Metrics VS Seed Number 35
Figure 18: Average Metrics of all Seeds 36

List of Tables

Table 1: MoSCoW 18
Table 2: Participant Data of Pre-Testing 20
Table 3: Participant Data of Final Testing 20

List of Code Snippets

Code 1: Control Random Forest Classifier (with no feature selection) 71
Code 2: Random Forest Classifier (on selected features) 71
Code 3: Control Hyper-parameter Tuning & CV Testing (to make sure model was accurate on all folds and not just one specific instance) 72
Code 4: Final Tuning (run on different random states to make sure accuracy is consistent throughout) 74

1.0 Introduction

Communication is a basic human right that is crucial for social inclusion. In this regard, the United Nations Convention on the Rights of Persons with Disabilities acknowledges sign languages as valid forms of communication and highlights the importance of accessible communication (*United Nations, 2006*). Despite the progress in technology, significant obstacles remain between sign language users and individuals who do not understand sign language. To address this communication divide, automated sign language translation has become an important area of research, utilizing recent advancements in artificial intelligence, wearable sensing, and mobile computing to transform gestures into real-time text and speech outputs.

The following section presents the problem statement, project scope, significance, aims and objectives that form the foundation of the SignLink project.

1.1 Problem Statement

According to the World Health Organization, more than 1.5 billion individuals worldwide suffer from hearing loss, with 430 million needing rehabilitation services *(WHO, 2026)*. Although sign language serves as the main mode of communication for many, the reliance on human interpreters is frequently obstructed by high costs and limited availability. This results in a continuous communication gap, compelling those who are deaf or hard-of-hearing to confront daily obstacles in educational, professional, and healthcare settings.

To address the environmental and lighting constraints associated with conventional vision-based translation systems *(Abdullahi & Chamnongthai, 2022)*, this project presents SignLink: an advanced translation system that employs wearable electromyography (EMG) sensing alongside machine learning (ML) to effectively monitor muscle activity during hand gestures. By combining this effective sensing technique with a supportive Android application, SignLink translates identified sign gestures into immediate text and speech outputs, thereby closing the communication divide with a practical and accessible solution.

1.2 Scope

The scope of this project encompasses the design, development and evaluation of SignLink and that includes the development of a ML-based gesture recognition framework, the implementation of a mobile application for communication assistance and the investigation of a wearable EMG sensing platform intended to support real-time gesture acquisition.

Due to limitations in time and resources, the machine learning model was trained and assessed using a publicly available Myo Armband EMG dataset for American Sign Language (ASL) fingerspelling alphabet gestures instead of the custom wearable prototype. Features like continuous recognition, real-time implementation on custom hardware, multilingual capabilities, and comprehensive user studies are beyond the current scope and are planned for future development.

1.3 Significance

The significance of this project lies in its potential to improve communication accessibility for individuals with hearing and speech impairments hence contributing to the field of assistive technology through the integration of ML, wearable sensing technologies and mobile computing. It also aligns with the United Nations Sustainable Development Goal 10 (Reduced Inequalities) by supporting accessibility and technological inclusion *(United Nations, 2015)*.

1.4 Project Aim and Objectives

The following aim and objectives define the intended outcomes of the project and provide a structured framework for its design, development and evaluation.

1.4.1 Project Aim

The aim of this project is to design, develop and evaluate SignLink, an intelligent sign language translation system that utilizes ML and EMG-based gesture recognition to translate sign language gestures into text and speech through an Android mobile application, thereby improving communication accessibility between sign language users and non-sign language users.

1.4.2 Project Objectives

Objective 1: Research and Requirements Analysis

- **Specific:** Investigate sign language translation systems, wearable sensing technologies, ML techniques and accessibility requirements relevant to assistive communication systems.
- **Measurable:** Produce a comprehensive literature review and define the functional and non-functional requirements of SignLink.
- **Achievable:** Utilize academic journals, conference papers and existing research studies to gather information and requirements.
- **Relevant:** Provides the theoretical and technical foundation necessary for the design and development of the proposed system.
- **Time-bound:** To be completed during the initial research and planning phase of the project.

Objective 2: User Interface Design and User Experience Validation

- **Specific:** Design UI prototypes and conduct usability testing with peers and volunteers to evaluate the accessibility, usability and effectiveness of the proposed application design.
- **Measurable:** Collect user feedback through surveys and observations and implement design improvements based on the results.
- **Achievable:** Utilize wireframes, mock-ups and interactive prototypes developed using standard UI/UX design principles.
- **Relevant:** Ensures the application is user-friendly and suitable for its intended audience before development begins.
- **Time-bound:** To be completed before the implementation phase of the mobile application.

Objective 3: Mobile Application Development

- **Specific:** Develop an Android mobile application capable of displaying translated sign language gestures as text and speech outputs.
- **Measurable:** Implement and test core functionalities including communication assistance and local data management.
- **Achievable:** Utilize Kotlin and Android Studio to develop and deploy the application.

- **Relevant:** Provides the primary platform through which users interact with the SignLink system.
- **Time-bound:** To be completed during the system implementation phase.

Objective 4: Wearable Hardware Development

- **Specific:** Design, develop and evaluate multiple iterations of a low-cost wearable EMG sensing prototype using commercially available components.
- **Measurable:** Produce and assess successive hardware revisions based on signal acquisition capability, usability and design improvements.
- **Achievable:** Utilize EMG sensors, microcontrollers, inertial sensing modules and battery-powered circuitry.
- **Relevant:** Supports the long-term objective of developing an affordable alternative to commercial EMG sensing devices.
- **Time-bound:** To be conducted throughout the hardware development phase of the project.

Objective 5: Machine Learning Model Development

- **Specific:** Develop and evaluate an ML model capable of classifying ASL fingerspelling alphabet gestures using EMG signal data.
- **Measurable:** Evaluate model performance using accuracy, precision, recall, F1-score and confusion matrix analysis.
- **Achievable:** Utilize a publicly available Myo Armband dataset for model training, validation and testing.
- **Relevant:** Provides the core gesture recognition functionality required for sign language translation.
- **Time-bound:** To be completed following dataset preparation and prior to final system evaluation.

Objective 6: System Testing and Evaluation

- **Specific:** Evaluate the developed mobile application, machine learning model and hardware prototype components to determine their functionality, usability and feasibility.
- **Measurable:** Assess application usability through user feedback, evaluate ML performance using classification metrics and analyze the effectiveness of the hardware prototypes through iterative testing.

- **Achievable:** Utilize testing procedures, usability surveys and machine learning evaluation metrics.
- **Relevant:** Ensures that the developed components meet the project requirements and identifies areas for future improvement and integration.
- **Time-bound:** To be completed during the final phase of the project.

2.0 Review of Existing Knowledge

As technology has progressed from initial configurations to contemporary portable solutions, the domain has broadened to encompass human-computer interaction, biomedical signal processing, and mobile computing. To provide context for these advancements, this chapter examines the associated literature.

2.1 Sign Language and Sign Language Translation Technologies

Sign language is a complete, grammatically complex, and natural visual-spatial language that enables communication through a synchronized combination of manual indicators such as hand gestures, wrist orientations and non-manual signals such as facial expressions, lip movements and torso postures *(Emmorey, 2002; Ambavane et al., 2020)*. It serves as the primary means of communication for many hard-of-hearing and mute individuals worldwide. However, communication difficulties often arise when interacting with people who do not understand sign language, creating barriers in education, healthcare, employment and everyday social interactions *(United Nations, 2006)*.

To alleviate these systemic inequities, contemporary human-computer interaction research has focused heavily on the engineering of automated sign language recognition and sign language translation systems. In the literature, these recognition mechanisms are broadly segmented into two primary architectural paradigms: **vision-based systems** and **sensor-based systems** *(Papatsimouli et al., 2022)*.

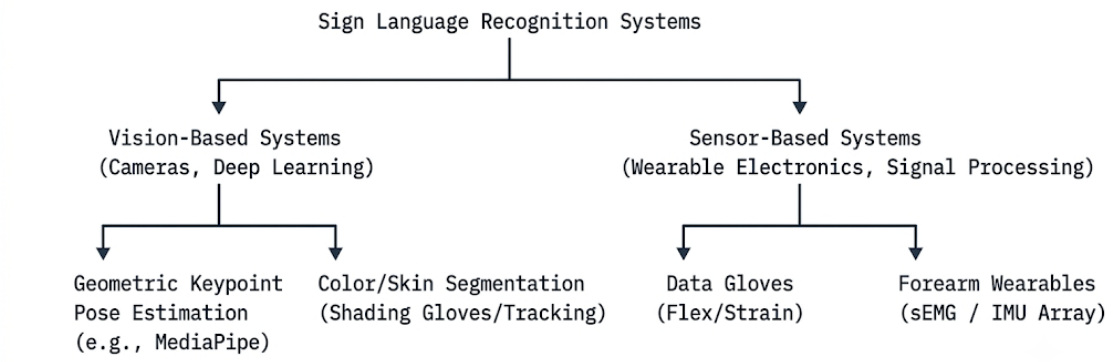


Figure 1: Vision-Based vs. Sensor-Based Approaches

2.2.1 Vision-Based Translation Systems

Vision-based architectures tend to utilize optical sensors, such as standard webcams, RGB cameras, or depth sensors, combined with computer vision and deep learning pipelines to capture and classify gestures. These systems offer distinct operational flexibility, as they natively allow for the extraction of non-manual linguistic features, including facial expressions, head tilts, and lip-reading cues (*Shukor et al., 2015; Akmeliawati et al., 2007*)

However, despite the advancements in deep convolutional networks, these systems are fundamentally bounded by the camera's field of view (FoV), rendering them highly sensitive to spatial geometry, background environmental noise, and variable lighting conditions *(Papatsimouli et al., 2022)*.

2.2.2 Sensor-Based Translation Systems

In contrast, sensor-based architectures utilize wearables placed directly on the hand or forearm to capture real-time kinematics and spatial orientation. These systems leverage diverse hardware matrix including IMUs for velocity tracking, flex sensors for finger articulation, and Time-of-Flight (ToF) or tactile sensors for surface contact. Since data collection occurs entirely onboard, they generate lightweight numeric data streams immune to external lighting and environmental noise.

To classify these signals, modern frameworks leverage deep learning; Convolutional Neural Networks (CNNs) extract spatial patterns from multi-channel data, while Recurrent Neural Networks (RNNs) with Long Short-Term Memory (LSTM) blocks map the temporal variations of continuous gestures (*Maharjan et al., 2024)*.

2.2 Electromyography (EMG) for Gesture Recognition

Electromyography (EMG) measures the electrical activity produced by skeletal muscles during contraction. **Surface electromyography (sEMG)** captures these signals non-invasively via skin-surface electrodes. The collected signals provide valuable information regarding muscle activation patterns and movement intentions, making EMG a widely used technology in rehabilitation, prosthetics, human-computer interaction and gesture recognition systems *(Chowdhury et al., 2013).*

For sign language translation, sEMG sensors are typically placed around the forearm to detect the distinct muscle contractions generated by specific hand and finger movements. Unlike vision-based systems, sEMG is immune to lighting variations, background noise, and hand occlusions. To optimize signal acquisition, electrodes must align precisely with the underlying musculature responsible for digit articulation.

Figures 2 and 3 illustrate the anatomical layout of these primary forearm muscle groups to demonstrate a multi-channel electrode placement strategy designed to isolate individual finger movements alongside a dedicated ground reference node.

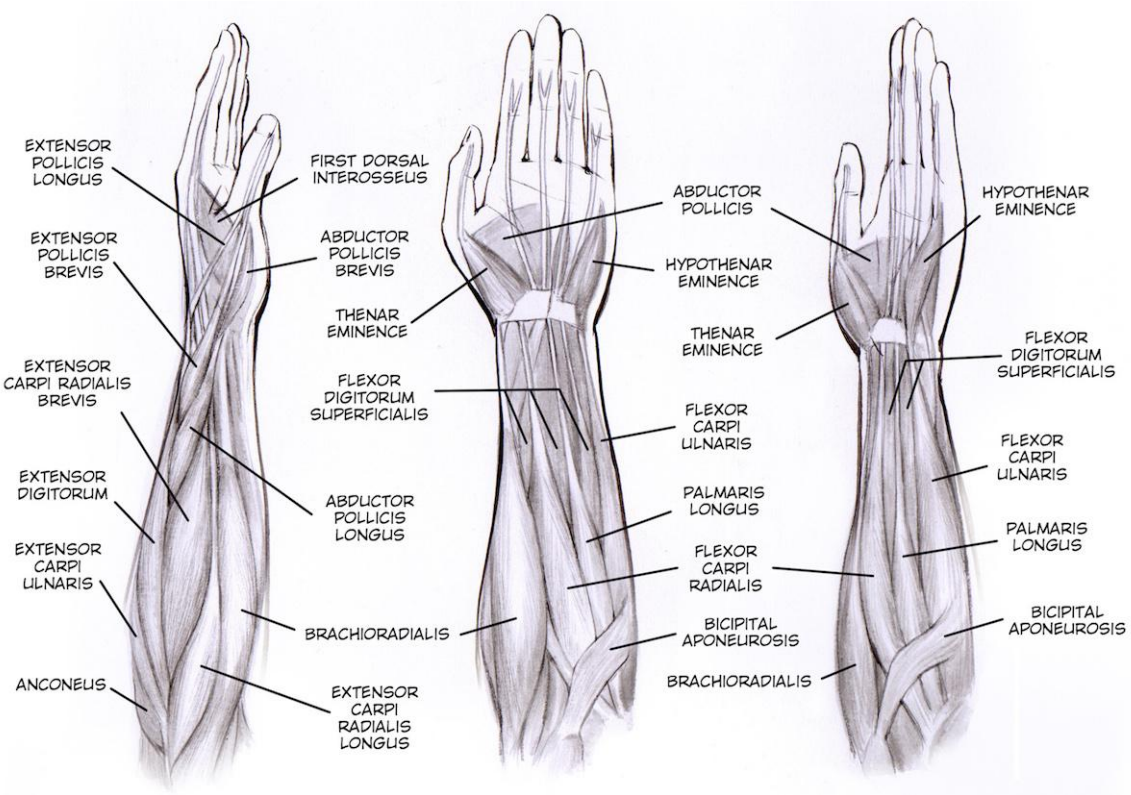


Figure 2: Hand Muscles (Retrieved from Backyard Brains: Signal Classification Experiment)

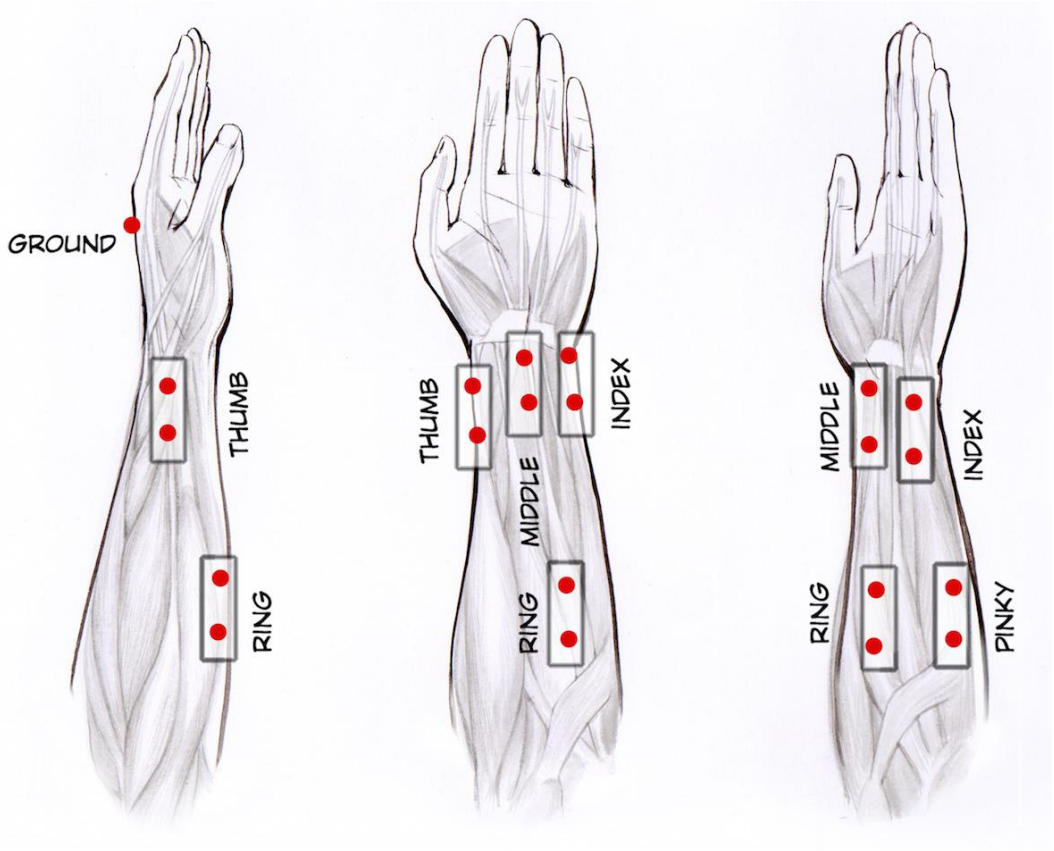


Figure 3: Example Electrode Placement for EMG Acquisition (Same source as figure 2)

The Myo Armband is a benchmark consumer-grade wearable featuring 8 surface Electromyography sensor channels and an integrated 9-axis Inertial Measurement Unit to simultaneously monitor muscle activity and spatial kinetics. Although discontinued, it continues to be a reference point in the literature on human-computer interaction due to its durable design, demonstrated classification effectiveness, and the extensive accessibility of high-quality, open-source gesture datasets, which provide essential baselines for contemporary research.

2.3 Machine Learning for Sign Language Recognition

Machine learning plays a crucial role in EMG-based gesture recognition by transforming raw muscle signals into gesture classifications through a series of processes including data collection, preprocessing, feature extraction, model training, and evaluation.

The algorithms frequently employed in this domain encompass Support Vector Machines (SVM), Random Forests, Artificial/Convolutional Neural Networks (ANNs/CNNs), and Gradient Boosting techniques, all of which have shown significant efficacy on EMG datasets *(Chowdhury et al., 2013).*

Among these algorithms, **Random Forest** stands out as one of the most prevalent baseline models utilized in gesture recognition studies. Research conducted by *Shakeel et al. (2020)(MAST)* and *Paudyal et al. (2017)(DyFav)* has compared custom algorithms against Random Forest baselines, yielding accuracies of 93% compared to 81.4% and 95.36% against 80.47%, respectively. The capability of Random Forest to manage structured numerical data, as well as its resilience to noise and inherent variations in EMG signals, makes it particularly effective for gesture recognition tasks. Consequently, a Random Forest classifier was chosen for the SignLink project and was trained using a publicly accessible Myo Armband EMG dataset that includes American Sign Language (ASL) fingerspelling gestures.

2.4 Research Gap

The current literature reveals a significant disconnect between high-accuracy laboratory models and practical, real-world utility. While camera-based translation systems offer linguistic flexibility, they remain fundamentally restricted by environmental lighting and background clutter. Conversely, sensor-based alternatives are heavily reliant on either expensive clinical equipment or discontinued consumer hardware like the Myo Armband. Furthermore, existing research focuses predominantly on maximizing offline machine learning accuracy on desktop computers, largely neglecting the integration challenges of deploying these models onto mobile devices.

This leaves a clear gap for an affordable, environment-immune system optimized for daily use and SignLink intends to address it with the system proposed.

3.0 Method and Implementation

The development of this project followed a structured and iterative chronological framework encompassing user-centred design planning, mobile application development, hardware research and design, and machine learning model training. The following sections describe each stage of the development process in detail, and the Gantt chart below provides an idea of the timeframe and time segregation.

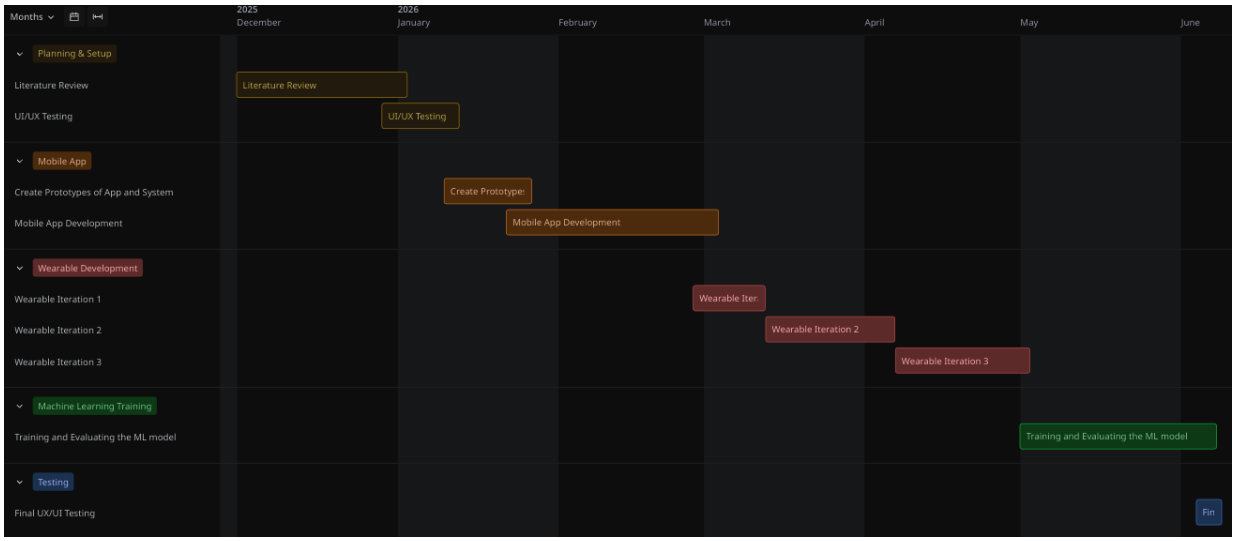


Figure 4: Gantt Chart

3.1 UI/UX Design and User Testing

The design of the SignLink mobile application was guided by a user-centered approach following **Nielsen's Heuristics**, with the primary objective of creating an accessible and intuitive communication platform for sign language users. Prior to implementation, the application's functionality and interface requirements were identified based on findings from the literature review, project objectives and anticipated user needs.

To prioritize development efforts, the **MoSCoW** prioritization technique was utilized to classify system features according to their importance as seen below. This approach enabled the identification of essential functionality required for the initial implementation while also establishing a roadmap for future enhancements.

Must Have

Essential (free) features required for basic app functionality and user accessibility.

- Profile Management
  - Profile includes user type (Deaf, hearing, learner, healthcare, etc.)
  - Adjustable text size, color contrast, vibration strength etc.
- EMG-acquiring hardware and translation
  - Pair and calibrate sign-language translating wristband to the app
- Text-to-Speech Output
  - Convert translated text to spoken audio in a default voice (English)
  - Basic offline speech synthesis
- Speech-to-Text Input
  - Capture microphone input and display it as text for the Deaf user
  - Trigger single vibration alert on band when new speech text appears
- Basic Chat History
  - Store last few conversations locally for reference

Should Have

Important but not strictly essential; adds comfort and usability.

- Basic Multi-language Support
  - Option to switch translation output language (e.g., English, Spanish, French)
  - Basic AI translation accuracy tuning
- Real-time EMG-to-Text Translation
  - Capture and process EMG signals from band sensors
  - Display recognized sign as text on-screen
- Real-time EMG Visualization
  - Display simple waveform or graph of detected EMG signals for learning feedback (learning feature is premium)
- Tutorial/Learning Mode
  - Built-in guide for band calibration and basic signing gestures
- Offline Mode (limited)
  - All functions without the need for internet

Could Have

These are premium features that would significantly boost the project.

- Advanced AI Sign Learning Mode
  - Personalized model training on user's unique EMG signal patterns
  - Accuracy feedback for the learning sign language feature.
- Extended Multi-language & Voice Library
  - Choose voice type (male/female/neutral)
  - Translate speech into multiple languages (e.g., English → Vietnamese, etc.)
- Contextual AI Translation
  - Context-aware sentence formation for smoother spoken output
  - Emotion tone adjustment in speech synthesis
- Conversation Logging & Cloud Sync
  - Securely store, label, and access full conversation history across devices
- Customizable Haptic Feedback Patterns
  - Define vibration patterns for different types of alerts (e.g., message received, translation complete)

Won't Have

Not in the current plan (of execution) but is a future prospect.

- Gesture-only interface for full navigation (no phone use required)
- VR/AR integration for 3D ASL practice
- Myo dual-band mode (two armbands for full hand gesture tracking)
- Advanced EEG integration for cognitive/ emotional context detection

Table 1: MoSCoW

Following feature prioritization, user flows and interface layouts were designed to visualize how users would interact with the system. A **Hierarchical Task Analysis (HTA)** was developed to model the primary user journey. Due to the size and complexity of the complete HTA diagram, a high-resolution version is available in the project repository: https://github.com/codingsanji/Honors_signLink/blob/main/designs/UI_UX/Prototypes/HTA.png

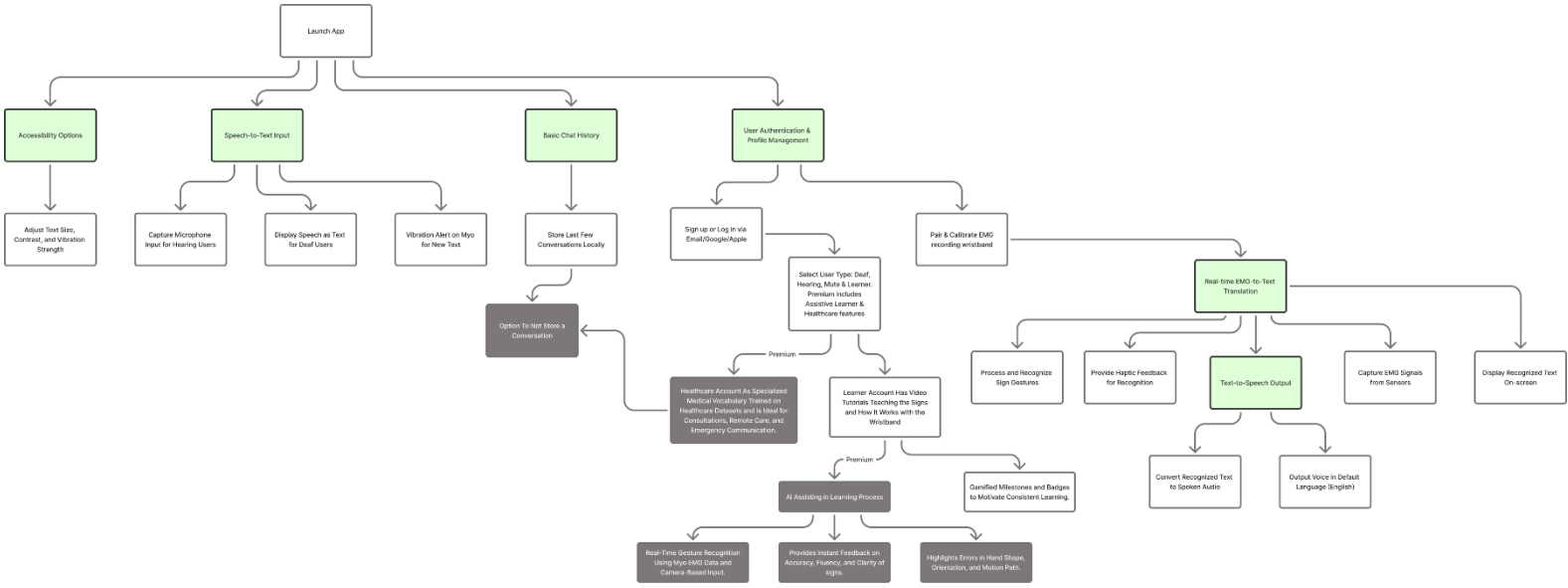


Figure 5: Hierarchical Task Analysis

Following the development of the HTA, a series of low-fidelity and high-fidelity prototypes were created to progressively refine the application's navigation flow, functionality and visual design.

3.1.1 Evaluation Framework and Participants

Both testing phases utilized a hybrid structure combining on-site in-person sessions (focusing on ergonomics, hand gestures, and physical environment interaction) and remote online sessions (focusing on navigation clarity and remote comprehension). Before each session, participants were briefed on system objectives and informed that evaluations targeted interface design rather than hardware sensor accuracy.

- **Phase 1 (Initial Usability Testing):** Conducted between December 29 and January 12 with 5 participants (3 in-person at the UIT Library, 2 remote via Google Meet using a Figma prototype) to evaluate initial navigation flow, Bluetooth pairing, and sensor calibration.

- **Phase 2 (Final UI/UX App Testing):** Conducted between June 5 and June 6 with 15 participants (9 in-person, 6 remote via video conferencing downloading the actual app on their phones) to evaluate the final refined iteration.

A diverse cohort of 20 total participants was recruited. To preserve privacy, precise identities are excluded; signed consent forms are available in the Appendix. The Phase 1 demographic baseline is outlined below:

Participant	Age Group	Type of User	Sign Language Familiarity
1	18-24	Hearing user (non-signer)	Not familiar at all
2	18-24	Sign Language Learner	Basic Knowledge
3	18-24	Hearing user (non-signer)	Not familiar at all
4	18-24	Sign Language Learner	Not familiar at all
5	25-34	Hearing user (non-signer)	Not familiar at all

Table 2: Participant Data of Pre-Testing

The Phase 2 demographic baseline is outlined below:

Participant	Age Group	Type of User	Sign Language Familiarity
1	18-24	Hearing user (non-signer)	Not familiar at all
2	18-24	Sign Language Learner	Not familiar at all
3	25-34	Hearing user (non-signer)	Not familiar at all
4	18-24	Hearing user (non-signer)	Not familiar at all
5	18-24	Hearing user (non-signer)	Not familiar at all
6	18-24	Sign Language Learner	Basic Knowledge
7	55-64	Hearing user (non-signer)	Not familiar at all
8	13-18	Sign Language Learner	Not familiar at all
9	55-64	Hearing user (non-signer)	Not familiar at all
10	25-34	Hearing user (non-signer)	Basic Knowledge
11	18-24	Hearing user (non-signer)	Not familiar at all
12	18-24	Hearing user (non-signer)	Not familiar at all
13	35-44	Hearing user (non-signer)	Not familiar at all
14	18-24	Sign Language Learner	Basic Knowledge
15	35-44	Hearing user (non-signer)	Not familiar at all

Table 3: Participant Data of Final Testing

3.1.2 Data Collection Methods

The evaluation incorporated a blend of qualitative and quantitative data collection methods:

- **Think Aloud Method (Qualitative):** Participants verbalized their thoughts and intentions while interacting with the interface, revealing whether the simulated Bluetooth pairing and sensor calibration workflows were intuitive.
- **System Usability Scale (Quantitative):** Following the tasks, users completed a standard 10-item System Usability Scale (SUS) questionnaire via Google Forms. This numerical metric provided a baseline to determine if the interface cleared the industry-average usability threshold score of 68.
- **Semi-Structured Interviews (Qualitative):** Each session concluded with an open-ended interview to probe deeper into observed usability "pain points," which was especially vital for gathering feedback from remote users navigating the framework without a physical device.

Due to the size of the raw prototype assets, detailed testing transcripts, participant feedback matrices, and granular evaluation results, all comprehensive supporting materials are included in the appendices.

3.2 Mobile Application Development

The SignLink mobile application, developed in Kotlin within Android Studio, serves as the primary interface for the translation system providing an accessible, modular platform designed to support communication assistance, hardware connectivity, and seamless model integration. In this section, the screens attached will be in “light” mode with “small” text font. The screens in other modes will be attached in section 7.2 of the appendix corner.

The workflow begins with the **Welcome Screen**, introducing users to the platform, followed by the **User Type Selection Screen**, where individuals identify their profile (e.g., Deaf user, hearing user, or learner). Following profile selection, users proceed to the **Device Pairing Screen** to establish communication between the application and the wearable EMG sensing device via Bluetooth Low Energy (BLE), as illustrated below.

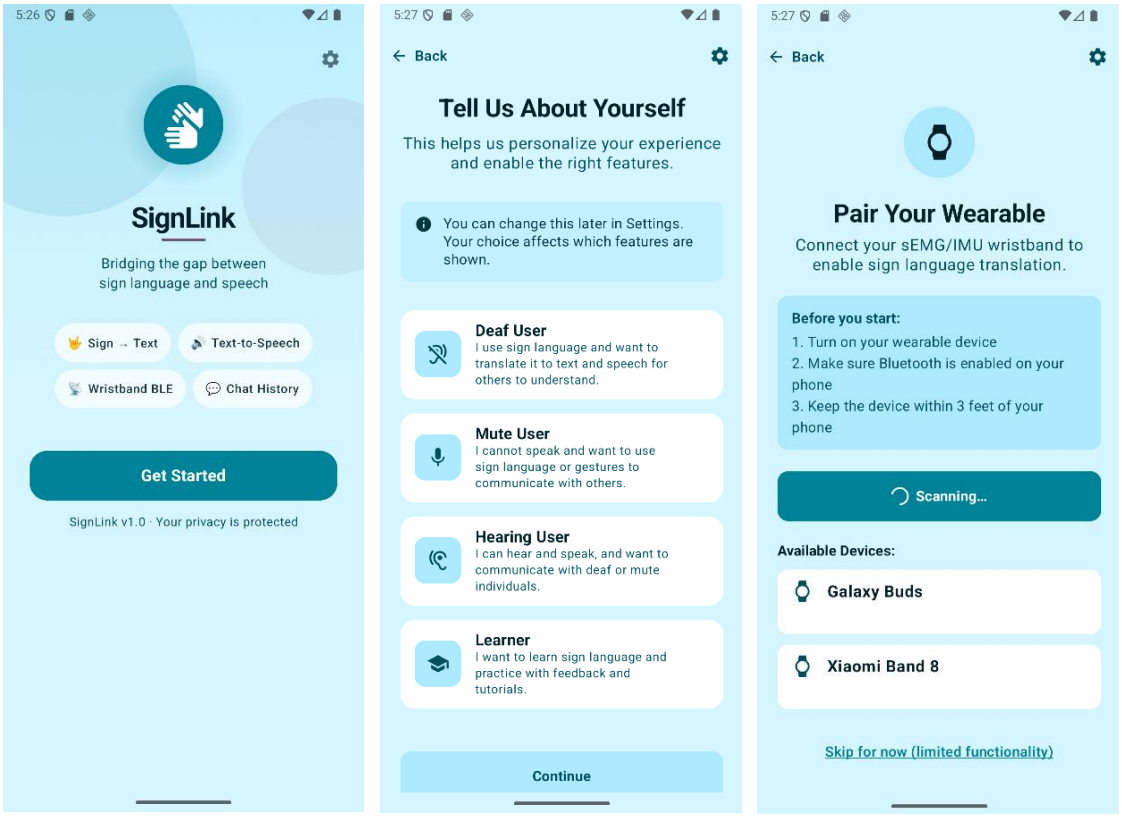


Figure 6: Screens set 1

Once a connection is established, users are guided to the **Calibration Screen** to prepare the wearable device for signal acquisition and future gesture recognition. This calibration workflow ensures more reliable sensor readings to improve the overall translation process. Upon successful calibration, users are directed to the **Home Screen**, which serves as the central navigation hub for all application features.

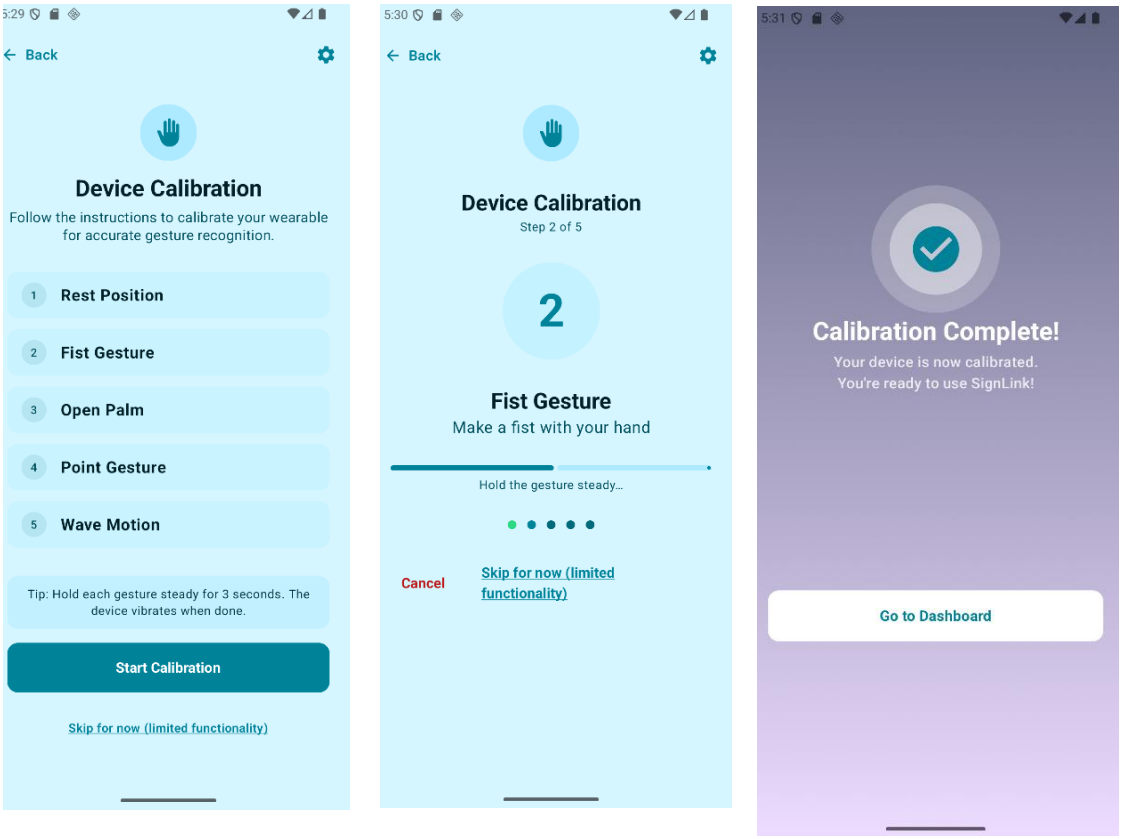


Figure 7: Screens set 2

The **Home Screen** provides access to the application's core communication modules: the **Translation Screen**, **Speech-to-Text Screen**, and **Text-to-Speech Screen**. The Translation module displays sign language outputs generated from recognized EMG gestures, while the Speech-to-Text and Text-to-Speech modules enable two-way communication between sign language and non-sign language users. Together, these components form the system's primary communication assistance functionality.

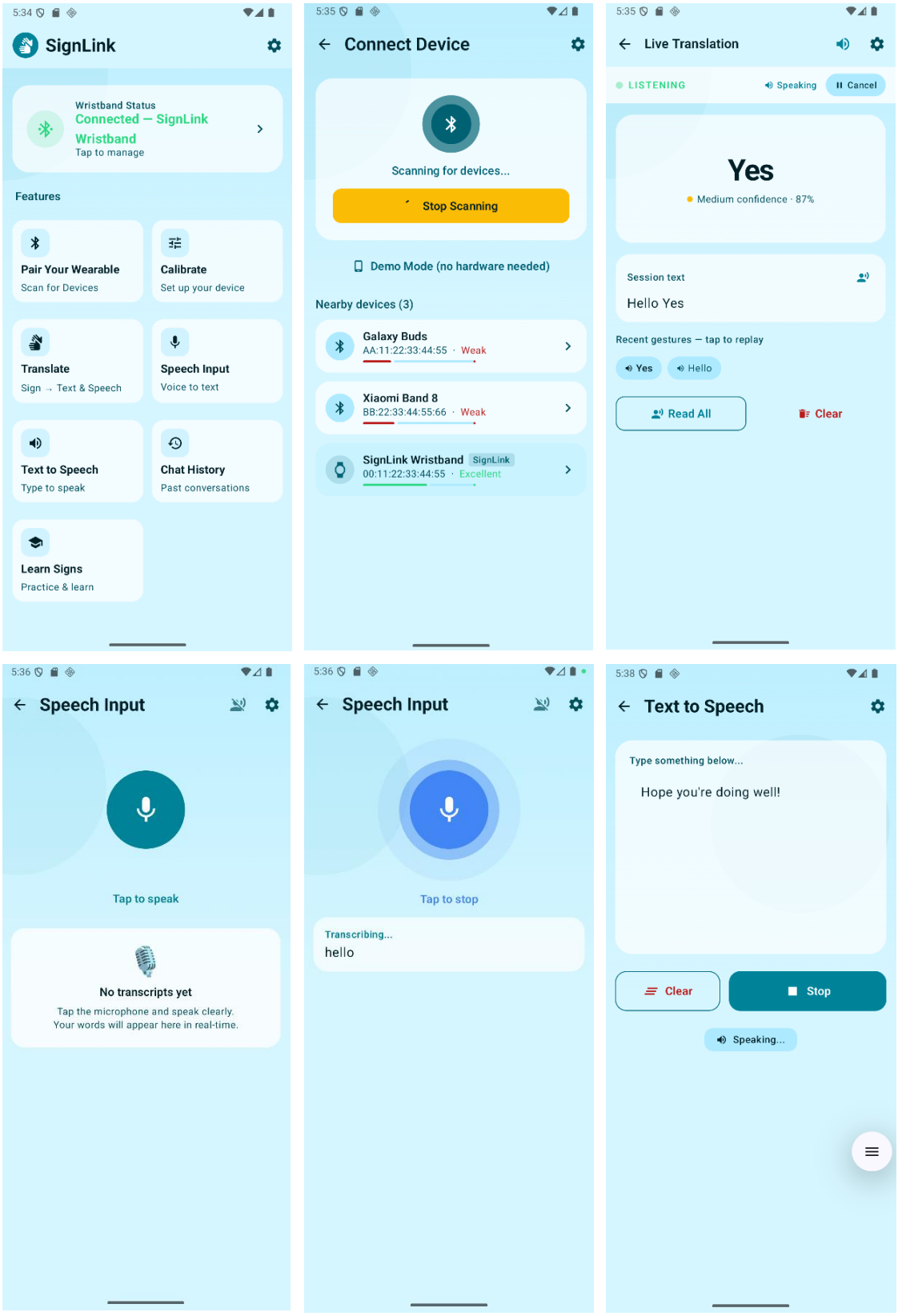


Figure 8: Screens set 3

Additional features include a **Chat History Screen** for reviewing previous conversations, a **Learning Screen** for future educational content and sign language practice, and a **Settings Screen** for accessibility and application configurations. These supporting modules were included to improve usability and enhance the overall user experience.

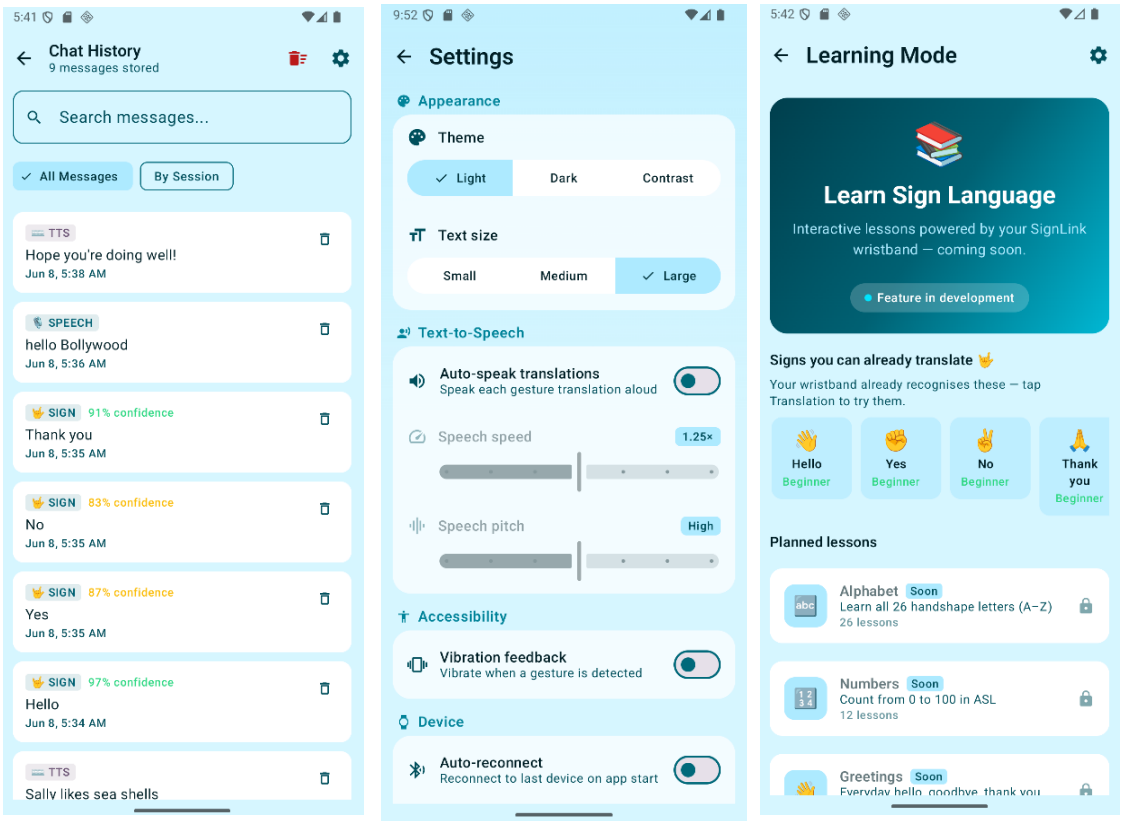


Figure 9: Screens set 4

3.3 Wearable Hardware Development

The SignLink wearable device was created to explore the potential of a cost-effective substitute for commercial EMG devices like the Myo Armband. Its aim was to record muscle activity produced during sign language gestures and relay the signals for processing within a mobile application. The hardware experienced three significant design revisions.

Version 1 (V1) featured a single EMG sensor linked to an Arduino Uno and was utilized to examine EMG signal acquisition, electrode positioning, and sensor functionality. However, this design was restricted to a single sensing channel and was

insufficient in capturing the intricate muscle activity associated with sign language gestures.

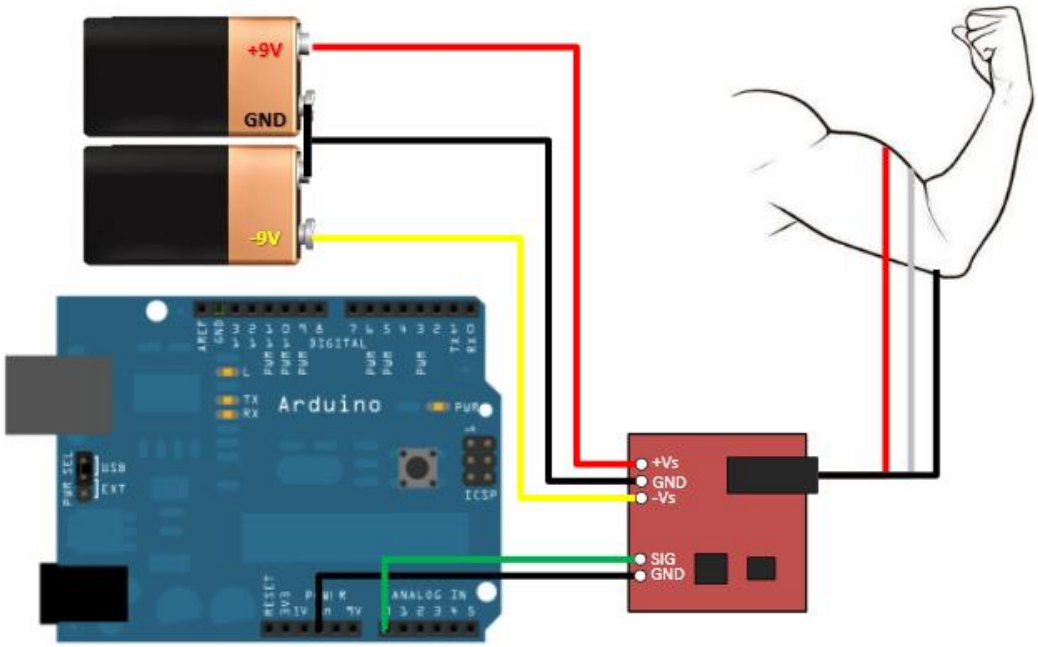


Figure 10: Setup of V1



Figure 11: V1 Signal

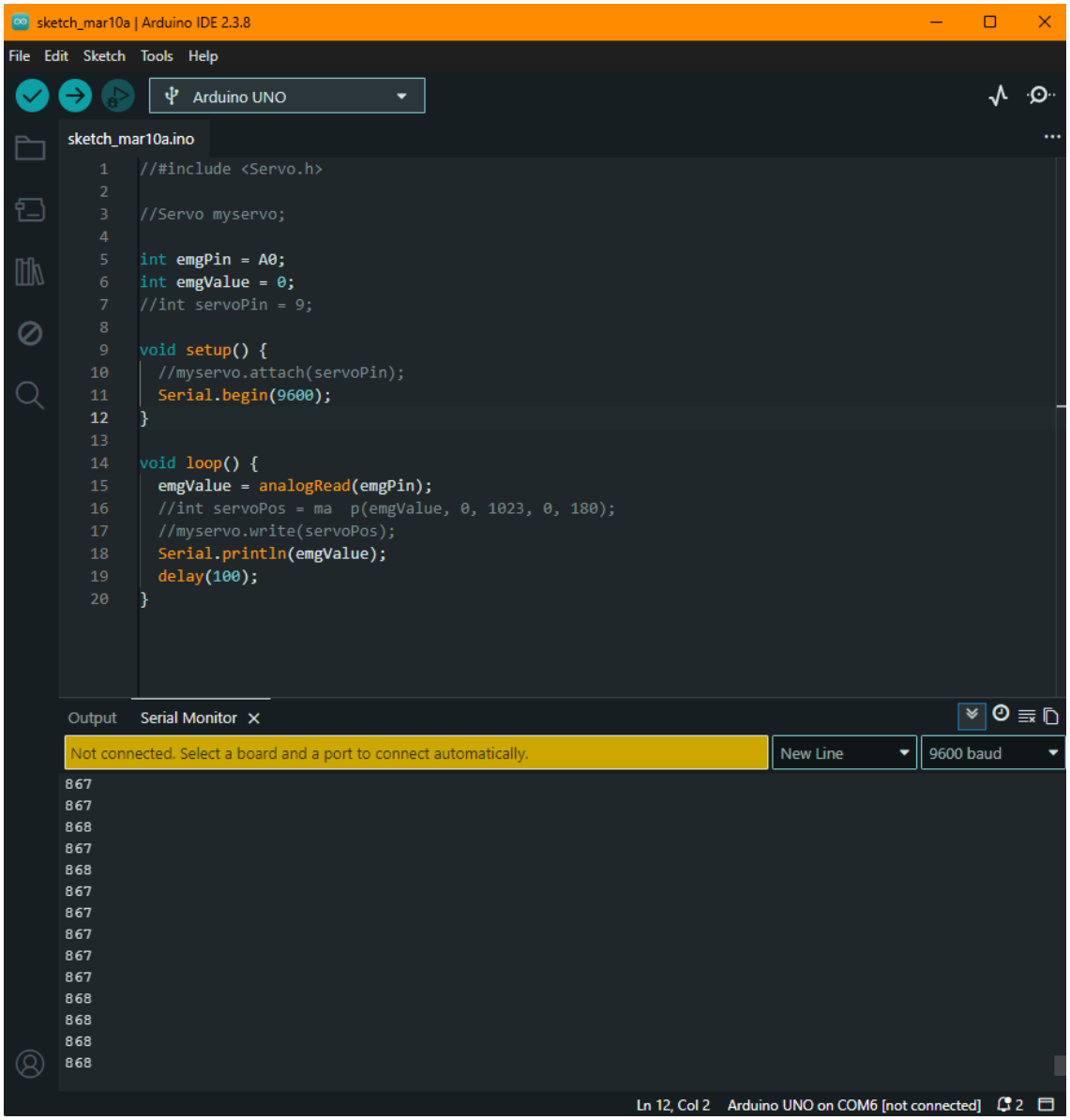


Figure 12: V1 Rough Code

To overcome this limitation, **Version 2 (V2)** enhanced the system by incorporating four EMG sensors connected to an Arduino Nano V3. Although this modification improved muscle coverage, the prototype faced issues with unstable signal acquisition due to loose jumper-wire connections. As a result, the design was modified, and soldered connections were implemented to enhance reliability.

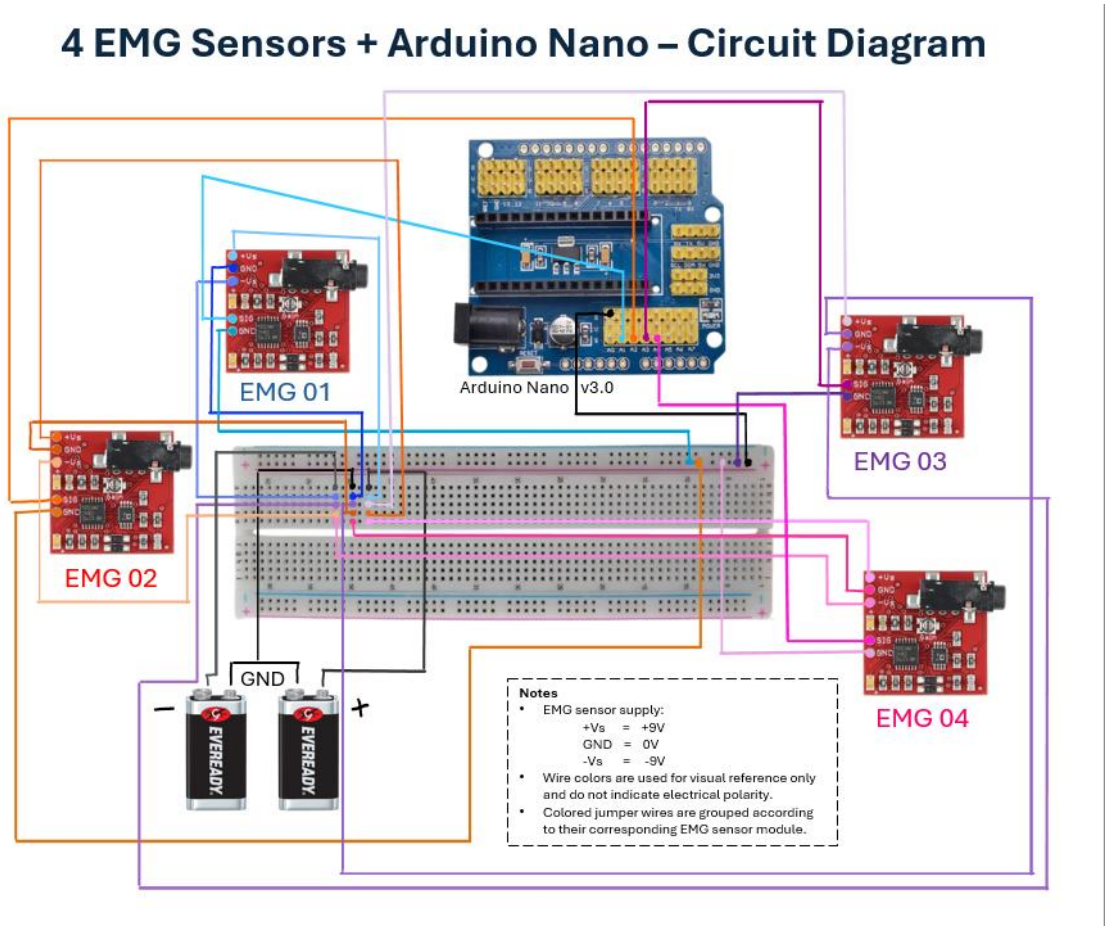


Figure 13: Setup of V2

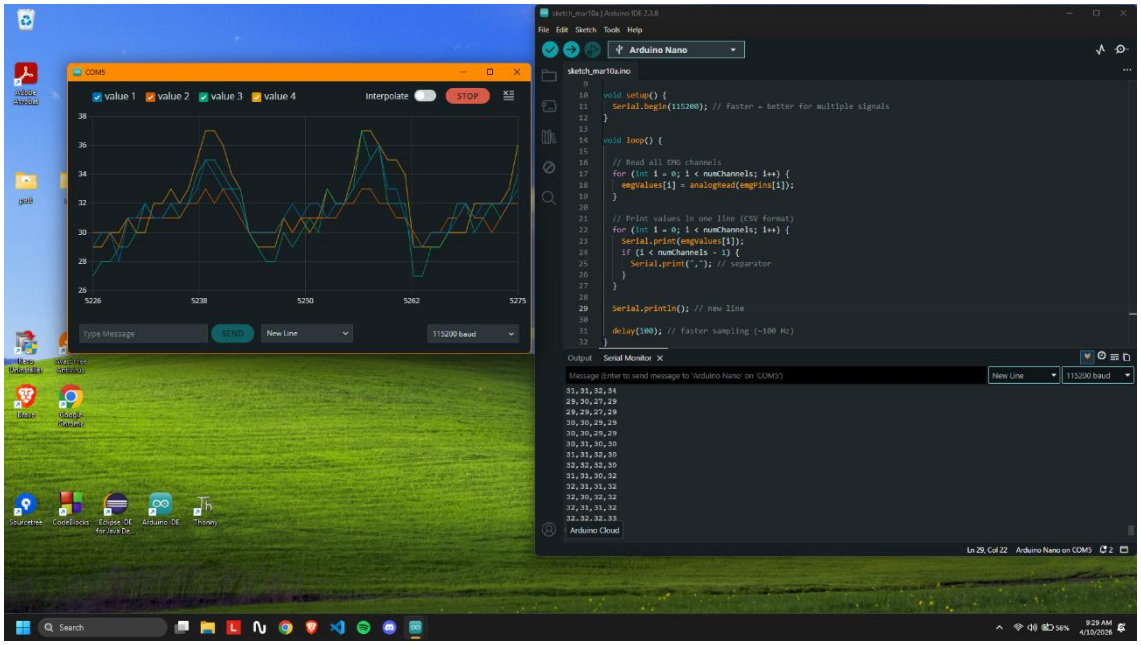


Figure 14: Code and Signal

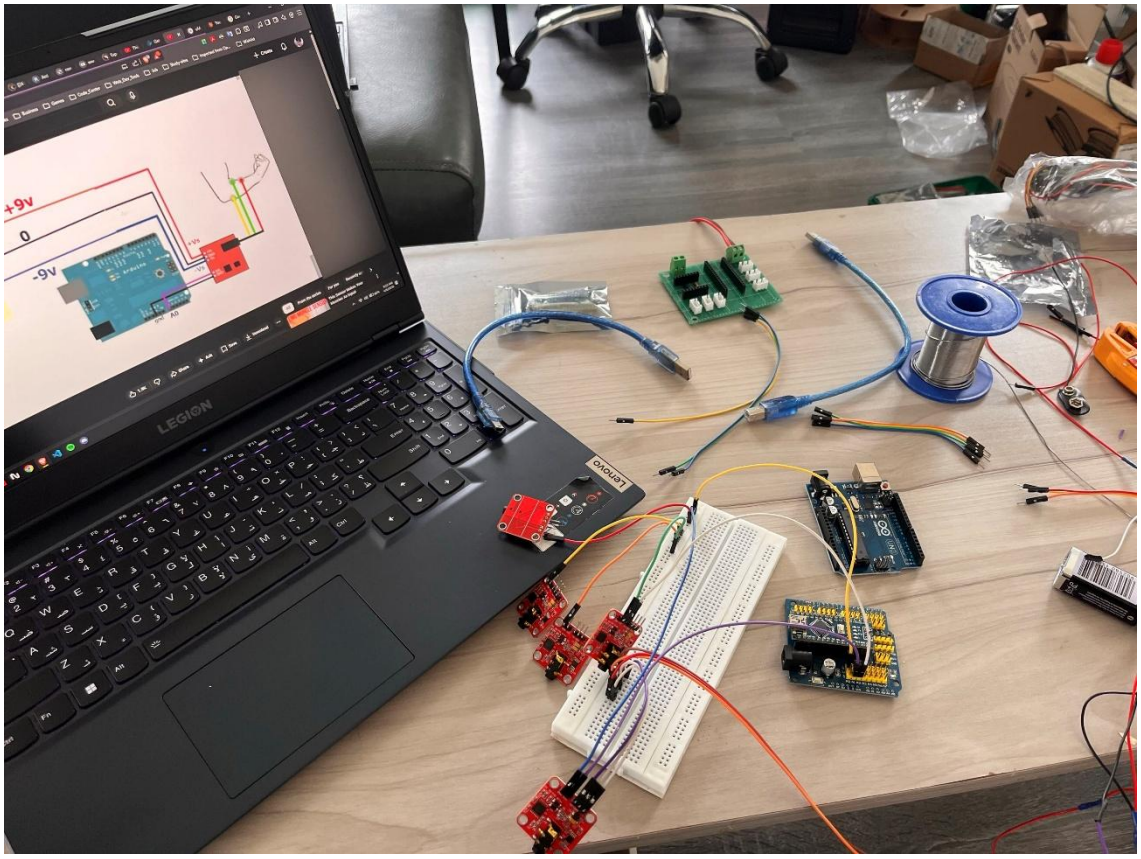


Figure 15: V2 In Real Life

Version 3 (V3) marked the most recent prototype and was still undergoing testing during this study. While it could accommodate four EMG sensors, only three were utilized to ensure circuit stability while validating the hardware architecture.

Despite the successful acquisition of EMG signals, obstacles such as signal noise, sensitivity to electrode placement, and the absence of a sufficiently large, labelled dataset hindered the custom hardware's application in training machine learning models.

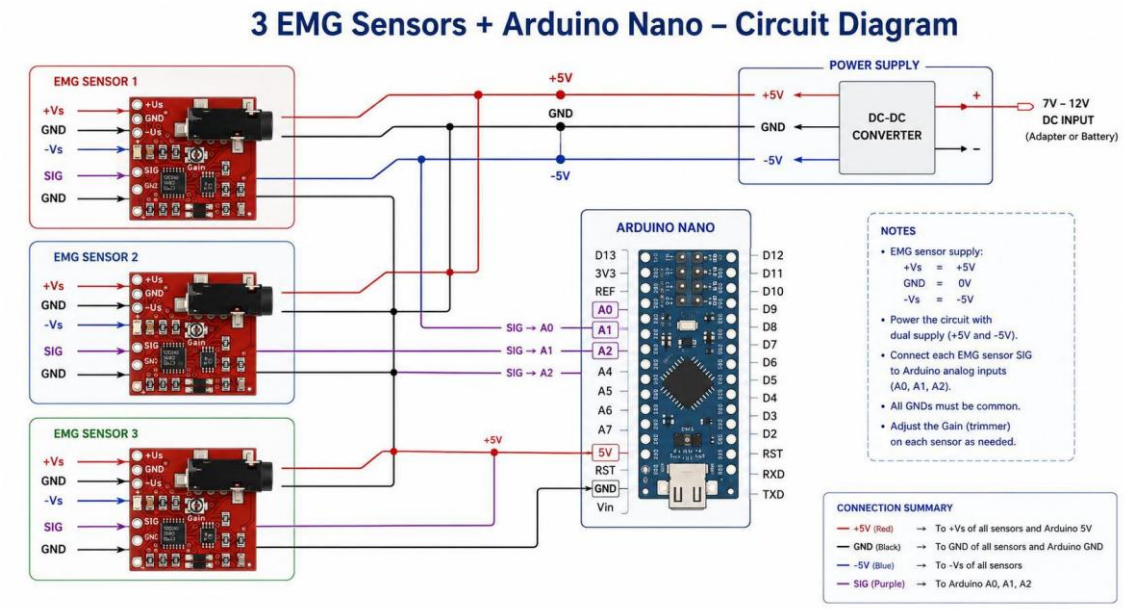


Figure 16: Setup of V3

Consequently, a publicly accessible Myo Armband ASL dataset was employed to train and assess the recognition framework under controlled conditions. Thus, the hardware functioned as a proof-of-concept platform and lays the groundwork for future endeavours focused on enhanced signal conditioning, wireless communication, and real-time integration with the SignLink application.

3.4 Machine Learning Development

3.4.1 Dataset Identification

A dataset released by the Arizona State University (**American Sign Language (ASL) Fingerspelling dataset for Myo Sensor**) was used to build and train the machine learning model. This dataset comprised of EMG and IMU sensor values from a Myo armband taken from **9 users** for **5 secs** for each letter of the ASL alphabet. The first 8 columns contain data for the 8 EMG pods/electrodes, the next 3 are for Accelerometer, the next 3 are for Gyroscope and the final 3 are for Orientation (Roll, Pitch and Yaw).

3.4.2 Data Cleaning and Preprocessing

The dataset was checked for duplicates, null/missing values and out of range values (the sensors in the Myo armband have specific ranges). The above issues were

handled accordingly while making sure the label balance was preserved. **Zeroing** of the EMG sensor values was done as explained by *Paudyal et al., 2017* to account for different starting positions and rest-level activations. It was found to be very fruitful since before zeroing and testing the limitation of the model, the accuracy turned out to be significantly low.

3.4.3 Exploratory Data Analysis

To understand the data on a deeper level, research papers using the dataset were studied and information gained from those papers were used to build and develop the model *(Paudyal et al.,2017; Paudyal et al., 2016)*.

3.4.4 Feature Selection and Engineering

The methodology for feature engineering was influenced by the works of *Shakeel et al. (2020)* and *Paudyal et al. (2017)*, both of which utilized comparable preprocessing methods. Initially, five statistical features - maximum, minimum, mean, standard deviation, and total energy were derived from each of the 17 parent features, culminating in a total of 85 engineered features.

In order to maintain temporal information, each 5-second recording was subsequently partitioned into five equal segments, and the same features were extracted from each segment, thereby elevating the feature count to 510.

When a baseline Random Forest classifier was tested on both the non-segmented and segmented datasets, the accuracies recorded were **73.3%** and **75.0%**, respectively, suggesting that segmentation enhanced performance. Subsequently, feature selection was conducted to further optimize the model.

Although dimensionality reduction techniques such as PCA have been investigated in prior research, they were deemed unnecessary due to the Random Forest's intrinsic capability to assess feature importance through metrics like Gini impurity and information gain. A baseline Random Forest model trained on all 510 features yielded an accuracy of 75.00%. By employing **SelectFromModel** to retain only the most significant half of the features, accuracy was improved to **77.97%**, thereby illustrating the advantages of feature selection.

3.4.5 Splitting of Data into Training and Testing

The first iteration of the train-test split started with Leave-One-Group-Out (LOGO) cross-validation, a common approach for sensor-based applications. However, accuracies remained low at approximately 20–25% per fold. This was expected because the model was trained on data from a subset of users and tested on a completely unseen user. Due to natural variations in muscle activity, arm size and armband placement, EMG signals differ significantly between individuals, making inter-user generalisation difficult.

A stratified 80–20 train-test split was therefore adopted, ensuring class balance while allowing data from every user to appear in both the training and testing sets. Under this approach, model accuracy increased to 75%, indicating that the model could successfully learn user-specific EMG patterns and recognise gestures from previously unseen samples of the same users.

These findings suggest that the model performs best as an intra-user classifier rather than an inter-user classifier. Consequently, the SignLink application incorporates a calibration phase, allowing the model to learn the EMG characteristics of a specific user before performing gesture recognition.

3.4.6 Training the model

A Random Forest Classifier was used with control parameters to ensure minimal overfitting. Even though overfitting is to be expected to an extent due to the smaller dataset. The code snippets are attached in the appendix section 7.3.

3.5 Final UI and UX App Testing

A final UI/UX testing phase was conducted between June 5 and June 6 to evaluate the refined iteration of the application interface. To ensure comprehensive feedback across diverse environments, a total of 15 participants were evaluated using a hybrid testing structure:

- **In-Person Testing:** Nine sessions were conducted on-site in a physical environment, allowing for direct observation of real-world usability, ergonomic comfort, and immediate user interaction with the application flow.

- **Remote Testing:** Six sessions were conducted online via video conferencing, where participants were guided on how to download and run the app on their phones to assess navigation clarity and remote user comprehension.

Before commencing each session, participants were briefed on the system's objectives. Similar to earlier testing phase, the evaluation focused strictly on the final application interface logic, visual layout consistency, and accessibility features. Qualitative feedback via the Think-Aloud method and quantitative metrics were gathered to *validate the final design enhancements.*

4.0 Evaluation

This chapter evaluates the effectiveness of SignLink through three stages of assessment: prototype usability testing, machine learning model evaluation and final application testing. The objective was to assess the usability, functionality and technical feasibility of the proposed system.

4.1 UI/UX Prototype Evaluation

4.1.1 Quantitative Results

Task Performance and Design Metrics (1–5 Likert Scale)

Initial Clarity: App purpose clarity scored **4.8/5.0**, with 100% confidence regarding home screen navigation.

Navigation Success: Locating the Bluetooth pairing menu scored **4.4/5.0** (P4 noted a slight delay in button discovery).

Calibration Logic: Comprehension of gesture instructions ("Fist" and "Wave") scored **4.8/5.0**.

Accessibility and Readability: Visual contrast and text size scored **4.6/5.0**, with 80% validating the accessibility-first design.

System Usability Scale (SUS)

The application achieved an average SUS score of **89.0**, placing it in the **Excellent (Grade A)** bracket. A minor friction point was identified by P4, whose lower score (72.5) stemmed from discomfort with the simulated calibration process.

Participant	SUS Score Calculation	SUS Score	Grade	Adjective Rating
P1	(18+16)*2.5 = 34*2.5	85.0	A	Excellent
P2	(18+20)*2.5 = 38*2.5	95.0	A+	Best Imaginable
P3	(18+19)*2.5 = 37*2.5	92.5	A+	Best Imaginable
P4	(14+15)*2.5 = 29*2.5	72.5	B	Good
P5	(20+20)*2.5= 40*2.5	100.0	A+	Best Imaginable
Avg		89.0	A	Excellent

4.3.2 Qualitative Results (Think Aloud & Interviews)

Think-Aloud Observations

Device Calibration: Interactions were generally positive (P2: *"Easy to use and quite intuitive"*). However, P4 found the wizard-driven setup somewhat abstract without the physical weight of the hardware band present.

Translation Mode: Navigation was fluid and confident. P5 noted that the interface layout was *"visually clear and uncluttered"* with zero points of navigation confusion.

4.2 Machine Learning Model Evaluation

Among all baseline models, the Random Forest that was trained on the 255 selected features (segmented data with feature selection) demonstrated the highest performance, achieving an accuracy of 77.97% prior to hyperparameter optimization and cross-validation.

Following the tuning process utilizing **RandomizedSearchCV** in conjunction with Stratified K-Fold cross-validation, the average cross-validation score was around 77%, whereas the final test accuracy rose to **83.9%**. Considering the relatively small dataset

and the unexpectedly elevated test accuracy, the experiment was replicated across various train-test random states to assess robustness.

Throughout all splits, the RF1, RF2, and cross-validation scores exhibited a degree of stability, while the optimized Random Forest consistently attained the highest accuracies, albeit with increased variability among test sets. This variability is likely attributable to the limited dataset size, where alterations in the train-test partition can considerably influence test difficulty, coupled with the model's capacity to leverage advantageous training partitions post hyperparameter optimization.

No indications of train-test leakage were detected, as feature selection, model training, and hyperparameter optimization were conducted solely on the training data. The consistency of the cross-validation scores across multiple random states further corroborates the dependability of the evaluation process.

The final hyperparameter-tuned model achieved an average accuracy of **84.67% ± 2.80%** across **17 random states**. This performance is within the range reported for Random Forest-based approaches by *Paudyal et al. (2017)* and *Shakeel et al. (2020)*, although direct comparisons should be interpreted cautiously due to potential differences in experimental methodology.

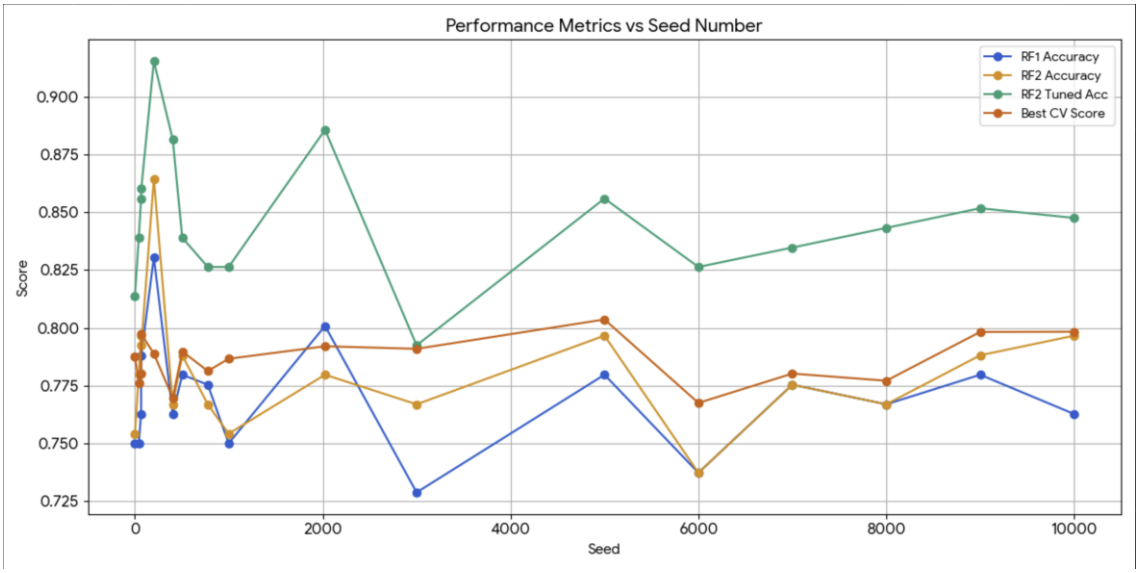


Figure 17: Performance Metrics VS Seed Number

Metric	RF1 Model	RF2 Model	RF2 Tuned	Best CV Score
Mean	0.7694	0.7807	0.8467	0.7862
Std Dev	0.0235	0.0267	0.0280	0.0100

Figure 18: Average Metrics of all Seeds

4.3 Final UI/UX Prototype Evaluation

4.3.1 Quantitative Results

Task Performance and Design Metrics (1–5 Likert Scale)

Initial Clarity: App purpose clarity scored **5.0/5.0**, with 100% confidence regarding home screen navigation.

Navigation Success: Locating the Bluetooth pairing menu scored **4.8/5.0** (P4 noted a slight delay in button discovery).

Calibration Logic: Comprehension of calibration gestures ("Fist" and "Wave") scored **4.7/5.0**.

Accessibility and Readability: Visual contrast and text size scored **4.9/5.0**, while the translated text readability scored 5.0/5.0.

System Usability Scale (SUS)

The application achieved an average SUS score of **96.8**, placing it in the **Best Imaginable (Grade A+)** bracket. Here is a table calculating all the SUS scores based on the Google Form data.

Participant	SUS Score Calculation	SUS Score	Grade	Adjective Rating
P1	39*2.5	97.5	A+	Best Imaginable
P2	38*2.5	95.0	A+	Best Imaginable
P3	39*2.5	97.5	A+	Best Imaginable
P4	39*2.5	97.5	A+	Best Imaginable
P5	38*2.5	95.0	A+	Best Imaginable
P6	40*2.5	100.0	A+	Best Imaginable
P7	38*2.5	95.0	A+	Best Imaginable
P8	38*2.5	95.0	A+	Best Imaginable

P9	38*2.5	95.0	A+	Best Imaginable
P10	40*2.5	100.0	A+	Best Imaginable
P11	38*2.5	95.0	A+	Best Imaginable
P12	40*2.5	100.0	A+	Best Imaginable
P13	39*2.5	97.5	A+	Best Imaginable
P14	39*2.5	97.5	A+	Best Imaginable
P15	38*2.5	95.0	A+	Best Imaginable
Avg		96.8	A+	Best Imaginable

4.3.2 Qualitative Results (Think Aloud & Interviews)

Think-Aloud Observations

Device Calibration: Interactions were generally positive but despite many finding the app still usable without the hardware, they thought it'd be better to have it.

Translation Mode: Navigation was fluid and confident.

4.4 Final Application Evaluation

The final application successfully implemented the core features identified during requirements analysis, including communication assistance, accessibility-focused design and support for future EMG integration. Combined with the machine learning results presented in Section 4.2, the project demonstrates the feasibility of an EMG-based sign language translation framework. While real-time integration with the custom hardware remains future work, the project achieved its primary objectives and provides a strong foundation for further development.

5.0 Conclusions

This project presented the design, development and evaluation of SignLink, an intelligent sign language translation system that combines wearable electromyography (EMG) sensing, machine learning and mobile application technologies. The project successfully demonstrated the feasibility of using EMG signals and machine learning techniques for sign language gesture recognition while also providing a functional mobile application capable of supporting communication between sign language users and non-sign language users.

The developed system incorporated a user-centred design approach, resulting in a mobile application that was evaluated positively during usability testing. Furthermore, the machine learning model achieved promising gesture classification performance using a publicly available Myo Armband dataset, demonstrating the potential of EMG-based sign language recognition. Although full integration between the wearable hardware, machine learning model and mobile application was not achieved within the project timeframe, the results provide a strong foundation for future development.

Overall, the findings suggest that wearable sensing technologies, artificial intelligence and mobile computing can be effectively combined to create accessible assistive communication solutions that help reduce communication barriers for individuals with hearing and speech impairments.

5.1 Limitations

Several limitations were encountered during the development of this project. The most significant limitation was the inability to collect a large-scale custom EMG dataset using the wearable prototype. As a result, the machine learning model was trained and evaluated using a publicly available Myo Armband dataset rather than data collected from the custom hardware platform.

Additionally, the wearable hardware prototype remained under development throughout the project and therefore was not sufficiently stable for extensive real-time testing and machine learning integration. Consequently, the complete end-to-end translation pipeline could not be fully validated using the custom hardware.

This Model also performs best for intra-user predictions as compared to inter-user predictions and hence a calibration system was implemented in the Application to make the App work best for a specific user.

The current implementation also focuses on recognizing individual American Sign Language (ASL) fingerspelling alphabet gestures rather than continuous words, sentences or conversational sign language. This limits the practical communication capabilities of the current system.

Finally, usability testing was conducted on a relatively small sample of participants due time and resource constraints. While the results provided valuable feedback, larger-scale testing would provide a more comprehensive evaluation of system usability and accessibility.

5.2 Future Work

Future development of SignLink should focus on completing and refining the wearable EMG sensing platform to enable reliable real-time data acquisition and direct integration with the mobile application. Collecting a dedicated EMG dataset from the custom hardware would also allow the machine learning model to be trained and evaluated using project-specific data.

Further improvements could include expanding gesture recognition beyond ASL fingerspelling alphabets to support words, phrases and continuous sign language translation. The integration of the trained machine learning model directly into the Android application would also enable real-time translation functionality.

An additional feature that could be explored is video-to-translation functionality, where users can upload or capture videos of sign language gestures using a smartphone camera. Computer vision and deep learning techniques could then be used to recognise signs directly from video input and convert them into text or speech. This would provide an alternative translation method for users who do not have access to the wearable device and would increase the accessibility and versatility of the SignLink platform.

Additional future enhancements may include multilingual translation support, cloud-based data synchronization, improved accessibility features and larger-scale user

testing involving Deaf and Hard of Hearing participants. These developments would further improve the practicality, accuracy and accessibility of the SignLink platform while supporting wider real-world adoption.

6.0 References

1. Abdullahi, S.B. and Chamnongthai, K., 2022. American Sign Language Words Recognition Using Spatio-Temporal Prosodic and Angle Features: A Sequential Learning Approach. IEEE Access, 10, pp.15911–15923. Available at: https://ieeexplore.ieee.org/document/9702061 [Accessed 5 November 2025].
2. United Nations, 2006. Convention on the Rights of Persons with Disabilities. New York: United Nations. Available at: https://www.un.org/development/desa/disabilities/convention-on-the-rights-of-persons-with-disabilities.html [Accessed 20 May 2026].
3. United Nations, 2015. Transforming our World: The 2030 Agenda for Sustainable Development. New York: United Nations. Available at: https://sdgs.un.org/2030agenda [Accessed 20 May 2026].
4. World Health Organization, 2026. Deafness and Hearing Loss. Available at: https://www.who.int/news-room/fact-sheets/detail/deafness-and-hearing-loss [Accessed 20 May 2026].
5. Papatsimouli, M., Kollias, K.F., Lazaridis, L., Maraslidis, G.S., Fragulis, G.F., Sarigiannidis, P. and Fragulis, G.F., 2022. Real Time Sign Language Translation Systems: A review study. In: 2022 11th International Conference on Modern Circuits and Systems Technologies (MOCAST). Bremen, Germany, 8-10 June 2022. New York: IEEE, pp. 1–4. doi:10.1109/MOCAST54814.2022.9837666. Available at: https://www.researchgate.net/publication/362331604_Real_Time_Sign_Language_Translation_Systems_A_review_study [Accessed 28 May 2026].
6. Ambavane, P., Karjavkar, R., Pathare, H., Relekar, S., Alte, B. and Sharma, N.K., 2020. A novel communication system for deaf and dumb people using gesture. ITM Web of Conferences, 32, p. 02003.

doi:10.1051/itmconf/20203202003. Available at: https://www.researchgate.net/publication/343284260_A_Novel_Communication_System_For_Deaf_And_Dumb_People_using_gesture [Accessed 25 May 2026].

7. Emmorey, K., 2002. Language, cognition, and the brain: Insights from sign language research. Mahwah, NJ: Lawrence Erlbaum Associates. Available at: https://staibabussalamsula.ac.id/wp-content/uploads/2024/03/LANGUAGE-COGNITION-THE-BRAIN-staibabussalamsula.ac_.id_.pdf [Accessed 28 May 2026].
8. Shukor, A.Z., Miskon, M.F., Jamaluddin, M.H., Ali@Ibrahim, F.b., Asyraf, M.F. and Bahar, M.B.b., 2015. A new data glove approach for Malaysian sign language detection. *Procedia Computer Science*, 76, pp.60–67. Available at: https://www.sciencedirect.com/science/article/pii/S1877050915037771 [Accessed 28 May 2026].
9. Akmeliawati, R., Ooi, M.P.L. and Kuang, Y.C., 2007. Real-time Malaysian sign language translation using colour segmentation and neural network. In: *2007 IEEE Instrumentation & Measurement Technology Conference IMTC 2007*. Warsaw, Poland, 1-3 May 2007. New York: IEEE, pp.1–6. Available at: http://ieeexplore.ieee.org/document/4258110/ [Accessed 28 May 2026].
10. Maharjan, S., Shrestha, S. and Fernando, S., 2024. Sign language detection and translation using smart glove. In: *International Conference on Computer Networking, Security & Data Analytics*. Springer, Singapore, pp. 419–431. doi:10.1007/978-981-97-3591-4_41. Available at: https://repository.londonmet.ac.uk/9015/1/Sign%20Language%20Detection%20and%20Translation%20using%20Smart%20Glove.pdf [Accessed 28 May 2026].
11. Shukor, A.Z., Miskon, M.F., Jamaluddin, M.H., Ali@Ibrahim, F.b., Asyraf, M.F. and Bahar, M.B.b., 2015. A new data glove approach for Malaysian sign language detection. Procedia Computer Science, 76, pp.60–67. Available at: https://www.sciencedirect.com/science/article/pii/S1877050915037771 [Accessed 28 May 2026].
12. Akmeliawati, R., Ooi, M.P.L. and Kuang, Y.C., 2007. Real-time Malaysian sign language translation using colour segmentation and neural network. In: 2007 IEEE Instrumentation & Measurement Technology Conference IMTC 2007.

Warsaw, Poland, 1-3 May 2007. New York: IEEE, pp.1–6. Available at: http://ieeexplore.ieee.org/document/4258110/ [Accessed 28 May 2026].

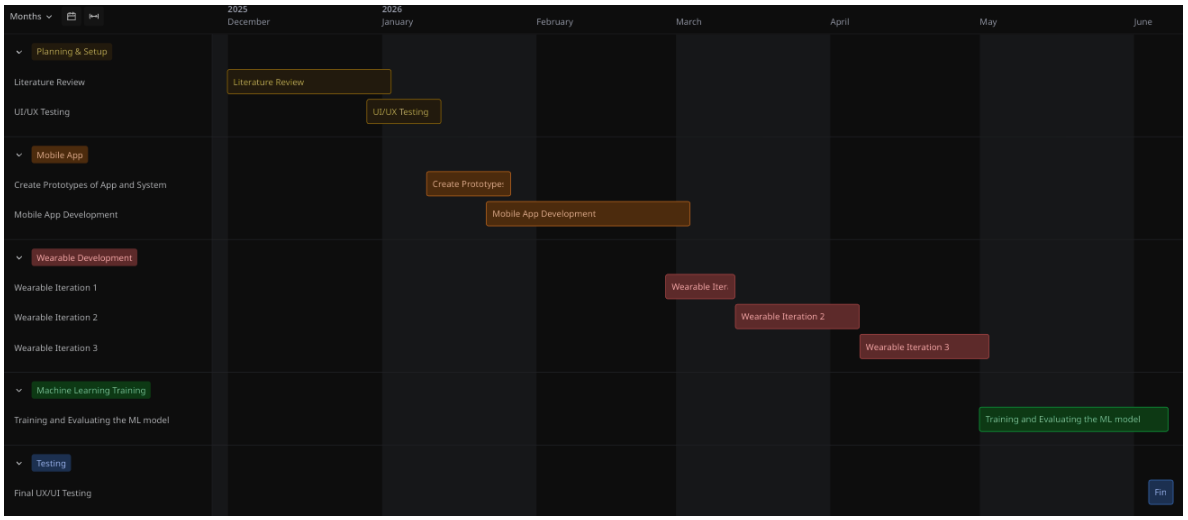
13. Shakeel, Z.M., So, S., Lingga, P. and Jeong, J.P., 2020. MAST: Myo Armband Sign-Language Translator for Human Hand Activity Classification. In: 2020 International Conference on Information and Communication Technology Convergence (ICTC). ICTC 2020. October 2020. DOI: 10.1109/ICTC49870.2020.9289153.Available at: https://iotlab.skku.edu/publications/international-conference/MAST-ICTC2020.pdf [Accessed: 30 May 2026].
14. Paudyal, P., Lee, J., Banerjee, A. and Sandeep, K.S. (2017) DyFAV: Dynamic Feature Selection and Voting for Real-time Recognition of Fingerspelled Alphabet using Wearables. In: Proceedings of the 22nd International Conference on Intelligent User Interfaces. New York, NY: ACM. Available at: https://doi.org/10.1145/3025171.3025216 [Accessed: 30 May 2026].
15. Paudyal, P., Banerjee, A. and Sandeep, K.S. (2016) SCEPTRE: a Pervasive, Non-Invasive, and Programmable Gesture Recognition Technology. In: Proceedings of the ACM Intelligent User Interfaces Conference (IUI 2016). New York, NY: ACM. Available at: https://doi.org/10.1145/2856767.2856794 [Accessed: 30 May 2026].
16. Chowdhury, R.H., Reaz, M.B.I., Ali, M.A.B.M., Bakar, A.A.A., Chellappan, K. and Chang, T.G. (2013) 'Surface Electromyography Signal Processing and Classification Techniques', Sensors, 13(9), pp. 12431–12466. doi: 10.3390/s130912431. Available at: https://pmc.ncbi.nlm.nih.gov/articles/PMC3821366/ [Accessed: 22 May 2026].
17. Chen, T. and Guestrin, C., 2016. XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp.785–794. Available at: https://dl.acm.org/doi/10.1145/2939672.2939785 [Accessed: 22 May 2026].
18. World Health Organization, 2022. *Global Report on Health Equity for Persons with Disabilities*. Geneva: World Health Organization. Available at: https://www.who.int/publications/i/item/9789240063600 [Accessed 23 May 2026].
19. Zhang, R., Hong, Y., Dong, H., Yang, X., Zhang, H. and Dang, L. 2025. Advances in electromyography armbands for gesture recognition and

multimodal fusion. iScience, 29(1), p. 114517. doi: 10.1016/j.isci.2025.114517. Available at: https://www.sciencedirect.com/science/article/pii/S2589004225027786 [Accessed: 22 May 2026].

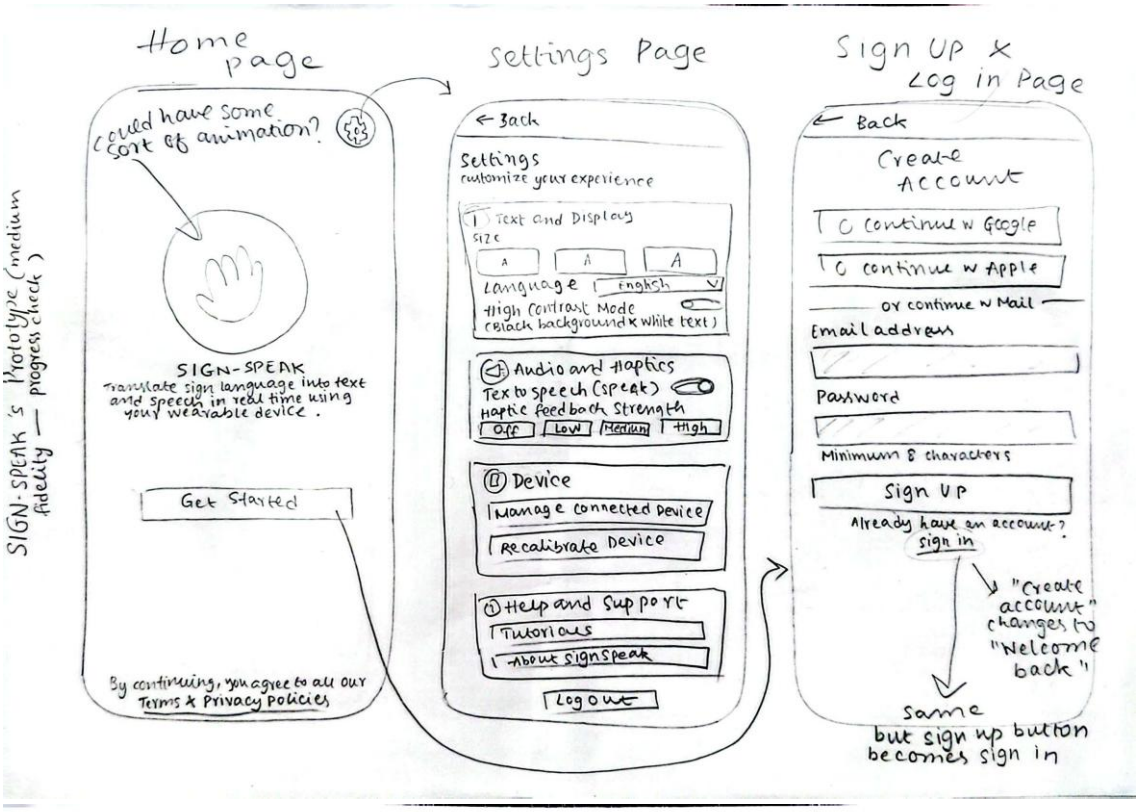
7.0 Appendix

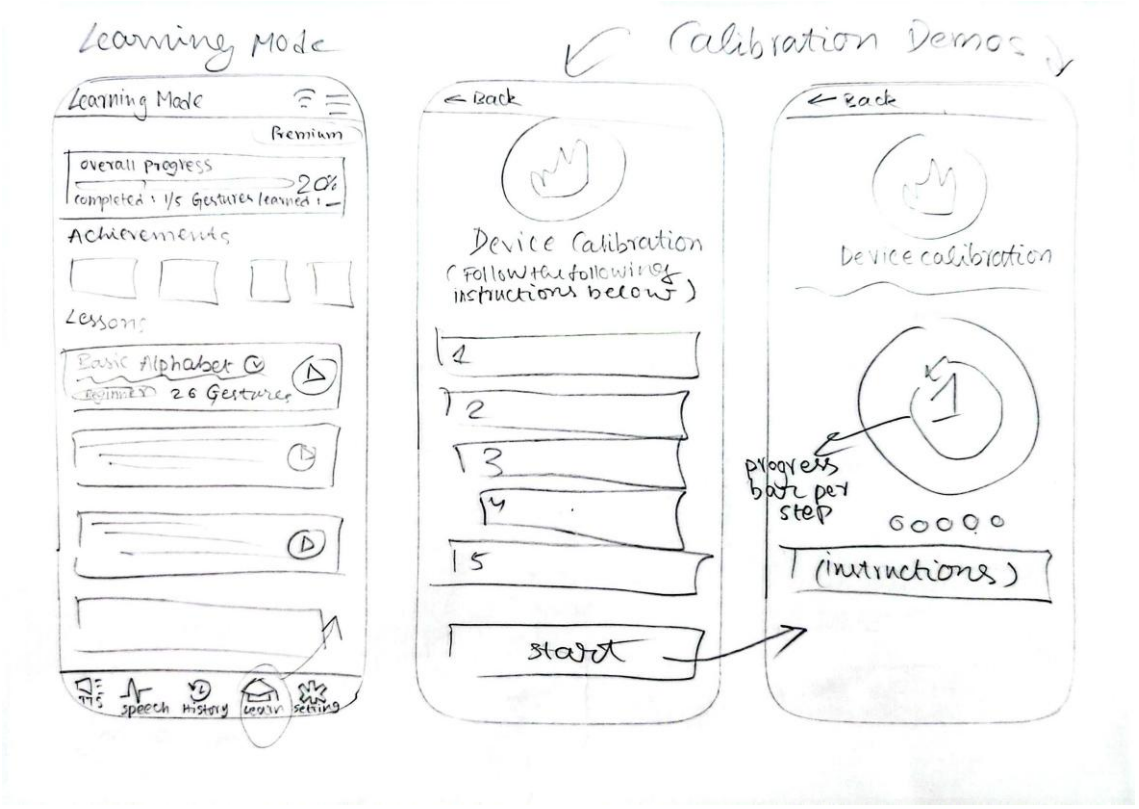
7.1 Appendices of UI/UX Design Section

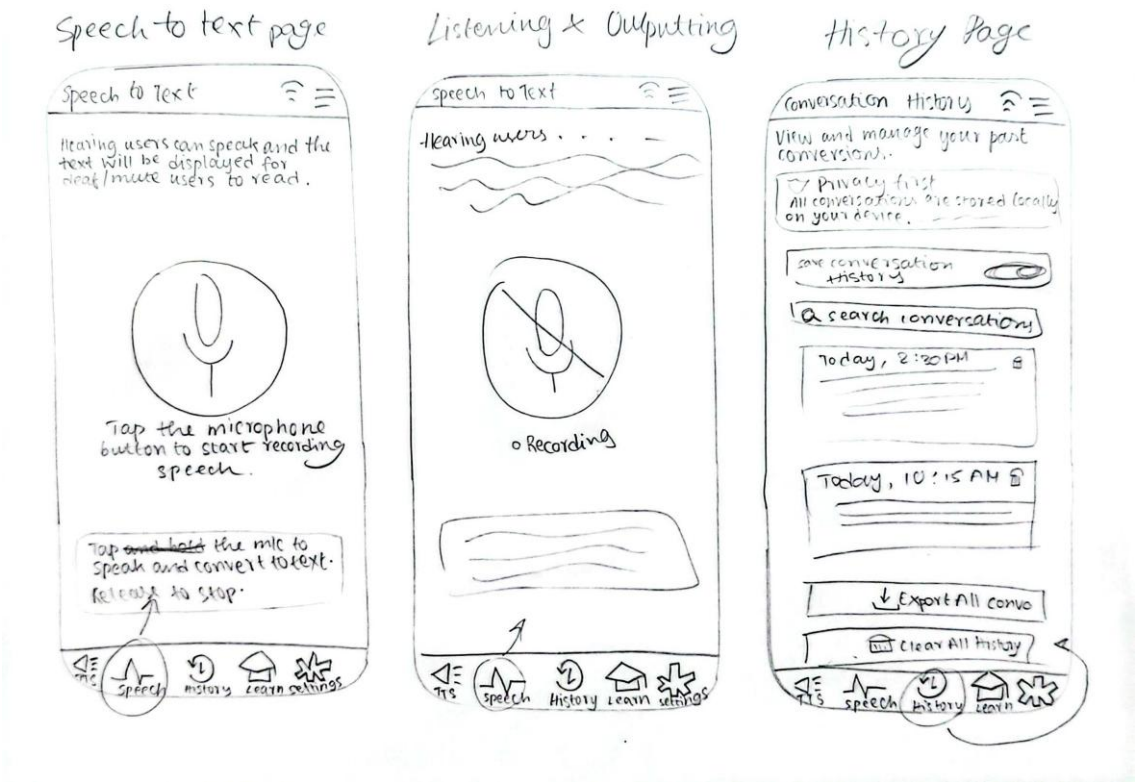
7.1.1 Gantt Chart



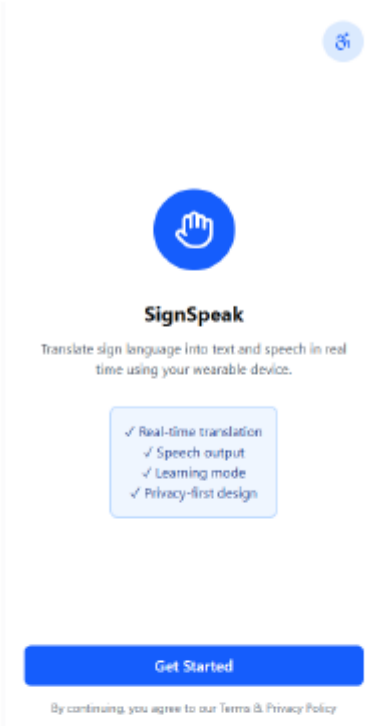
7.1.2 Low Fidelity Diagrams

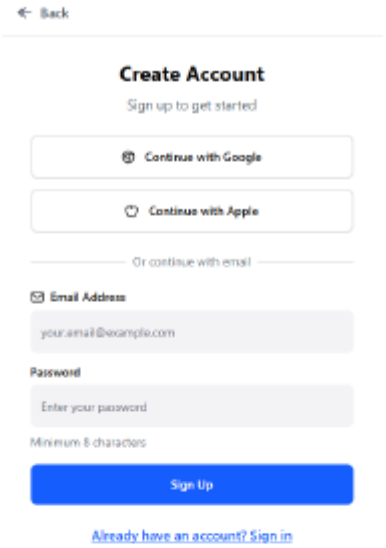


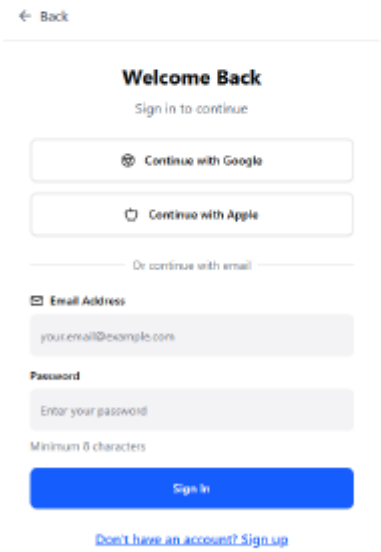

Learning Mode
Calibration Demo
Learning Mode
Premium
overall Progress
20%
completed : 1/5 Gestures learned : -
Achievements
Lessons
Basic Alphabet
Beginner 26 Gestures
TTS
speech
History
Learn
setting
Back
Device Calibration
(Follow the following instructions below)
1
2
3
4
5
start
Back
Device calibration
1
progress bar per step
(instructions)

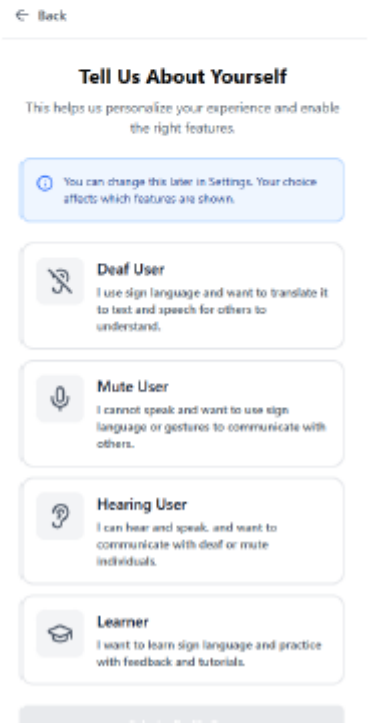

Speech-to text page
Listening & Outputting
History Page
Speech to Text
Hearing users can speak and the text will be displayed for deaf/mute users to read.
Tap the microphone button to start recording speech.
Tap the mic to speak and convert to text.
Release to stop.
TTS
Speech
History
Learn
settings
Speech to Text
Hearing users . . .
Recording
TTS
Speech
History
Learn
settings
Conversation History
View and manage your past conversions.
Privacy first
All conversations are stored locally on your device.
save conversation history
search conversations
Today, 2:30 PM
Today, 10:15 AM
Export All convo
Clear All history
TTS
Speech
History
Learn

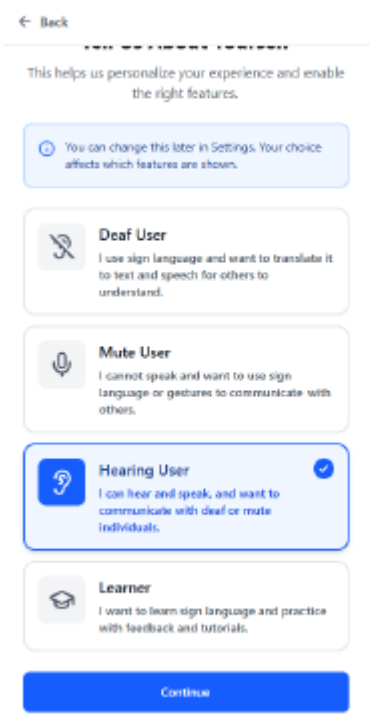
7.1.3 High Fidelity Diagrams



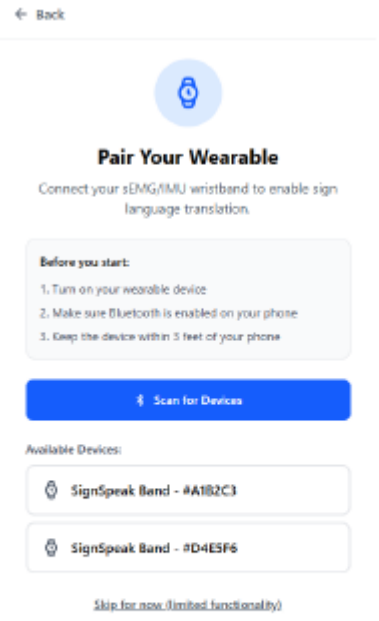


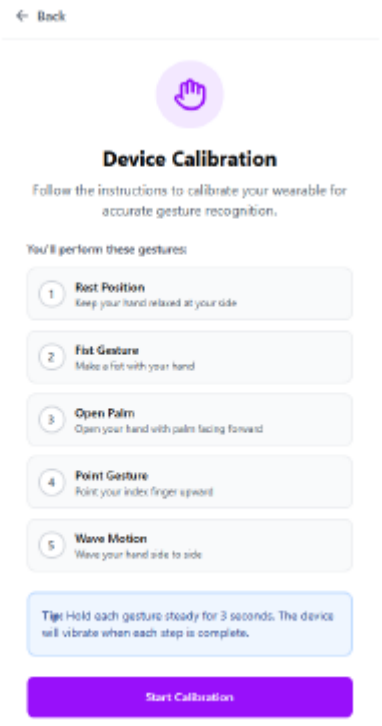


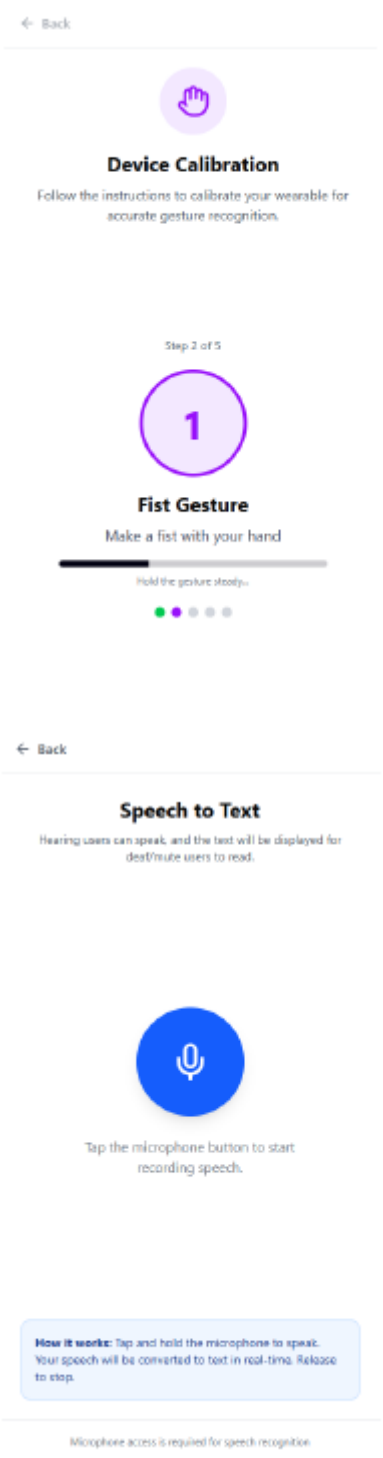


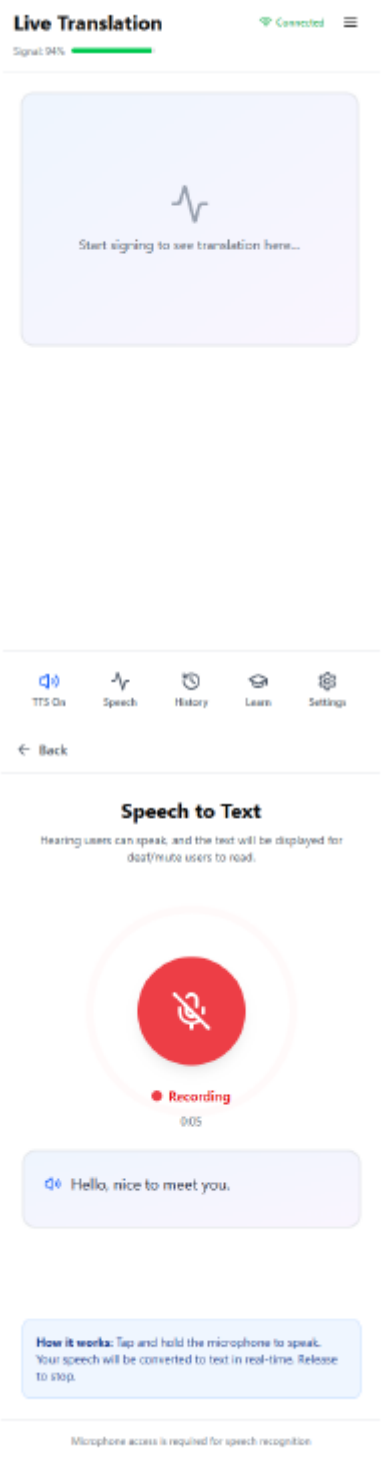

Back
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to understand.
Mute User
I cannot speak and want to use sign language or gestures to communicate with others.
Hearing User
I can hear and speak, and want to communicate with deaf or mute individuals.
Learner
I want to learn sign language and practice with feedback and tutorials.
Continue

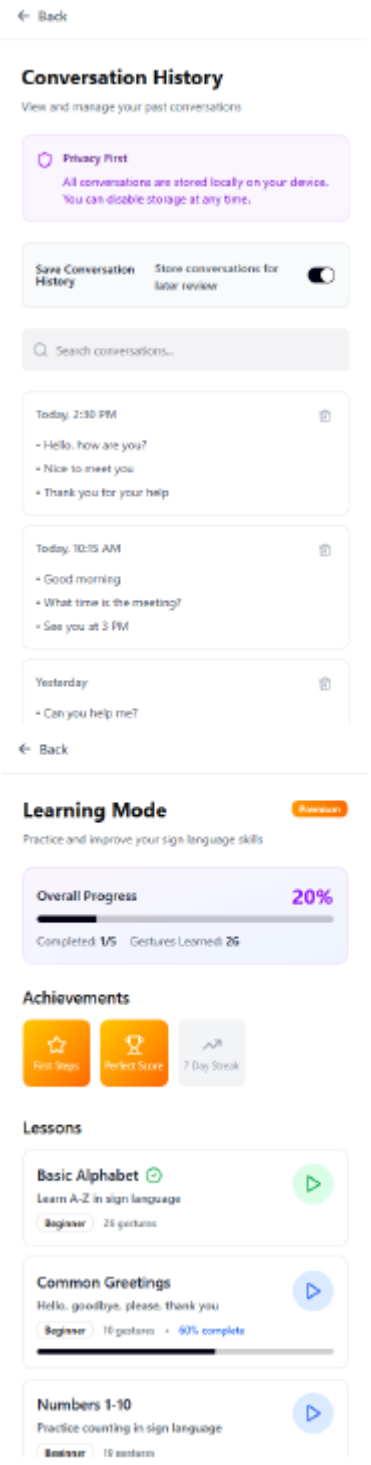

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scan for Devices
Skip for now (limited functionality)

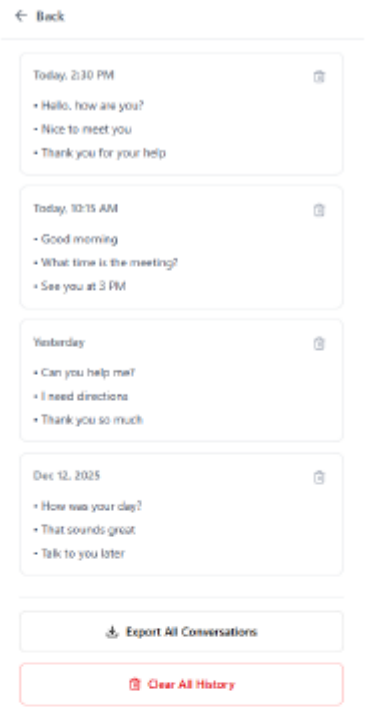

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scan for Devices
Available Devices:
SignSpeak Band - #A1B2C3
SignSpeak Band - #D4E5F6
Skip for now (limited functionality)


Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
You'll perform these gestures:
1 Rest Position
Keep your hand relaxed at your side
2 Fist Gesture
Make a fist with your hand
3 Open Palm
Open your hand with palm facing forward
4 Point Gesture
Point your index finger upward
5 Wave Motion
Wave your hand side to side
Tip: Hold each gesture steady for 3 seconds. The device will vibrate when each step is complete.
Start Calibration

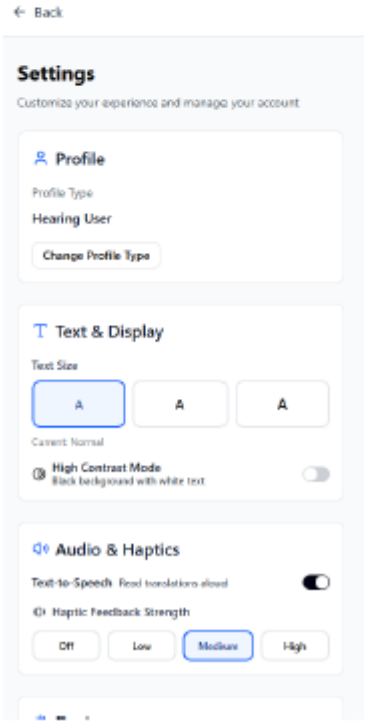

Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
Step 2 of 5
1
Fist Gesture
Make a fist with your hand
Hold the gesture steady...
Back
Speech to Text
Hearing users can speak, and the text will be displayed for deaf/mute users to read.
Tap the microphone button to start recording speech.
How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop.
Microphone access is required for speech recognition

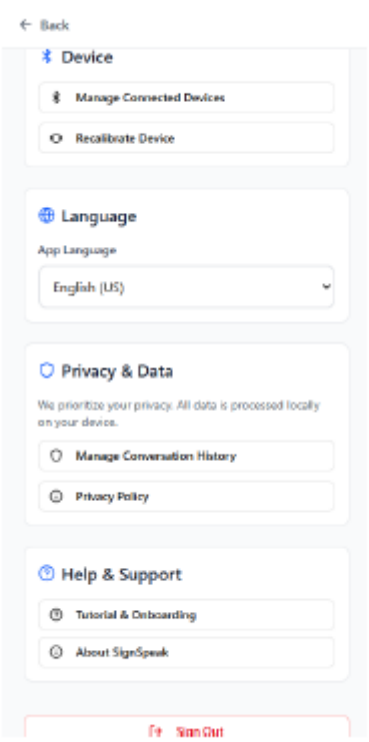

Live Translation
Connected
Signal: 94%
Start signing to see translation here...
TTS On
Speech
History
Learn
Settings
Back
Speech to Text
Hearing users can speak, and the text will be displayed for deaf/mute users to read.
Recording
0:05
Hello, nice to meet you.
How it works: Tap and hold the microphone to speak. Your speech will be converted to text in real-time. Release to stop.
Microphone access is required for speech recognition

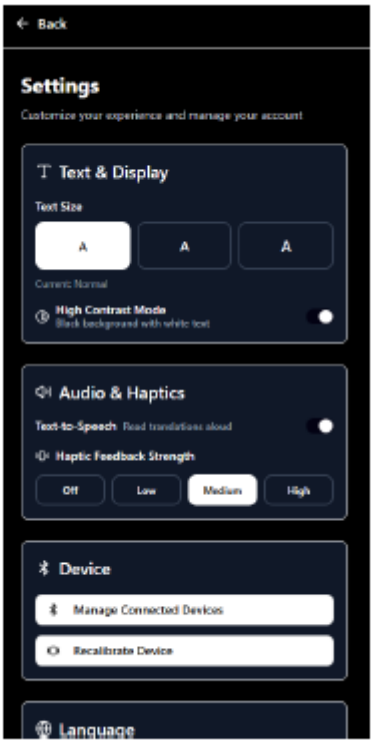

Back
Conversation History
View and manage your past conversations
Privacy First
All conversations are stored locally on your device.
You can disable storage at any time.
Save Conversation History
Store conversations for later review
Search conversations...
Today, 2:30 PM
• Hello, how are you?
• Nice to meet you
• Thank you for your help
Today, 10:15 AM
• Good morning
• What time is the meeting?
• See you at 3 PM
Yesterday
• Can you help me?


Back
Today, 2:30 PM
• Hello, how are you?
• Nice to meet you
• Thank you for your help
Today, 10:15 AM
• Good morning
• What time is the meeting?
• See you at 3 PM
Yesterday
• Can you help me?
• I need directions
• Thank you so much
Dec 12, 2025
• How was your day?
• That sounds great
• Talk to you later
Export All Conversations
Clear All History

Back
Learning Mode
Premium
Practice and improve your sign language skills
Overall Progress
20%
Completed: 1/5
Gestures Learned: 26
Achievements
First Steps
Perfect Score
7 Day Streak
Lessons
Basic Alphabet
Learn A-Z in sign language
Beginner
26 gestures
Common Greetings
Hello, goodbye, please, thank you
Beginner
10 gestures
60% complete
Numbers 1-10
Practice counting in sign language


Back
Settings
Customize your experience and manage your account
Profile
Profile Type
Hearing User
Change Profile Type
Text & Display
Text Size
A
A
A
Current: Normal
High Contrast Mode
Black background with white text
Audio & Haptics
Text-to-Speech
Haptic Feedback Strength
Off
Low
Medium
High


Back
Device
Manage Connected Devices
Recalibrate Device
Language
App Language
English (US)
Privacy & Data
We prioritize your privacy. All data is processed locally on your device.
Manage Conversation History
Privacy Policy
Help & Support
Tutorial & Onboarding
About SignSpeak
Sign Out


Back
Settings
Customize your experience and manage your account
Text & Display
Text Size
A
A
A
Current: Normal
High Contrast Mode
Black background with white text
Audio & Haptics
Text-to-Speech Read translations aloud
Haptic Feedback Strength
Off
Low
Medium
High
Device
Manage Connected Devices
Recalibrate Device
Language

7.1.4 Testing Consent Forms

TESTING INFORMATION SHEET

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Duration: December 29,2025 - January 5,2026

Researcher: Sombatla Truc Deydeepya Lakshmi

Institution: Vietnam National University of Information Technology (UIT)

You are invited to participate in a research study evaluating the user interface of **SignSpeak**, a mobile application designed to translate sign language into text and speech. Before you decide, it is important to understand why the research is being done and what it will involve. Please take the time to read the following information.

What is the purpose of this research? This research aims to evaluate the navigation logic and user experience of the SignSpeak app. While the final product will eventually include a wearable sensor (sEMG/IMU wristband), this test focuses exclusively on how intuitive the mobile software is for onboarding, calibration, and translation tasks.

Why have I been invited? You have been invited because you are either a student at UIT or a potential user who can provide feedback on the app's logic. We are looking for 5 participants to help identify friction points in the design.

What does participation involve? If you agree to take part, you will be asked to:

- Complete 3-4 specific tasks using the **Figma** prototype on a smartphone (in-person) or via screen-share (remote).
- Use the "**Think Aloud**" method (verbalize your thoughts as you navigate).
- Complete a 10-item **System Usability Scale (SUS)** survey on a Google Form.
- Participate in a brief 5-minute exit interview. The entire session will last approximately **30 minutes**.

Is this compulsory to take part in? Your participation is entirely voluntary. You can withdraw at any time during the session without giving a reason and without any penalty.

How will my data be used? All data collected (notes, SUS scores, and interview responses) will be **anonymized**. You will be referred to as "Participant X" (e.g., P1, P2) in the final report. No personal identification will be shared.

Who can I contact for further information? If you have questions, you can contact the researcher at: ddpyawork@gmail.com

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombattla Truc Daydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES
I agree to take part in the above study.	YES

Participant Name: [redacted]

Participant Age (if under 18 years of age): [redacted]

Date: January 3rd , 2026

Signature:



CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 30 / 12/ 2025

Signature: _____

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

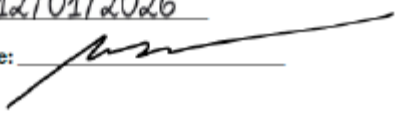
Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	(YES) / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	(YES) / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	(YES) / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	(YES) / NO
I agree to take part in the above study.	(YES) / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 12/01/2026

Signature: 

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 01/01/2026

Signature: Satwik Biswas

CONSENT FORM

Project Title: Usability Evaluation of the SignSpeak Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 3/01/2026

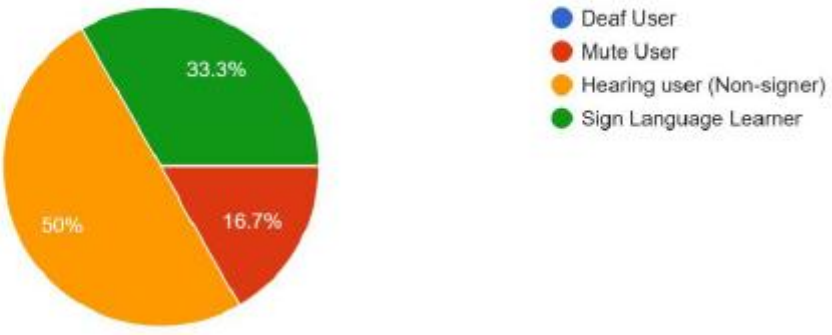
Signature:



7.1.5 Results from Form

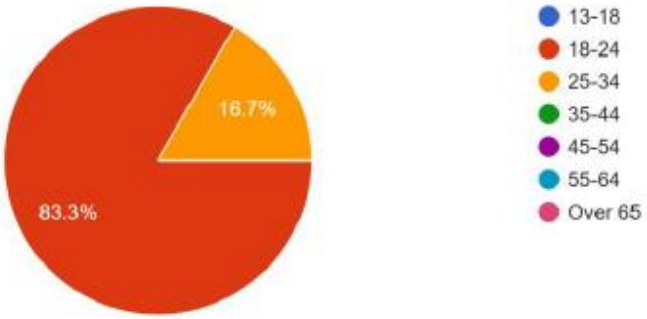
Please select the category that you fall under.

6 responses



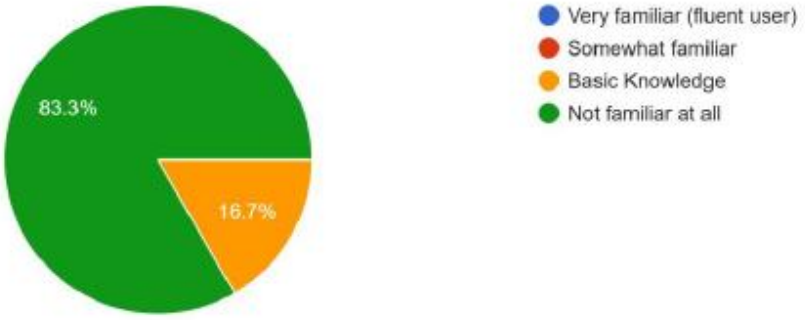
Age

6 responses



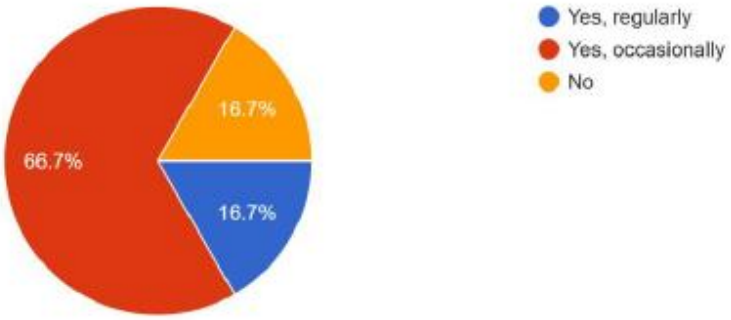
How familiar are you with sign language?

6 responses



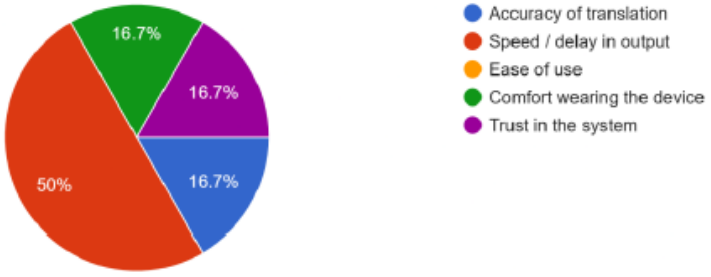
Have you ever used assistive or translation technology before?

6 responses



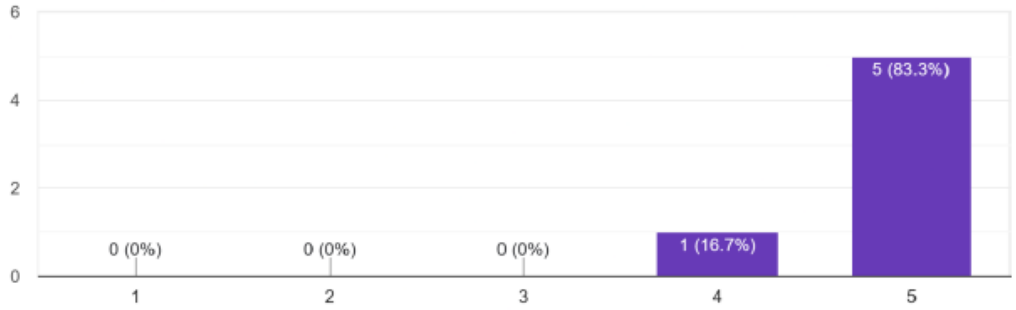
In what context would you most likely use a sign language translation app? (e.g., daily communication, education, public services, emerge...enge when using a sign language translation app?

6 responses



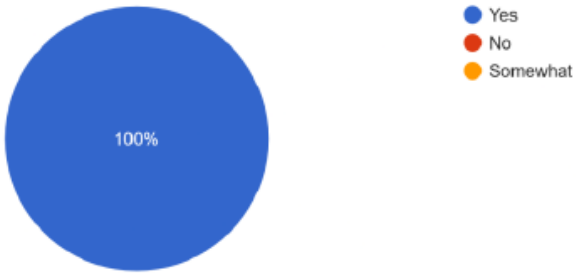
How clear was the purpose of the app when you first opened it? [1 (Not clear at all) – 5 (Very clear)]

6 responses



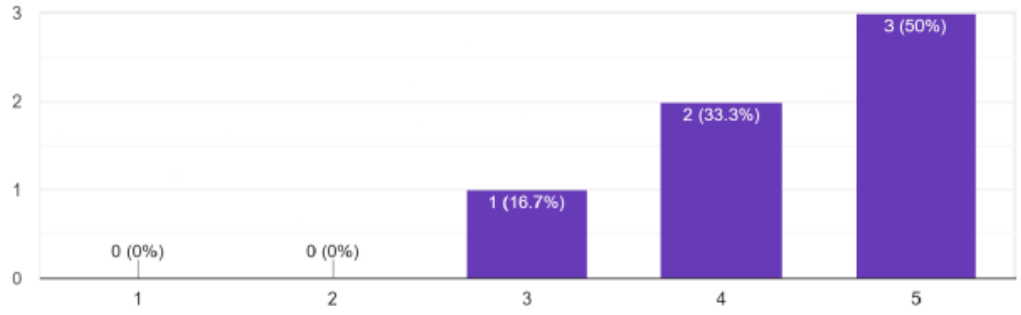
Did you feel confident about what to do next after seeing the home screen?

6 responses



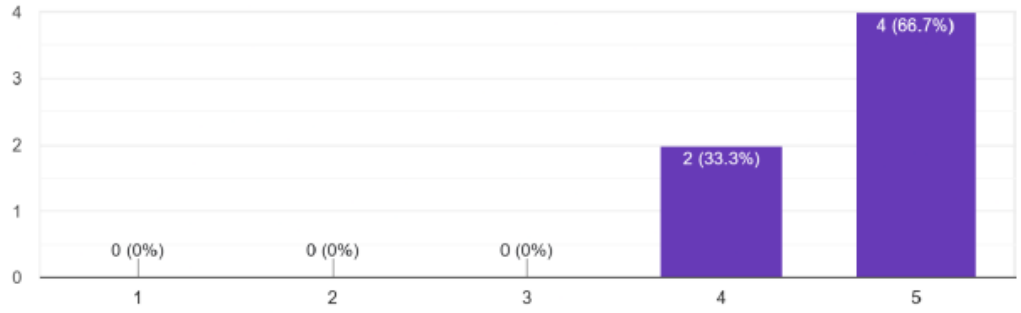
How easy was it to find and use the option to connect the wearable device? [1 (Very difficult) – 5 (Very easy)]

6 responses



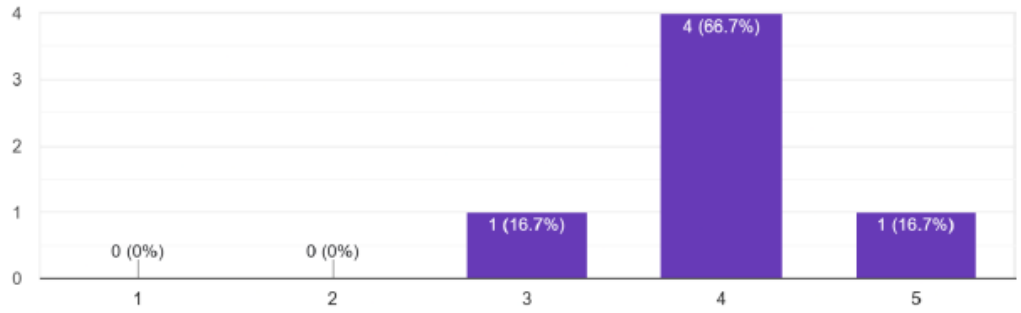
How easy was it to understand the calibration instructions? [1 (Very difficult) – 5 (Very easy)]

6 responses



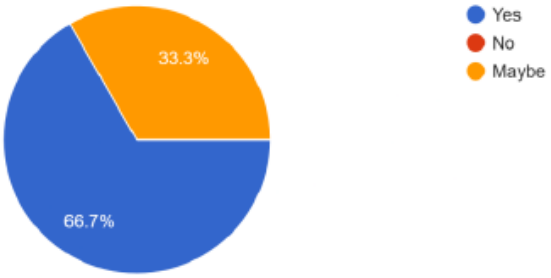
How easy was it to switch from text output to speech output? [1 (Very difficult) – 5 (Very easy)]

6 responses



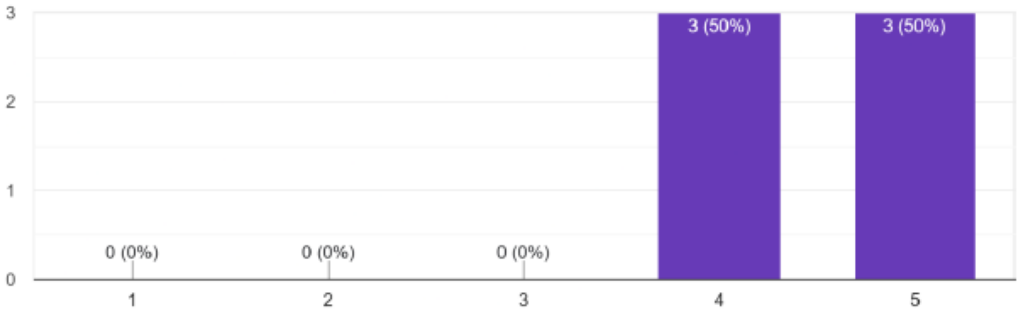
Did you feel the app was designed with accessibility in mind?

6 responses



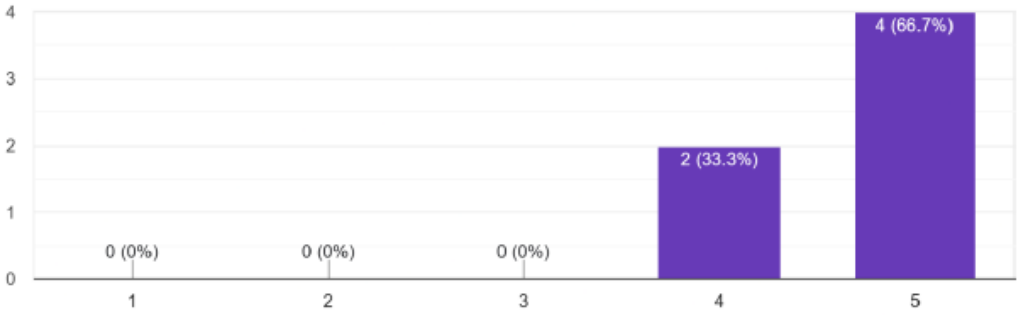
How readable was the text size and contrast? [1 (Very difficult) – 5 (Very easy)]

6 responses



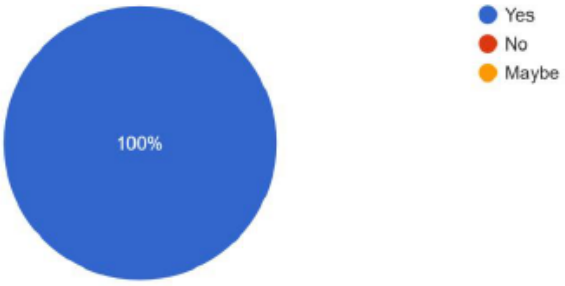
How clear and readable was the translated text on screen? [1 (Very difficult) – 5 (Very easy)]

6 responses



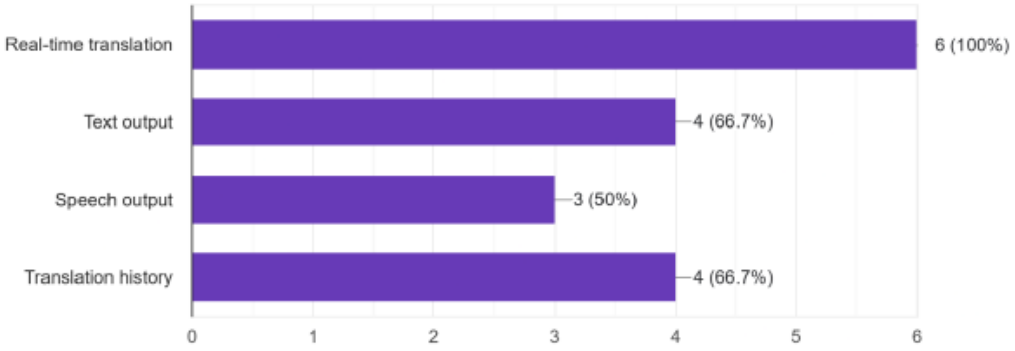
Did the interface feel visually clear and uncluttered?

6 responses



Which features did you find most valuable? (Select all that apply)

6 responses



What additional feature would most improve your experience?

2 responses

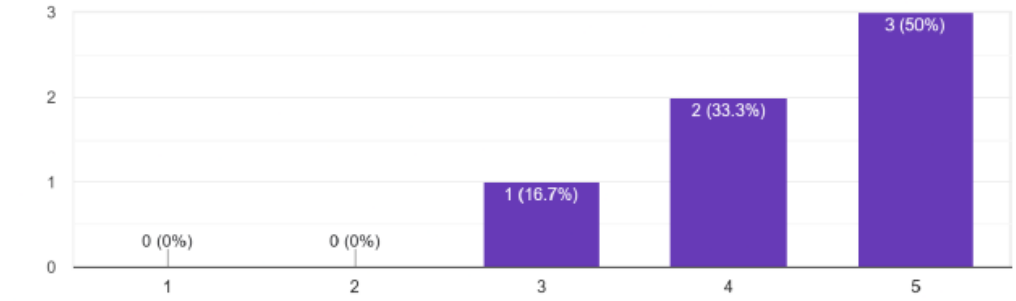
Currently unsure :/

I want to input a full story or text passage and see it translated into a sequence of 3D hand signs. This allows me to see how a continuous narrative looks in sign language, rather than just isolated words.

SUS:

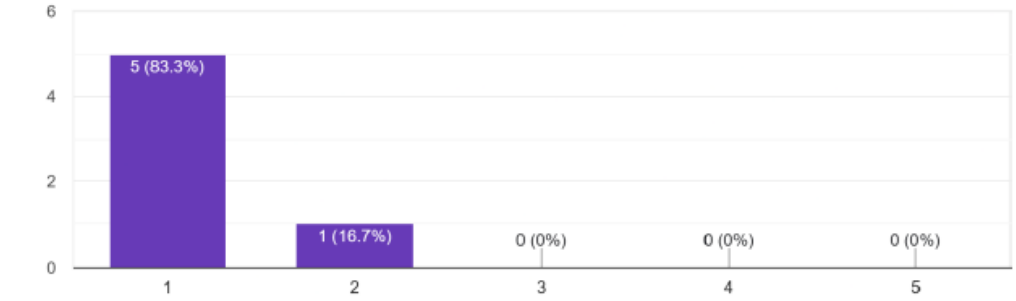
I think that I would like to use SignSpeak frequently.

6 responses



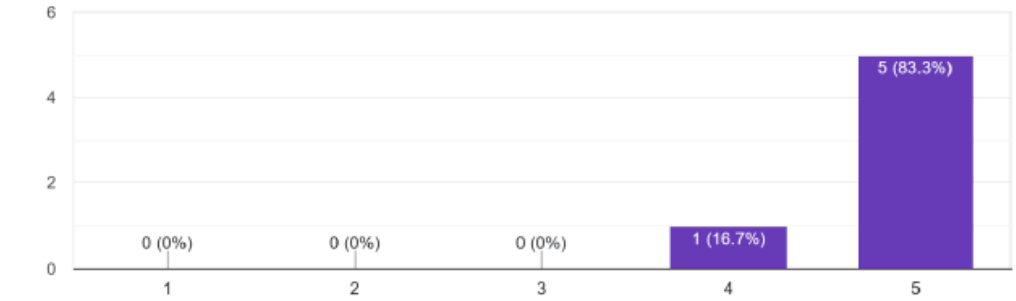
I found the app unnecessarily complex.

6 responses



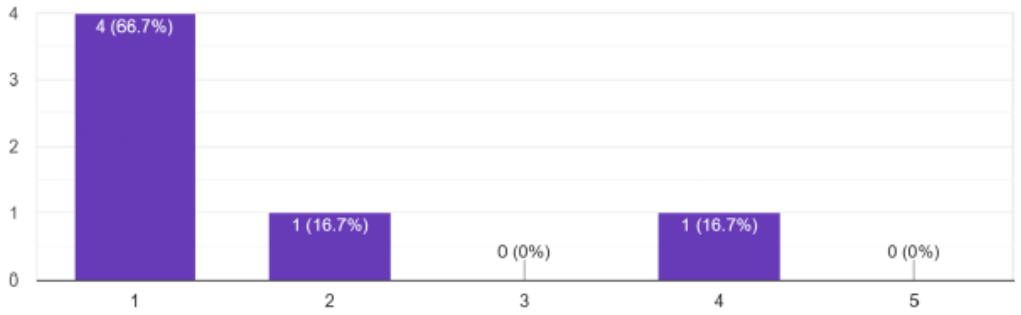
I thought the app was easy to use.

6 responses



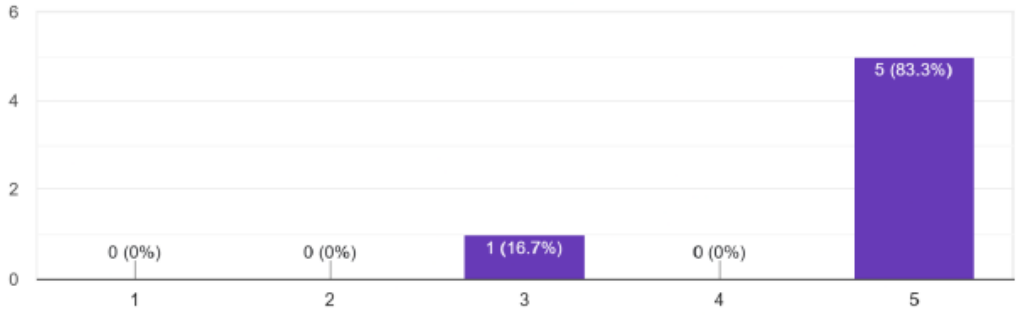
I think that I would need the support of a technical person to be able to use this app.

6 responses



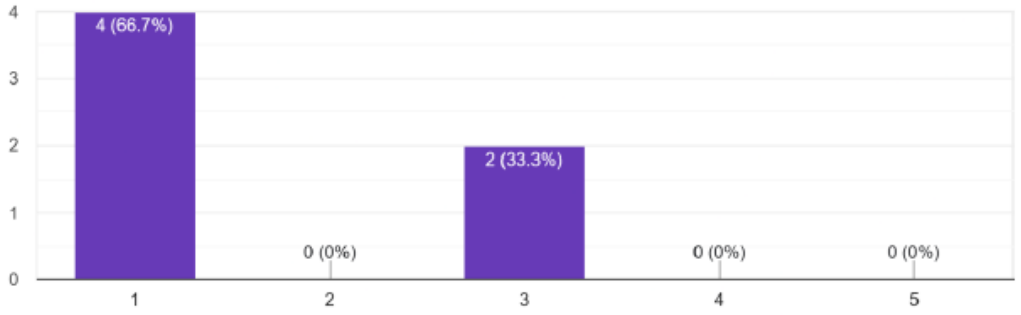
I found the various functions in this app were well integrated.

6 responses



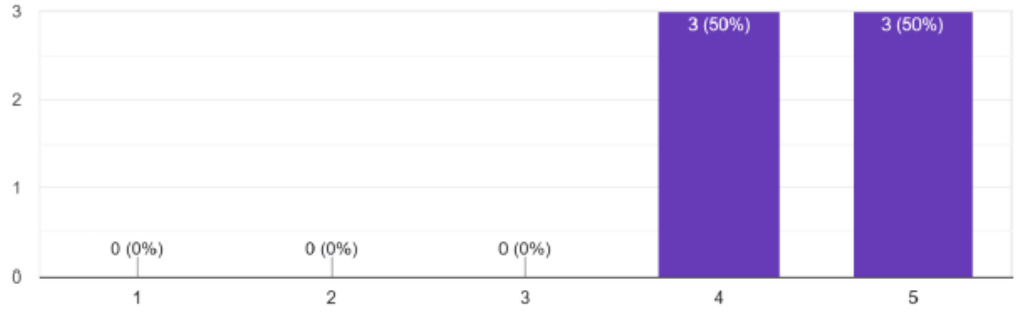
I thought there was too much inconsistency in this app.

6 responses



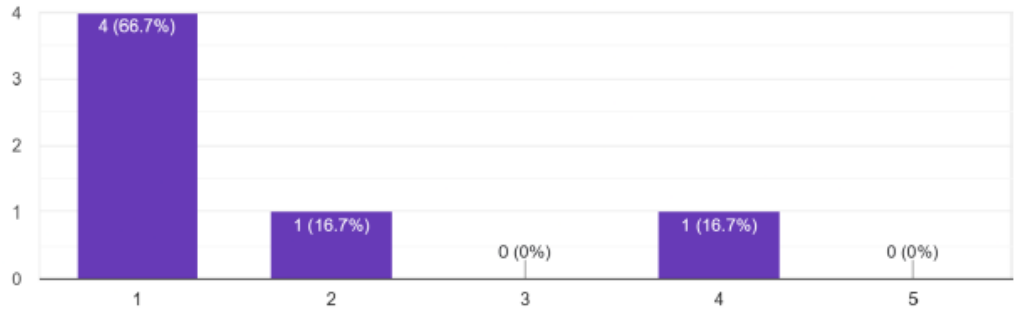
I would imagine that most people would learn to use this app very quickly.

6 responses



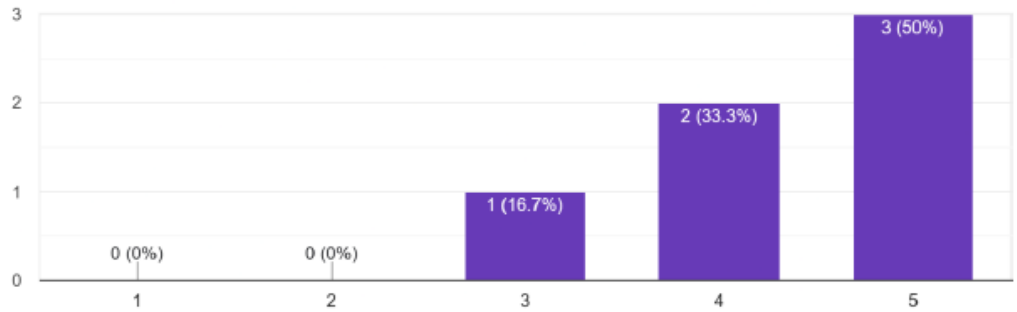
I found the app very cumbersome/awkward to use.

6 responses



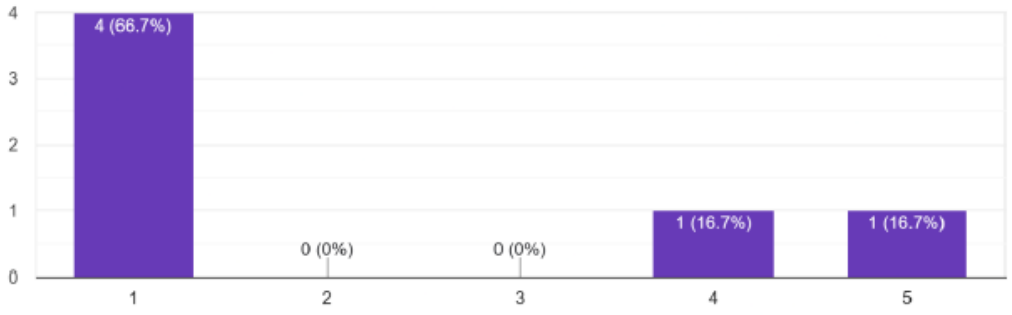
I felt very confident using the app.

6 responses



I needed to learn a lot of things before I could get going with this app.

6 responses



What was the most confusing part of the navigation?

4 responses

Nothing

At the moment, I feel okay. :)))

It was easy to use and quite intuitive.

I dont feel confused

How did you feel about the calibration process without the physical wristband?

5 responses

Interesting

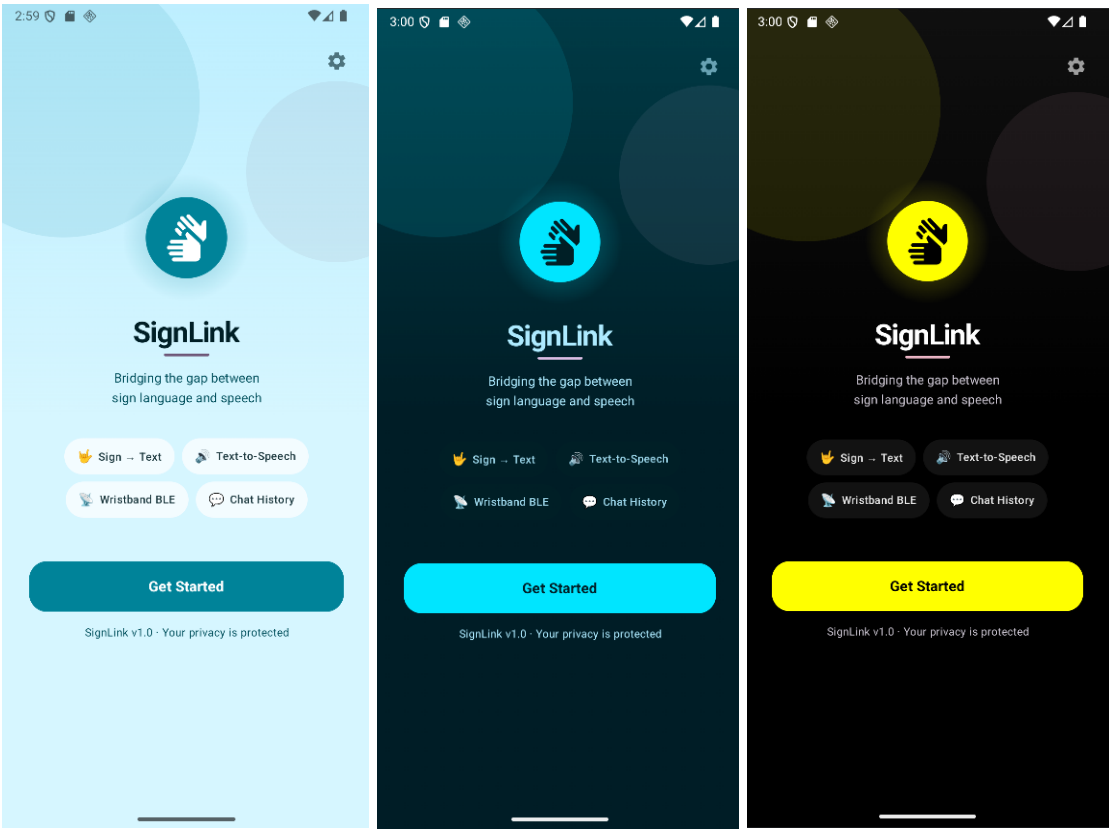
good

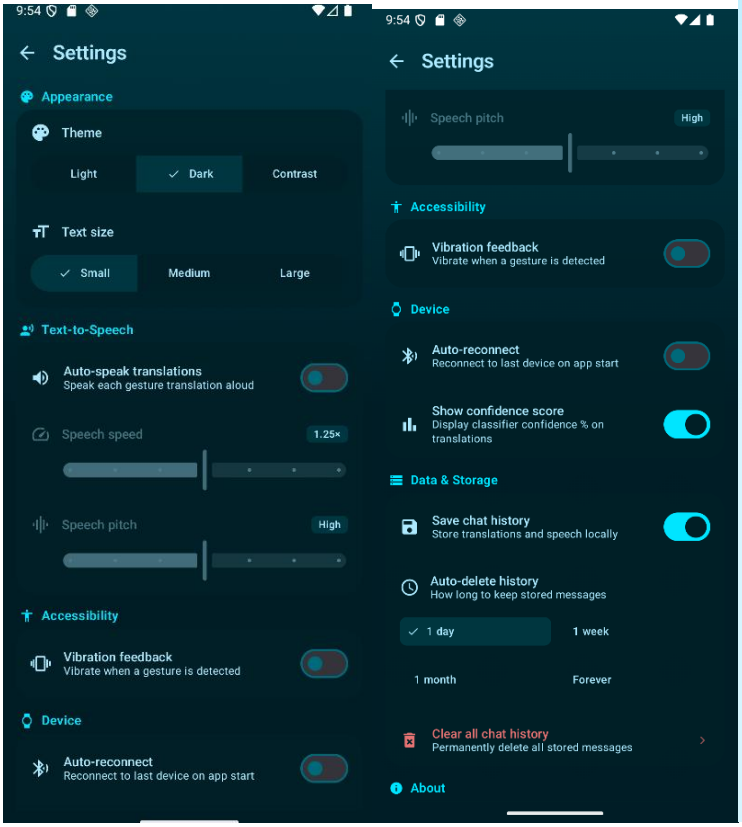
I think it's fine if it's not have, but it's even better if it is.

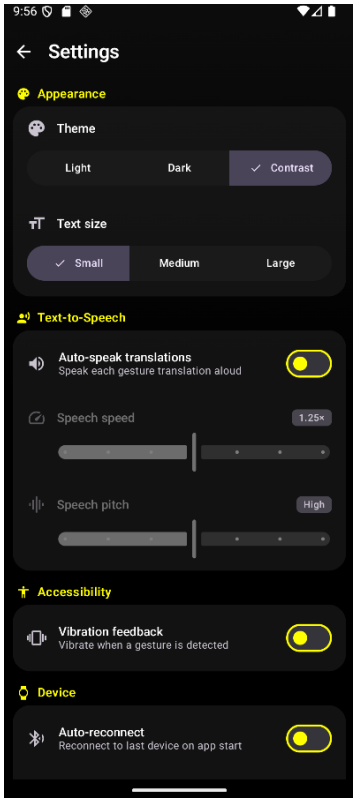
Would be better if the band was available too but it was alright.

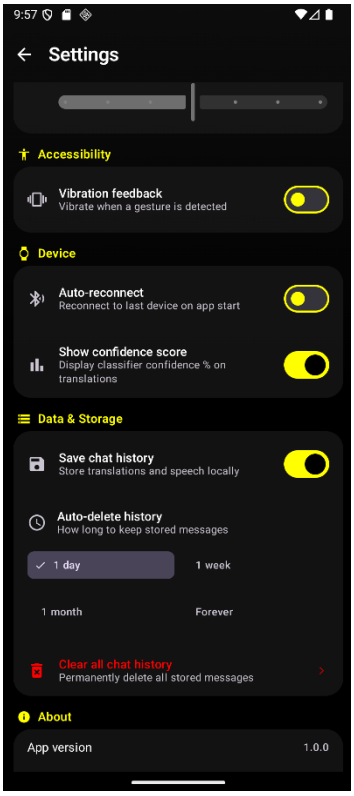
feel uncomfortable

7.2 App Screens

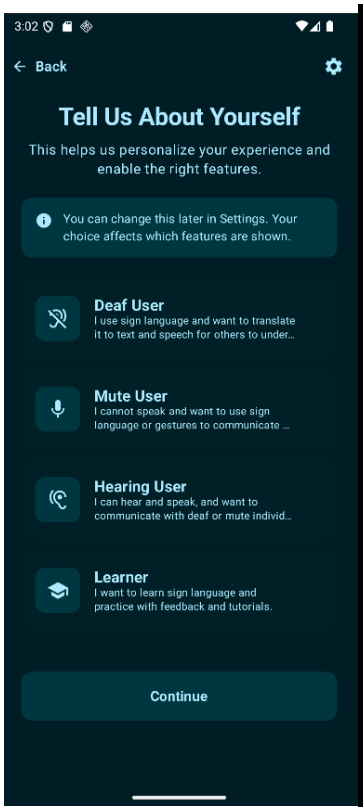


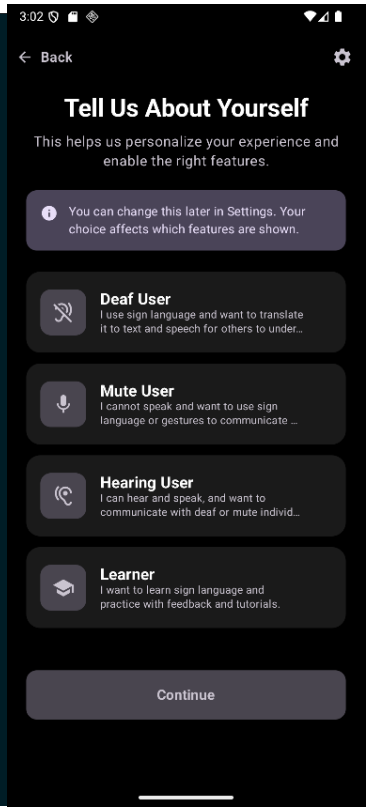


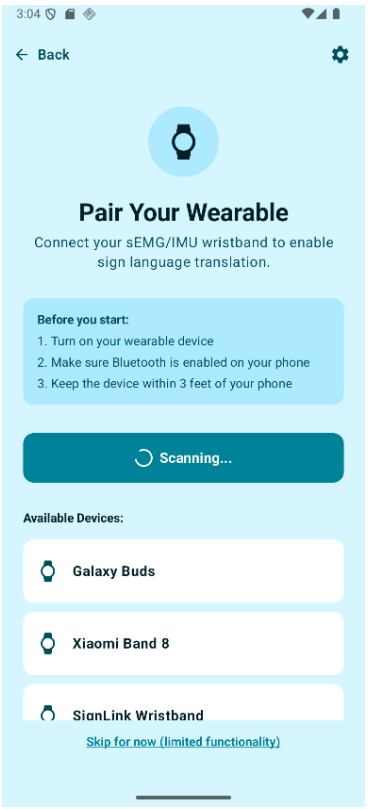

9:56
Settings
Appearance
Theme
Light
Dark
Contrast
Text size
Small
Medium
Large
Text-to-Speech
Auto-speak translations
Speak each gesture translation aloud
Speech speed
1.25×
Speech pitch
High
Accessibility
Vibration feedback
Vibrate when a gesture is detected
Device
Auto-reconnect
Reconnect to last device on app start

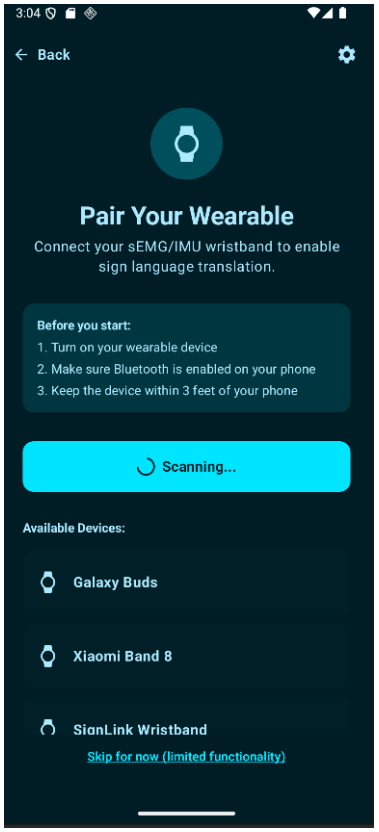

9:57
Settings
Accessibility
Vibration feedback
Vibrate when a gesture is detected
Device
Auto-reconnect
Reconnect to last device on app start
Show confidence score
Display classifier confidence % on translations
Data & Storage
Save chat history
Store translations and speech locally
Auto-delete history
How long to keep stored messages
1 day
1 week
1 month
Forever
Clear all chat history
Permanently delete all stored messages
About
App version
1.0.0

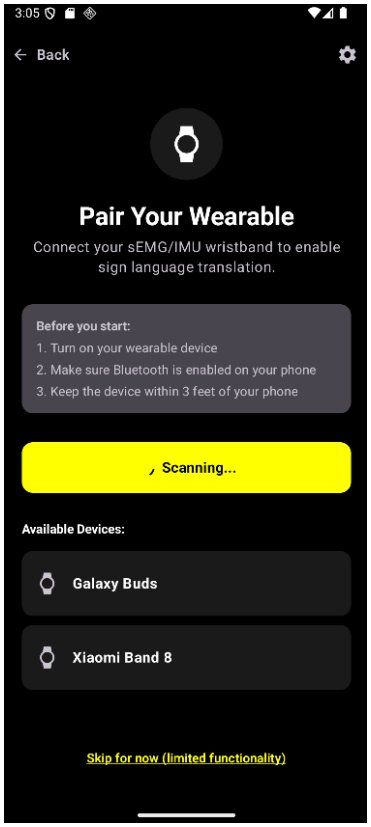

3:01
Back
Tell Us About Yourself
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to under...
Mute User
I cannot speak and want to use sign language or gestures to communicate ...
Hearing User
I can hear and speak, and want to communicate with deaf or mute individ...
Learner
I want to learn sign language and practice with feedback and tutorials.
Continue

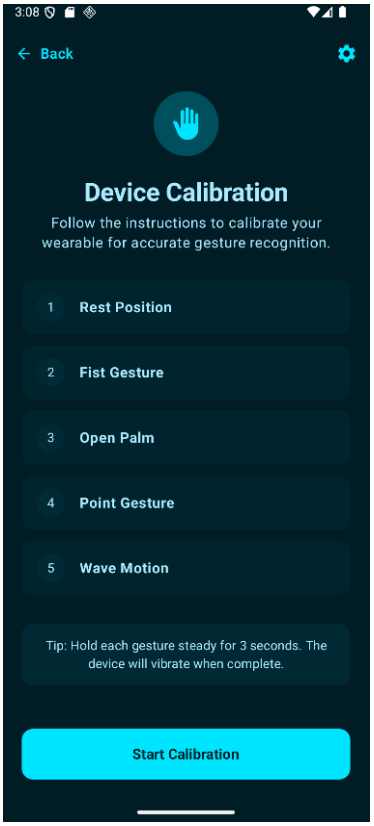

3:02
Back
Tell Us About Yourself
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to under...
Mute User
I cannot speak and want to use sign language or gestures to communicate ...
Hearing User
I can hear and speak, and want to communicate with deaf or mute individ...
Learner
I want to learn sign language and practice with feedback and tutorials.
Continue

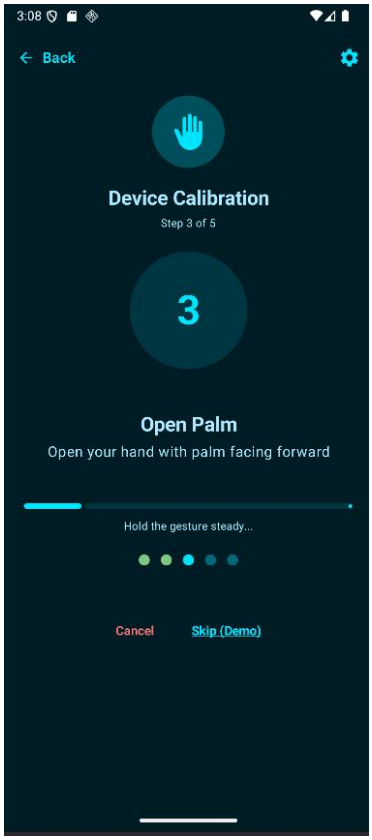

3:02
Back
Tell Us About Yourself
This helps us personalize your experience and enable the right features.
You can change this later in Settings. Your choice affects which features are shown.
Deaf User
I use sign language and want to translate it to text and speech for others to under...
Mute User
I cannot speak and want to use sign language or gestures to communicate ...
Hearing User
I can hear and speak, and want to communicate with deaf or mute individ...
Learner
I want to learn sign language and practice with feedback and tutorials.
Continue

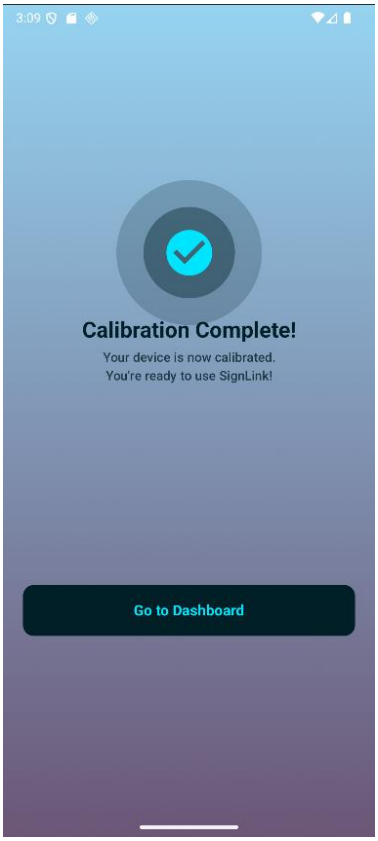

3:04
Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scanning...
Available Devices:
Galaxy Buds
Xiaomi Band 8
SignLink Wristband
Skip for now (limited functionality)

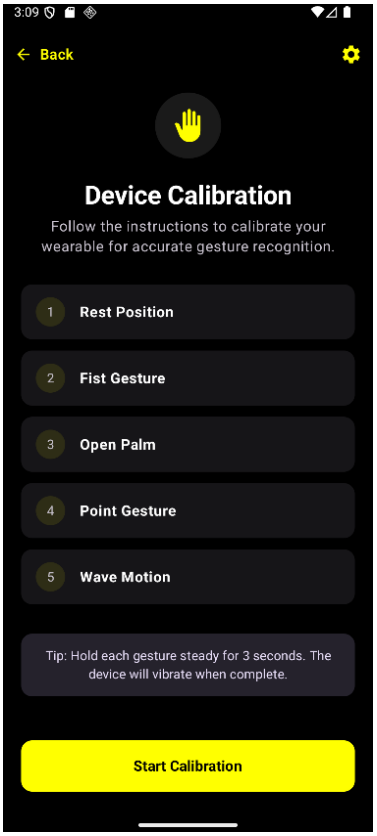

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scanning...
Available Devices:
Galaxy Buds
Xiaomi Band 8
SignLink Wristband
Skip for now (limited functionality)

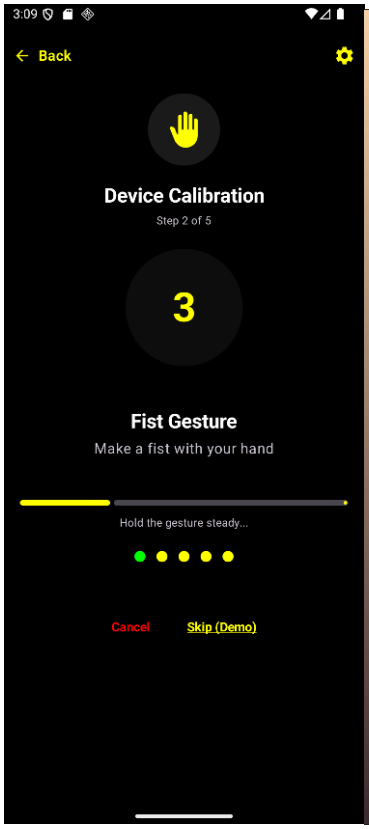

Back
Pair Your Wearable
Connect your sEMG/IMU wristband to enable sign language translation.
Before you start:
1. Turn on your wearable device
2. Make sure Bluetooth is enabled on your phone
3. Keep the device within 3 feet of your phone
Scanning...
Available Devices:
Galaxy Buds
Xiaomi Band 8
Skip for now (limited functionality)

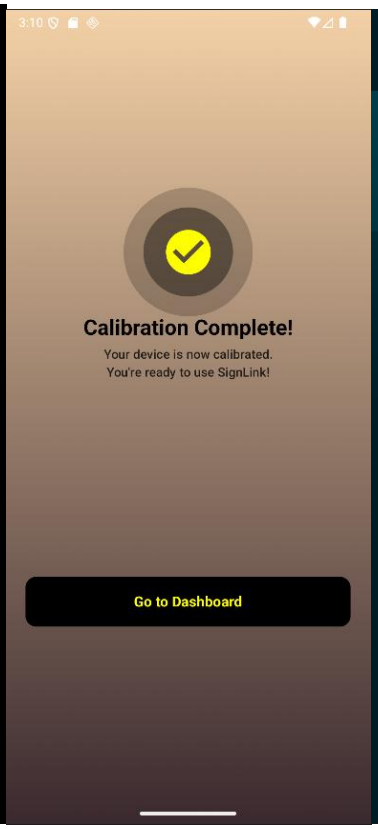

Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
1 Rest Position
2 Fist Gesture
3 Open Palm
4 Point Gesture
5 Wave Motion
Tip: Hold each gesture steady for 3 seconds. The device will vibrate when complete.
Start Calibration

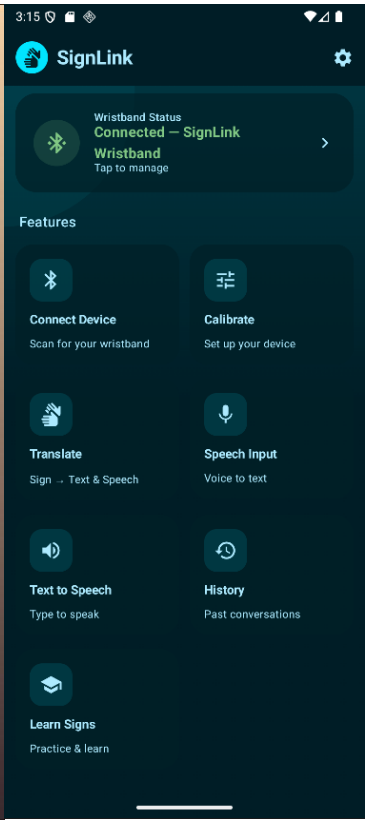

Back
Device Calibration
Step 3 of 5
3
Open Palm
Open your hand with palm facing forward
Hold the gesture steady...
Cancel
Skip (Demo)

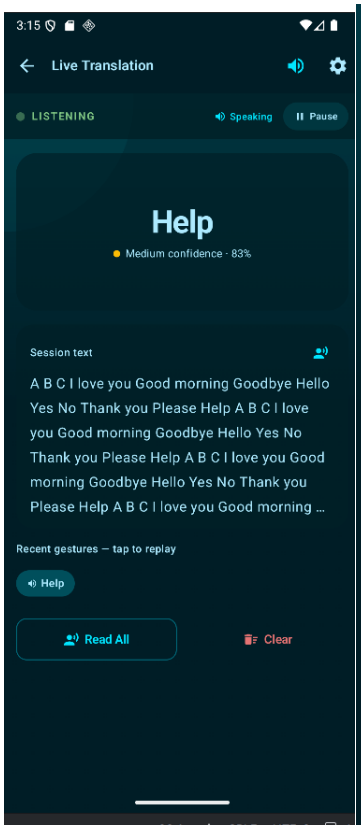

Calibration Complete!
Your device is now calibrated.
You're ready to use SignLink!
Go to Dashboard

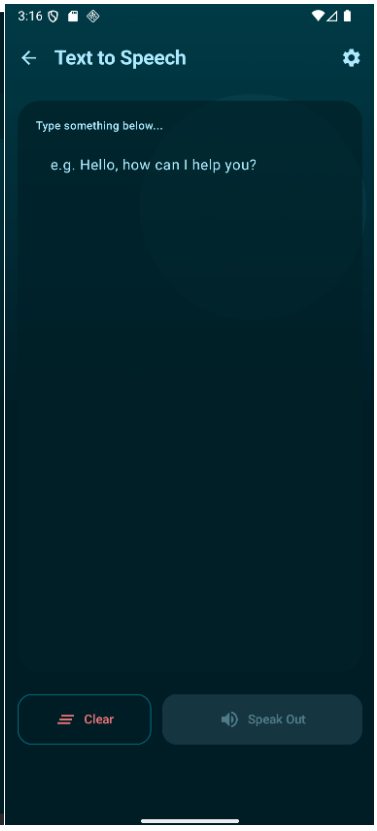

Back
Device Calibration
Follow the instructions to calibrate your wearable for accurate gesture recognition.
1 Rest Position
2 Fist Gesture
3 Open Palm
4 Point Gesture
5 Wave Motion
Tip: Hold each gesture steady for 3 seconds. The device will vibrate when complete.
Start Calibration

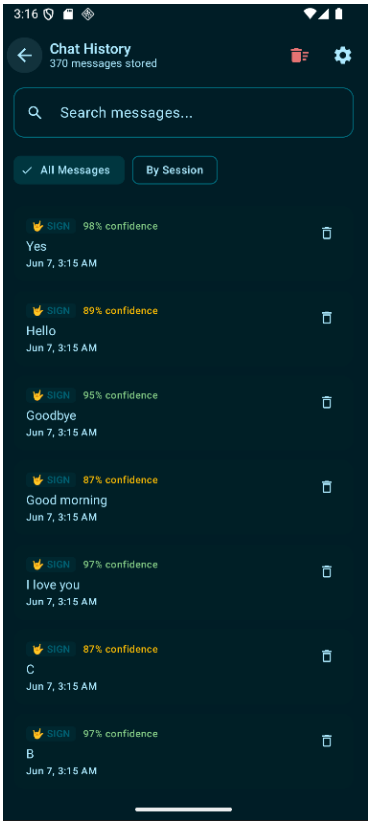

3:09
Back
Device Calibration
Step 2 of 5
3
Fist Gesture
Make a fist with your hand
Hold the gesture steady...
Cancel
Skip (Demo)

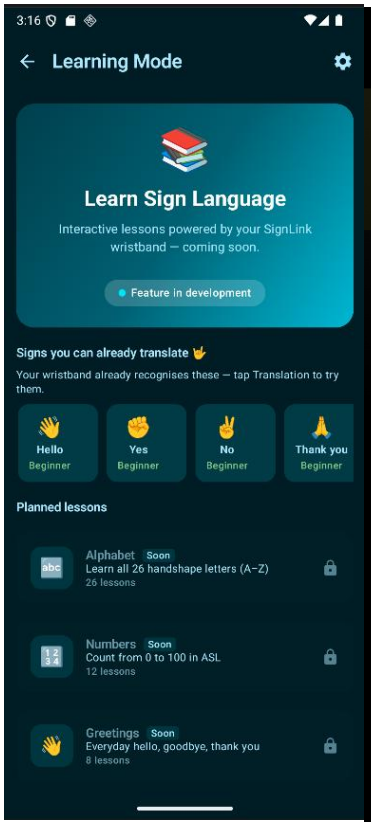

3:10
Calibration Complete!
Your device is now calibrated.
You're ready to use SignLink!
Go to Dashboard

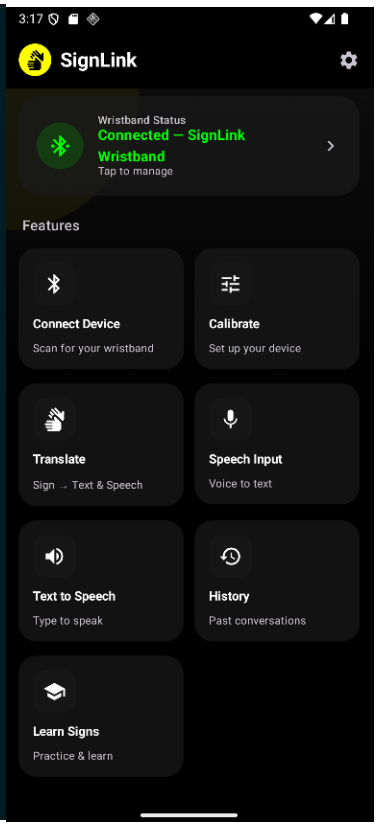

3:15
SignLink
Wristband Status
Connected — SignLink
Wristband
Tap to manage
Features
Connect Device
Scan for your wristband
Calibrate
Set up your device
Translate
Sign → Text & Speech
Speech Input
Voice to text
Text to Speech
Type to speak
History
Past conversations
Learn Signs
Practice & learn

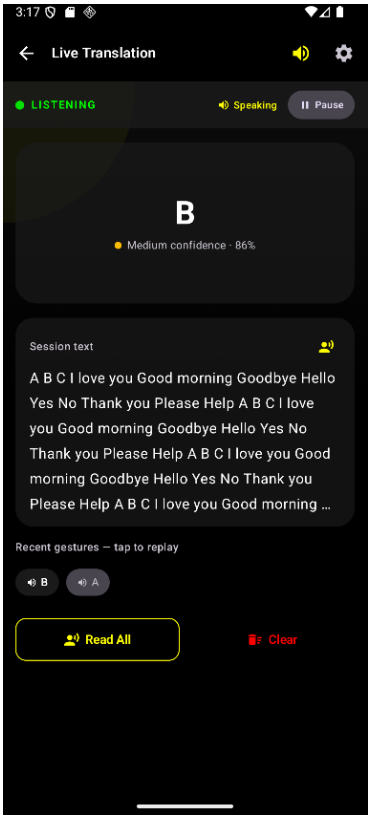

3:15
Live Translation
LISTENING
Speaking
Pause
Help
Medium confidence · 83%
Session text
A B C I love you Good morning Goodbye Hello Yes No Thank you Please Help A B C I love you Good morning Goodbye Hello Yes No Thank you Please Help A B C I love you Good morning Goodbye Hello Yes No Thank you Please Help A B C I love you Good morning ...
Recent gestures — tap to replay
Help
Read All
Clear


3:16
Text to Speech
Type something below...
e.g. Hello, how can I help you?
Clear
Speak Out

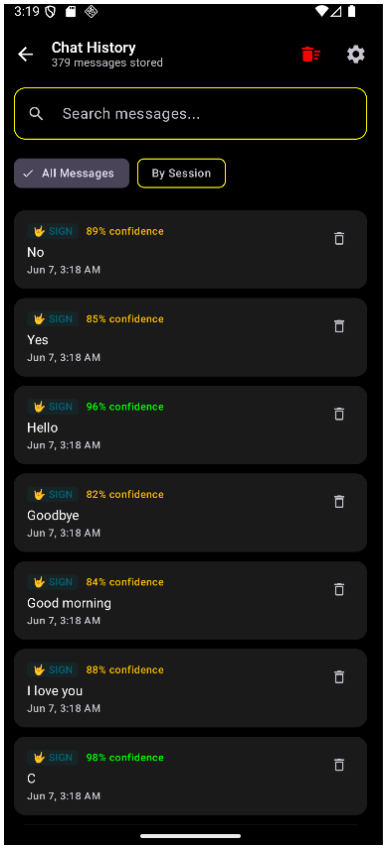

3:16
Chat History
370 messages stored
Search messages...
All Messages
By Session
SIGN 98% confidence
Yes
Jun 7, 3:15 AM
SIGN 89% confidence
Hello
Jun 7, 3:15 AM
SIGN 95% confidence
Goodbye
Jun 7, 3:15 AM
SIGN 87% confidence
Good morning
Jun 7, 3:15 AM
SIGN 97% confidence
I love you
Jun 7, 3:15 AM
SIGN 87% confidence
C
Jun 7, 3:15 AM
SIGN 97% confidence
B
Jun 7, 3:15 AM











and many more…

7.3 Model Training Code

```
rf = RandomForestClassifier(
    random_state=42,
    n_estimators=1000,
    criterion="gini",
    max_depth=None,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features="sqrt",
    bootstrap=True,
     n_jobs=-1
)

rf.fit(X_train, y_train)

```

Code 1: Control Random Forest Classifier (with no feature selection)

```
rf2 = RandomForestClassifier(
    random_state=42,
    n_estimators=1000,
    criterion="gini",
    max_depth=None,
    min_samples_split=5,
    min_samples_leaf=2,
    max_features="sqrt",
    bootstrap=True,
     n_jobs=-1
)
selector = SelectFromModel(
     rf,
     threshold='median',
     prefit=True
)
X_train_sel = selector.transform(X_train)
X_test_sel = selector.transform(X_test)

rf2.fit(X_train_sel, y_train)

```

Code 2: Random Forest Classifier (on selected features)

```
param_distributions = {
    'n_estimators': [300,500,800,1000,1500],
    'criterion': ['gini','entropy'],
    'max_depth': [None,20,30,40],
    'min_samples_split': [2,3,5,8,10],
    'min_samples_leaf': [1,2,3,5],
    'max_features': ['sqrt','log2']
}

cv = StratifiedKFold(
     n_splits=5,
     shuffle=True,
     random_state=42
)

# Initialize the RandomizedSearchCV object
# We use the rf2 model as the estimator
random_search = RandomizedSearchCV(
     estimator=rf2,
     param_distributions=param_distributions,
     n_iter=100,
     cv=cv,
     n_jobs=-1,
```

```
    verbose=2,
    random_state=42,
    scoring='accuracy'
)

# Fit the random search to the data
print("Starting Randomized Search for optimal hyperparameters...")
random_search.fit(X_train_sel, y_train)
print("Randomized Search complete.")

# Print the best parameters and the best score
print(f"Best parameters found: {random_search.best_params_}")
print(f"Best cross-validation accuracy: {random_search.best_score_:.4f}")

# You can access the best model directly
best_rf2_model_random = random_search.best_estimator_

# Evaluate the best model on the test set
best_preds_random = best_rf2_model_random.predict(X_test_sel)
best_acc_random = accuracy_score(y_test, best_preds_random)
print(f"Test accuracy with best parameters from Randomized Search: {best_acc_random:.4f}")

```

Code 3: Control Hyper-parameter Tuning & CV Testing (to make sure model was accurate on all folds and not just one specific instance)

```
seeds = [
    1,
    42,
    67,
    69,
    203,
    404,
    505,
    777,
    999,
    2025,
    3000,
    5000,
    6000,
    7000,
    8000,
    9000,
    10000,
]

# Define the parameter distributions to sample from for RandomizedSearch
param_distributions = {
    "n_estimators": [300, 500, 800, 1000, 1500],
    "criterion": ["gini", "entropy"],
    "max_depth": [None, 20, 30, 40],
    "min_samples_split": [2, 3, 5, 8, 10],
    "min_samples_leaf": [1, 2, 3, 5],
    "max_features": ["sqrt", "log2"],
}

# Lists to store accuracies for final statistics
rf1_accuracies = []
rf2_accuracies = []
rf2_tuned_accuracies = []
cv_scores = []

print("Running iterations over seeds...\n")
print(
    f"{'Seed':<8} | {'RF1 Accuracy':<14} | {'RF2 Accuracy':<14} | {'RF2 Tuned Acc':<14} | {'Best CV Score':<14}"
```

```
)
print("-" * 82)

for seed in seeds:
    # 1. Train/Test Split (Only changing this random_state)
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, stratify=y, random_state=seed
    )

    # 2. First Random Forest Model
    rf = RandomForestClassifier(
        random_state=42,
        n_estimators=1000,
        criterion="gini",
        max_depth=None,
        min_samples_split=5,
        min_samples_leaf=2,
        max_features="sqrt",
        bootstrap=True,
        n_jobs=-1,
    )

    rf.fit(X_train, y_train)
    preds = rf.predict(X_test)
    acc = accuracy_score(y_test, preds)
    rf1_accuracies.append(acc)

    # 3. Feature Selection & Second Random Forest Model
    rf2 = RandomForestClassifier(
        random_state=42,
        n_estimators=1000,
        criterion="gini",
        max_depth=None,
        min_samples_split=5,
        min_samples_leaf=2,
        max_features="sqrt",
        bootstrap=True,
        n_jobs=-1,
    )

    selector = SelectFromModel(rf, threshold="median", prefit=True)
    X_train_sel = selector.transform(X_train)
    X_test_sel = selector.transform(X_test)

    rf2.fit(X_train_sel, y_train)
    preds2 = rf2.predict(X_test_sel)
    acc2 = accuracy_score(y_test, preds2)
    rf2_accuracies.append(acc2)

    # 4. Hyperparameter Tuning on RF2 via RandomizedSearchCV
    cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)

    random_search = RandomizedSearchCV(
        estimator=rf2,
        param_distributions=param_distributions,
        n_iter=100,
        cv=cv,
        n_jobs=-1,
        verbose=0,  # Turned verbose down to 0 to keep the loop table readable
        random_state=42,
         scoring="accuracy",
    )

    random_search.fit(X_train_sel, y_train)

    # Evaluate the best randomized search model on the test set
```

```
    best_rf2_model_random = random_search.best_estimator_
    best_preds_random = best_rf2_model_random.predict(X_test_sel)
    best_acc_random = accuracy_score(y_test, best_preds_random)
    rf2_tuned_accuracies.append(best_acc_random)
    best_cv_score = random_search.best_score_
    cv_scores.append(best_cv_score)

    # Print individual results for the current seed
    print(f"{seed:<8} | {acc:<14.4f} | {acc2:<14.4f} | {best_acc_random:<14.4f} | {best_cv_score:<14.4f}")

print("-" * 82)

# 5. Calculate and display Mean and Standard Deviation
print(
    f"\n{'Metric':<10} | {'RF1 Model':<14} | {'RF2 Model':<14} | {'RF2 Tuned':<14} | {'Best CV Score':<14}"
)
print("-" * 82)
print(
    f"{'Mean':<10} | {np.mean(rf1_accuracies):<14.4f} | {np.mean(rf2_accuracies):<14.4f} | {np.mean(rf2_tuned_accuracies):<14.4f} | {np.mean(cv_scores):<14.4f}"
)
print(
    f"{'Std Dev':<10} | {np.std(rf1_accuracies):<14.4f} | {np.std(rf2_accuracies):<14.4f} | {np.std(rf2_tuned_accuracies):<14.4f} | {np.std(cv_scores):<14.4f}"
)

```

Code 4: Final Tuning (run on different random states to make sure accuracy is consistent throughout)

7.4 Appendices of Final UI/UX Design Section

7.4.1 Testing Consent Forms

TESTING INFORMATION SHEET

Project Title: Usability Evaluation of the SignLink Mobile Interface

Duration: June 5,2026 - June 7,2026

Researcher: Sombatla Truc Deydeepya Lakshmi

Institution: Vietnam National University of Information Technology (UIT)

You are invited to participate in a research study evaluating the user interface of **SignLink**, a mobile application designed to translate sign language into text and speech. Before you decide, it is important to understand why the research is being done and what it will involve. Please take the time to read the following information.

What is the purpose of this research? This research aims to evaluate the navigation logic and user experience of the SignLink app. While the final product will eventually include a wearable sensor (sEMG/IMU wristband), this test focuses exclusively on how intuitive the mobile software is for onboarding, calibration, and translation tasks.

Why have I been invited? You have been invited because you are either a student at UIT or a potential user who can provide feedback on the app's logic. We are looking for 5 participants to help identify friction points in the design.

What does participation involve? If you agree to take part, you will be asked to:

- Complete 3-4 specific tasks in real time with the demo-application.
- Use the **"Think Aloud"** method (verbalize your thoughts as you navigate).
- Complete a 10-item **System Usability Scale (SUS)** survey on a Google Form.
- Participate in a brief 5-minute exit interview. The entire session will last approximately **30 minutes**.

Is this compulsory to take part in? Your participation is entirely voluntary. You can withdraw at any time during the session without giving a reason and without any penalty.

How will my data be used? All data collected (notes, SUS scores, and interview responses) will be **anonymized**. You will be referred to as "Participant X" (e.g., P1, P2) in the final report. No personal identification will be shared.

Who can I contact for further information? If you have questions, you can contact the researcher at: ddpyawork@gmail.com

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES
I agree to take part in the above study.	YES

Participant Name: [redacted]

Participant Age (if under 18 years of age): [redacted]

Date: 5/6/2026

Signature:



BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	(YES) / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	(YES) / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	(YES) / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	(YES) / NO
I agree to take part in the above study.	(YES) / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 05/06/2026

Signature: Andrew

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age): [redacted]

Date: 5/6/2026

Signature:



BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 6/6/2026

Signature:



BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	(YES) / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	(YES) / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	(YES) / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	(YES) / NO
I agree to take part in the above study.	(YES) / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age): [redacted]

Date: 06/06/2026

Signature: 

BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 6/6/2026

Signature:

Dr. Lasya

BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

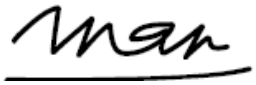
Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: June 6th 2026

Signature: 

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 6/6/2026

Signature:



BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

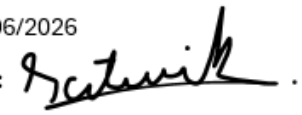
Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 06/06/2026

Signature: 

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 6 June 2026

Signature: SDSheetty

BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 05/06/2026

Signature: 

BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

Date: 6th June, 2026

Signature:



CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	YES / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	YES / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 6/6/2026

Signature: 

BIRMINGHAM CITY University

CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

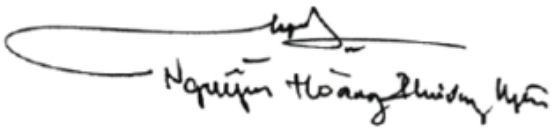
Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	**YES** / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	**YES** / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	**YES** / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	**YES** / NO
I agree to take part in the above study.	**YES** / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) : [redacted]

Date: 6th June 2026

Signature: [redacted]



CONSENT FORM

Project Title: Usability Evaluation of the SignLink Mobile Interface

Researcher: Sombatla Truc Deydeepya Lakshmi

Please tick/circle each box to confirm your agreement:

Statement	Option
I confirm that I have read and understood the Information Sheet for the above study.	~~YES~~ / NO
I have had the opportunity to consider the information, ask questions, and have had these answered satisfactorily.	YES / NO
I understand that my participation is voluntary and that I am free to withdraw at any time without giving a reason.	YES / NO
I agree to the researcher taking notes during the session and understand that my comments will be anonymized.	YES / NO
I agree to take part in the above study.	~~YES~~ / NO

Participant Name: [redacted]

Participant Age (if under 18 years of age) :

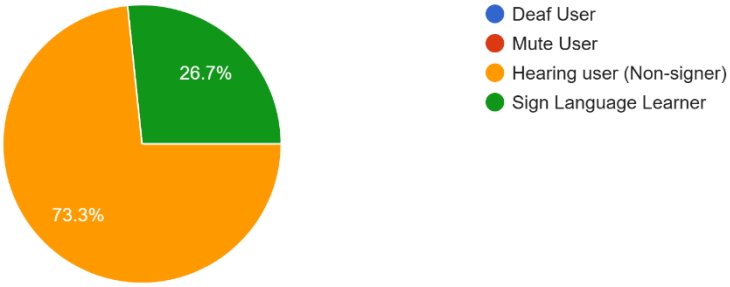
Date: 05/06/2026

Signature:

7.1.5 Results from Form

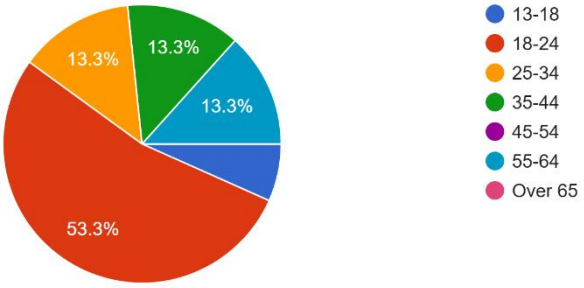
Please select the category that you fall under.

15 responses



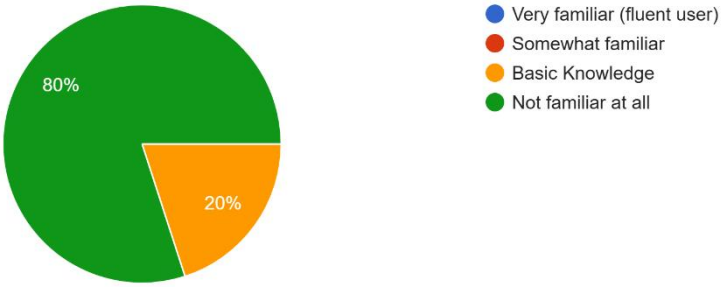
Age

15 responses



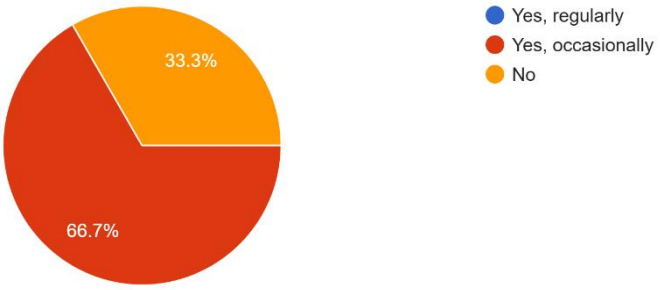
How familiar are you with sign language?

15 responses



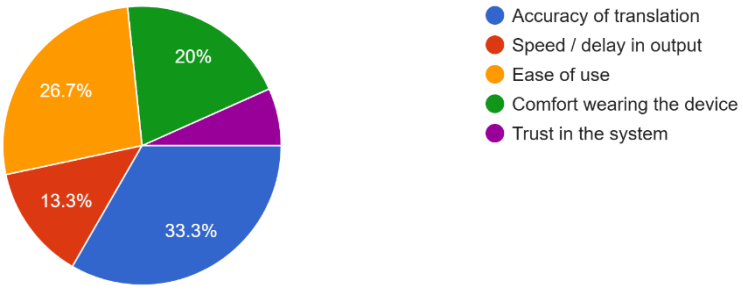
Have you ever used assistive or translation technology before?

15 responses



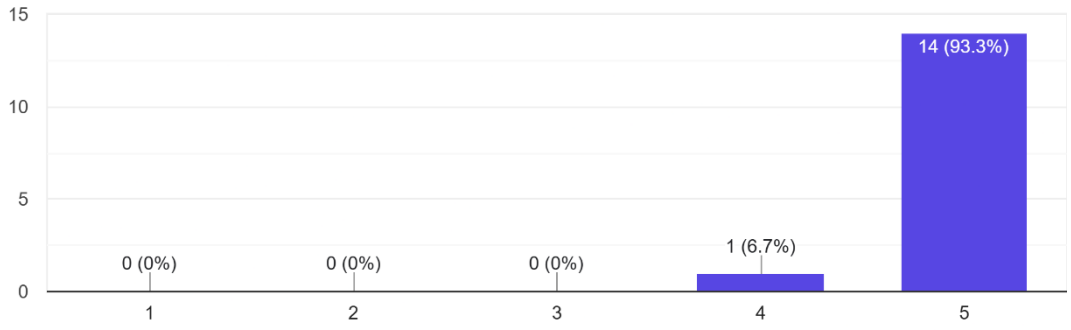
In what context would you most likely use a sign language translation app? (e.g., daily communication, education, public services, emerge...enge when using a sign language translation app?

15 responses



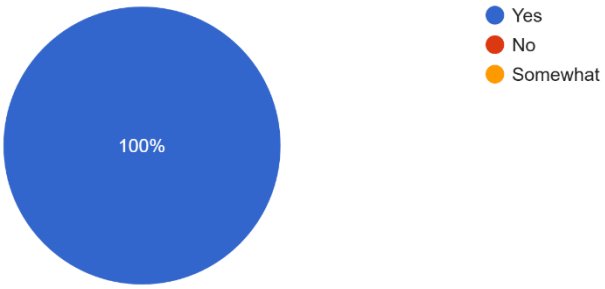
How clear was the purpose of the app when you first opened it? [1 (Not clear at all) – 5 (Very clear)]

15 responses



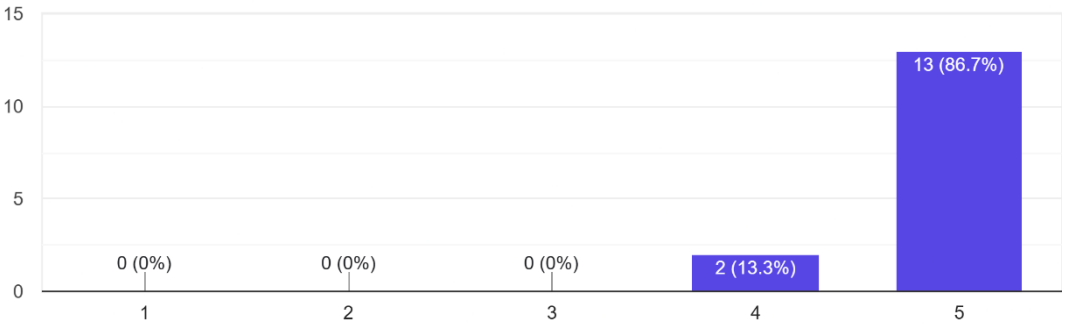
Did you feel confident about what to do next after seeing the home screen?

15 responses



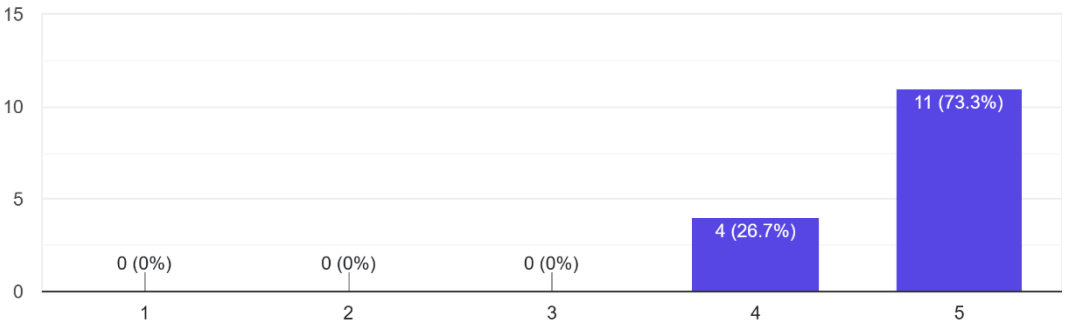
How easy was it to find and use the option to connect the wearable device? [1 (Very difficult) – 5 (Very easy)]

15 responses



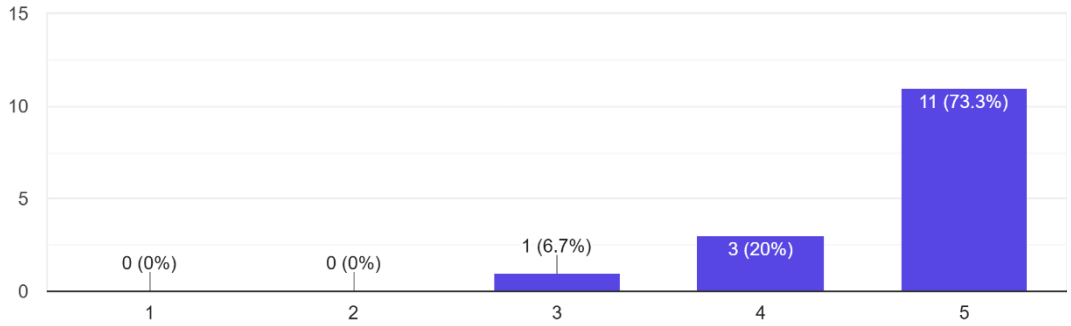
How easy was it to understand the calibration instructions? [1 (Very difficult) – 5 (Very easy)]

15 responses



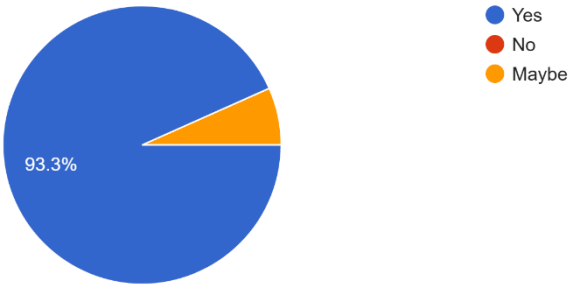
How easy was it to switch from text output to speech output? [1 (Very difficult) – 5 (Very easy)]

15 responses



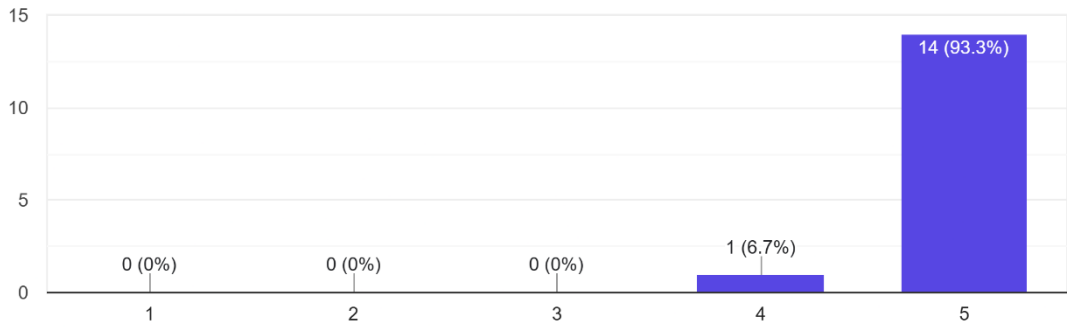
Did you feel the app was designed with accessibility in mind?

15 responses



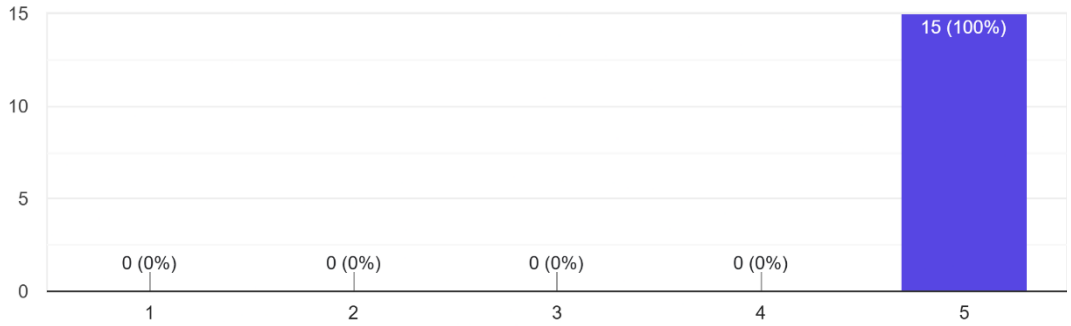
How readable was the text size and contrast? [1 (Very difficult) – 5 (Very easy)]

15 responses



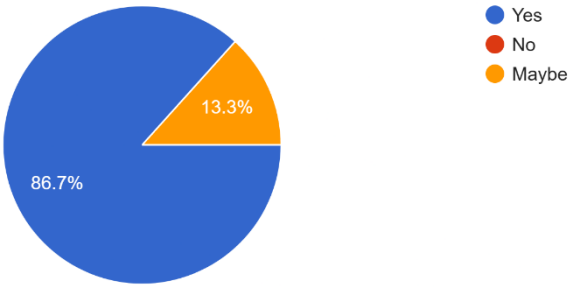
How clear and readable was the translated text on screen? [1 (Very difficult) – 5 (Very easy)]

15 responses



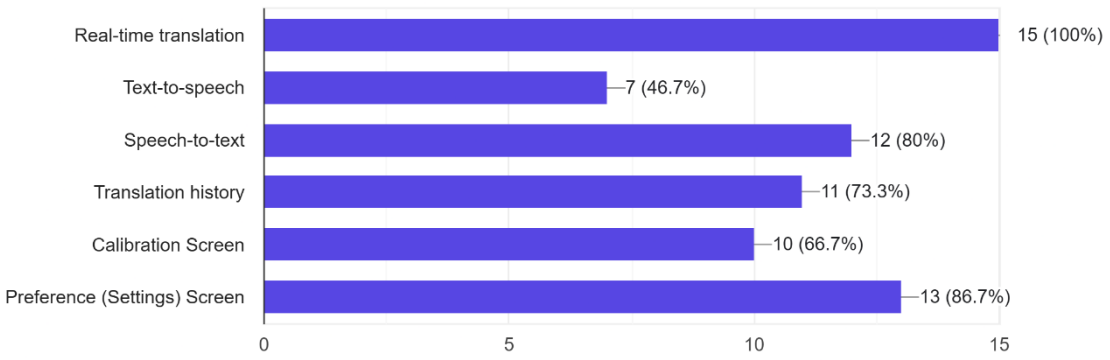
Did the interface feel visually clear and uncluttered?

15 responses



Which features did you find most valuable? (Select all that apply)

15 responses



What additional feature would most improve your experience?

7 responses

Translation to many languages

No ideas at the moment

What if you had a feature to translate into many different languages and also one place with videos of the sign

I want to be able to put in a sentence and see it translate that into a sequence of 3D hand signs in sign language. This would both help me learn and I think would be more inclusive.

Multi-language translation

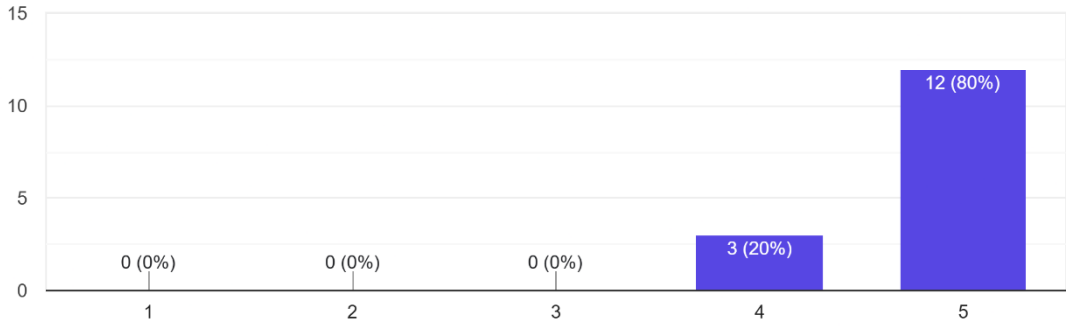
many languages translation and types of sign lang

No ideas yet

- SUS

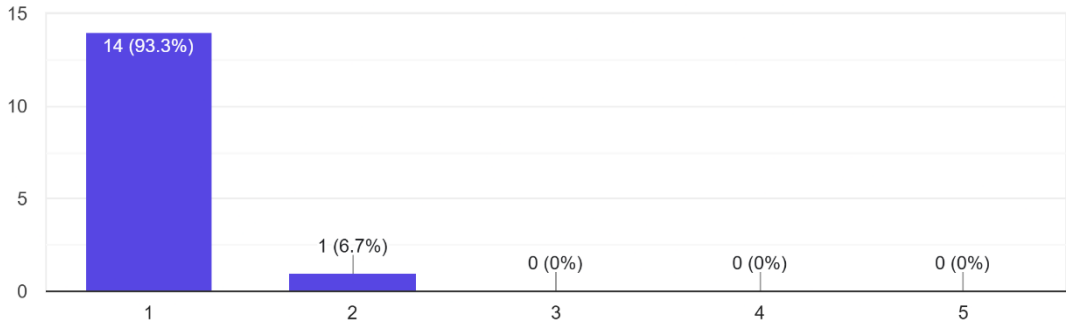
I think that I would like to use SignLink every time the need for translating sign language arises.

15 responses



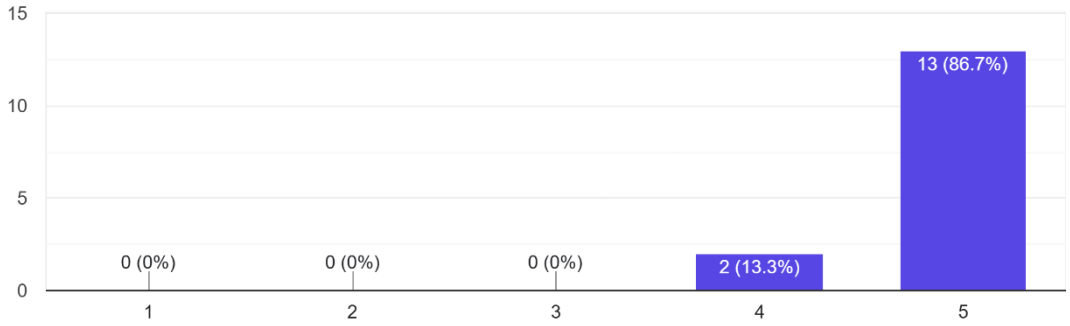
I found the app unnecessarily complex and didn't know where to find what I needed.

15 responses



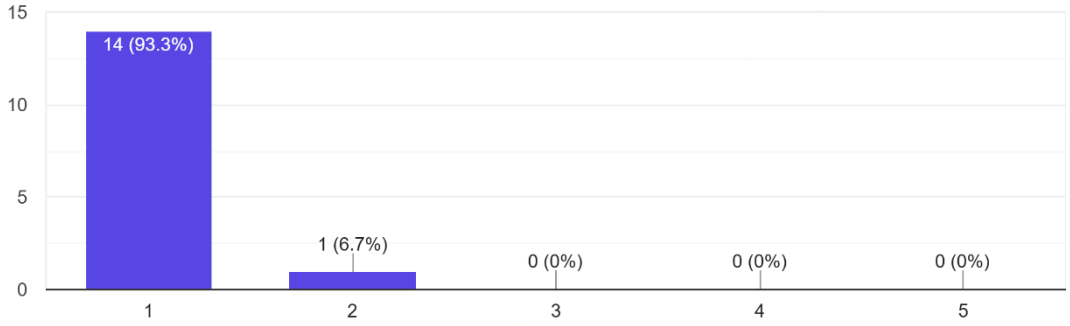
I thought the app was easy to use and was quite intuitive.

15 responses



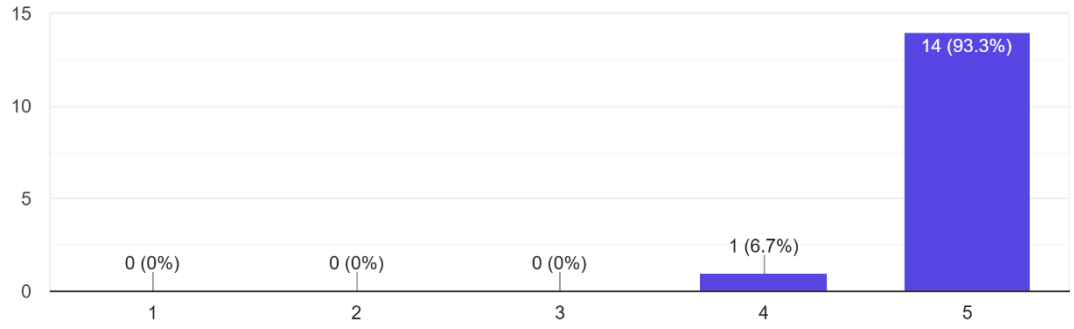
I think that I would need the support of a technical person to be able to use this app.

15 responses



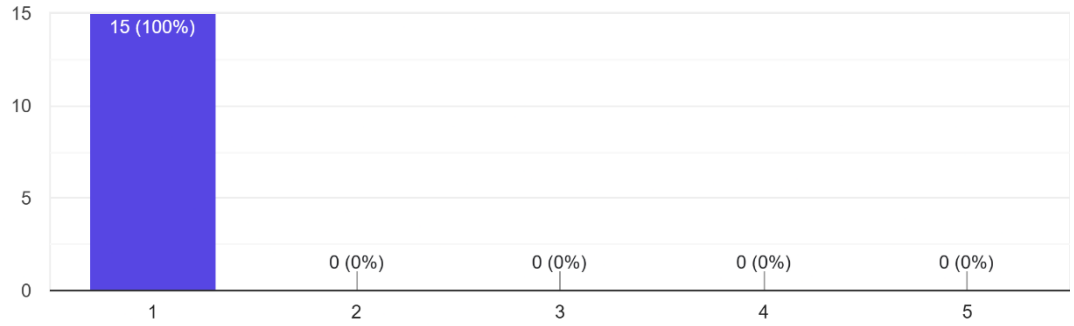
I found the various functions in this app were well integrated.

15 responses



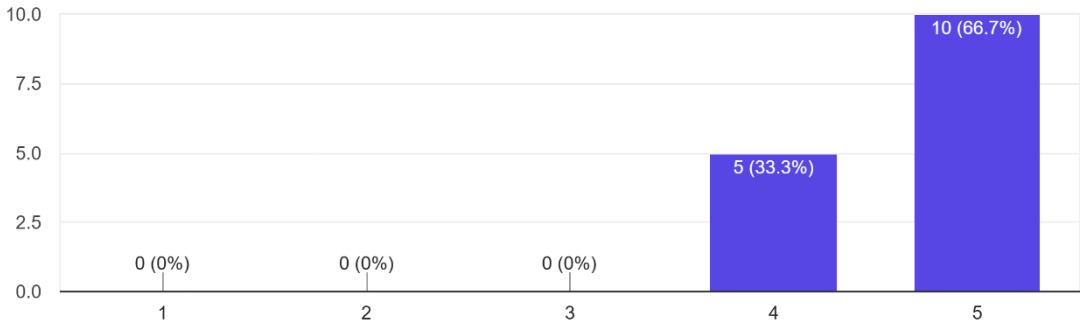
I thought there was too much inconsistency in this app.

15 responses



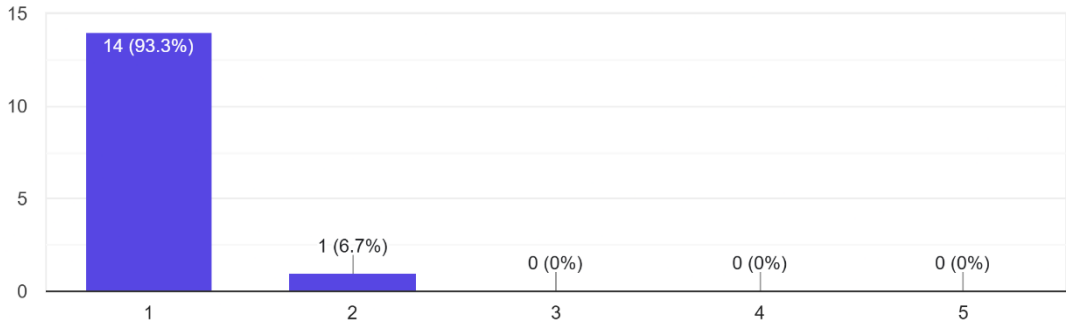
I would imagine that most people would learn to use this app very quickly.

15 responses



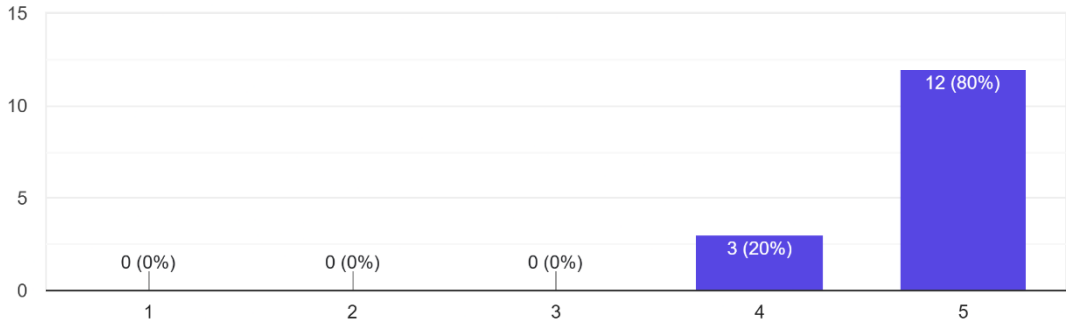
I found the app very cumbersome/awkward to use.

15 responses



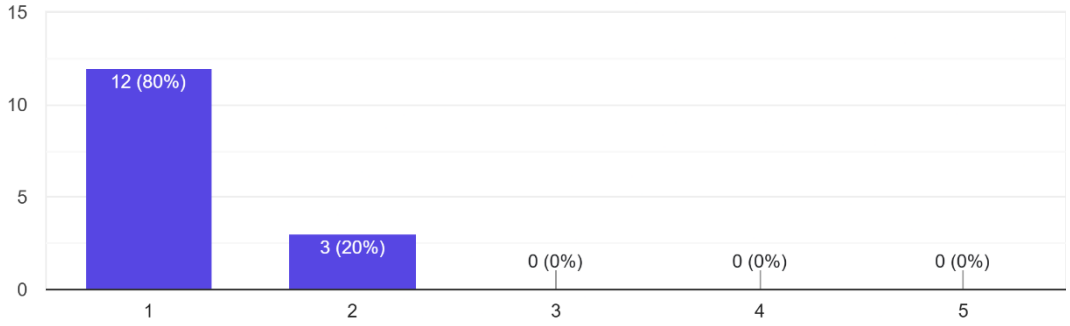
I felt very confident using the app.

15 responses



I needed to learn a lot of things before I could get going with this app.

15 responses



What was the most confusing part of the navigation?

10 responses

I am not confused

NA

Nice work

I feel okay :))

I think it was interesting and easy to use

Not confused

None to be honest. It was easy to find the things I need if I was to communicate with someone using sign language

good

its good

How did you feel about the calibration process without the physical wristband?

9 responses

its quite alright

It was okay because I understand but would be even better if it is there.

Ok

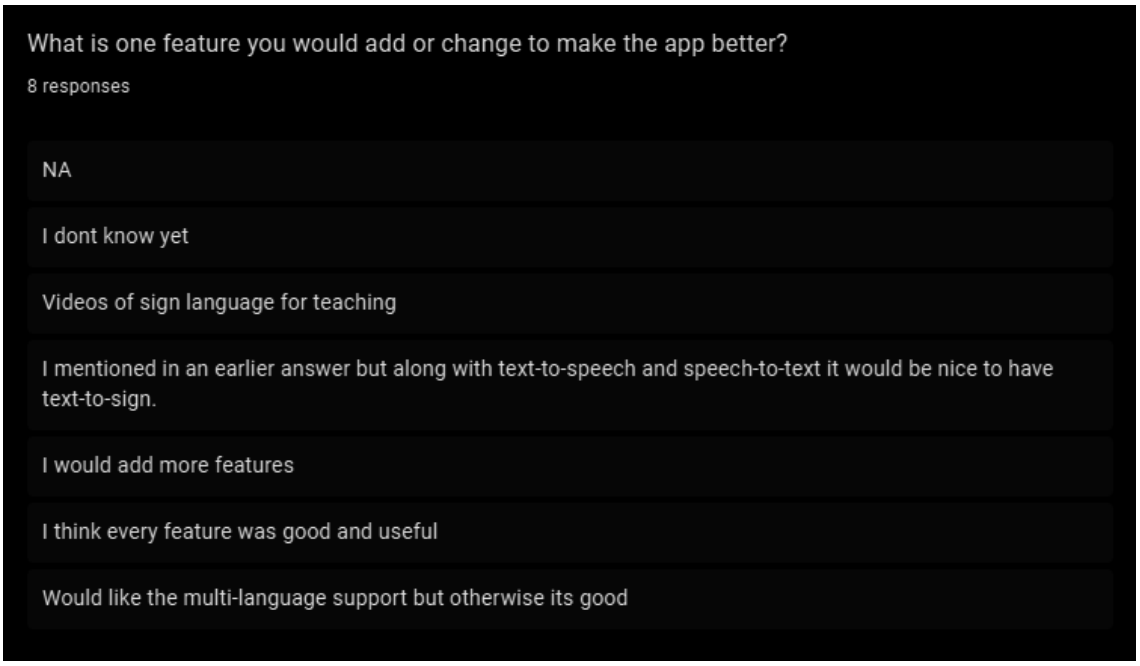
A little weird

I think it would be interesting with the wristband but it is okay because the app seems to still have a purpose without the item.

Interesting

I could visualize it but I would have loved to try the gadget with the app. However, the interface made it clear what would happen if in an ideal situation where I would have it

its okay


What is one feature you would add or change to make the app better?
8 responses
NA
I dont know yet
Videos of sign language for teaching
I mentioned in an earlier answer but along with text-to-speech and speech-to-text it would be nice to have text-to-sign.
I would add more features
I think every feature was good and useful
Would like the multi-language support but otherwise its good